A DTV Profile for Uncompressed High Speed Digital Interfaces

1 SCOPE

CTA-861 establishes protocols, requirements, and recommendations for the utilization of uncompressed digital interfaces by consumer electronics devices such as Digital Televisions (DTVs), digital cable, satellite, or terrestrial set-top boxes (STBs), and related peripheral devices including, but not limited to DVD players/recorders, and other related Sources or Sinks.

CTA-861 is applicable to a variety of standard DTV-related high-speed digital physical interfaces — such as Digital Visual Interface (DVI) 1.0 [4], Open LVDS Display Interface (LDI) [8], and High-Definition Multimedia Interface (HDMI) [110] specifications. Protocols, requirements, and recommendations that are defined include Video Formats and waveforms; colorimetry and quantization; transport of compressed and uncompressed, as well as Linear Pulse Code Modulation (L-PCM), audio; carriage of auxiliary data; and implementations of the Video Electronics Standards Association (VESA*) Enhanced Extended Display Identification Data Standard* (E-EDID) [9], which is used by Sinks to declare display capabilities and characteristics.

CTA-861 adopters are strongly encouraged to implement High-bandwidth Digital Content Protection (HDCP) [3] content protection, defined by the Digital Content Protection, LLC (DCP) method, in order to be compatible with digital cable STBs as authorized by 47 C.F.R. § 76.602 [108] and 47 C.F.R. §76.640 [109]. HDCP [3] permits viewing of high-value content that may be available from other video Sources in a home network.

2 GENERAL

2.1 References

CTA-861 includes mechanisms that allow a digital video Source (such as a cable, satellite or terrestrial STB, digital VCR, or DVD player) to supply displayable, baseband, digital video to High Definition Television (HDTV) devices, as well as peripheral devices such as repeaters, switchers, and recorders.

2.1.1 Normative References

The following standards contain provisions that, through reference in this text, constitute normative provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision. Users of this standard are cautioned that a newer edition might or might not be compatible.

2.1.1.1 Normative Reference List

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2. SMPTE ST 274:2008, *Television — 1920 x 1080 Image Sample Structure, Digital Representation and Digital Timing Reference Sequences for Multiple Picture Rates*, https://www.smpte.org/standards/document-index/st.
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6. Recommendation ITU-R BT.601-7, *Studio Encoding parameters of Digital Television for standard 4:3 and wide-screen 16:9 aspect ratios*, March 2011, https://www.itu.int/rec/R-REC-BT.601.
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8. *Open LVDS Display Interface (Open LDI) Specification,* Version 0.95, May 1999, https://inova-semiconductors.de/files/daten/pdf/openldi.pdf (accessed October 29, 2020).
9. *VESA Enhanced Extended Display Identification Data (E-EDID) Standard*, Release A, Revision 1, February 2000, https://vesa.org/vesa-standards/.

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11. ATSC Standard A/52:2018, *Digital Audio Compression (AC-3, E-AC-3*), January 2018, https://www.atsc.org/documents/.

12. IEC 60958-3:2021, *Digital Audio Interface — Part 3: Consumer Applications*, https://webstore.iec.ch/.

13. IEC 61909:2000, *Audio recording — Minidisc system*, https://webstore.iec.ch/.

14. IEC 61937-3:2017, *Digital audio — Interface for non-linear PCM encoded audio bitstreams applying IEC 60958 — Part 3: Non-linear PCM bitstreams according to the AC-3 and enhanced AC-3 formats with AMD1:2020*, https://webstore.iec.ch/.

15. IEC 61937-4:2003, *Digital audio — Interface for non-linear PCM encoded audio bitstreams applying IEC 60958 — Part 4: Non-linear PCM bitstreams according to the MPEG audio formats*, https://webstore.iec.ch/.

16. IEC 61937-5:2006, *Digital audio — Interface for non-linear PCM encoded audio bitstreams applying IEC 60958 — Part 5: Non-linear PCM bitstreams according to the DTS (Digital Theater Systems) format(s) with AMD1:2019*, https://webstore.iec.ch/.

17. IEC 61937-6:2006, *Digital audio — Interface for non-linear PCM encoded audio bitstreams applying IEC 60958 — Part 6: Non-linear PCM bitstreams according to the MPEG-2 AAC and MPEG-4 AAC audio formats with AMD1:2014*, https://webstore.iec.ch/.

18. IEC 61937-7:2004, *Digital audio — Interface for non-linear PCM encoded audio bitstreams applying IEC 60958 — Part 7: Non-linear PCM bitstreams according to the ATRAC, ATRAC2/3 and ATRAC-X formats with AMD1:2016*, https://webstore.iec.ch/.

19. IEC 61937-8:2006, *Digital audio — Interface for non-linear PCM encoded audio bitstreams applying IEC 60958 — Part 8: Non-linear PCM bitstreams according to the Windows Media Audio (WMA) Professional format*, https://webstore.iec.ch/.

20. IEC 61937-9:2017, *Digital audio — Interface for non-linear PCM encoded audio bitstreams applying IEC 60958 — Part 9: Non-linear PCM bitstreams according to the MAT format*, https://webstore.iec.ch/.

21. IEC 61937-11:2021, *Digital audio — Interface for non-linear PCM encoded audio bitstreams applying IEC 60958 — Part 11: MPEG-4 AAC and its extensions and MPEG-D USAC in LATM/LOAS*, https://webstore.iec.ch/.

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23. ISO/IEC 11172-3:1993, *Information Technology — Coding of moving pictures and associated audio for digital storage media at up to about 1.5 Mbit/sec, Part 3: Audio*, https://webstore.iec.ch/.

24. ISO/IEC 13818-3:1998, *Information Technology — Generic coding of moving pictures and associated audio information, Part 3: Audio*, https://webstore.iec.ch/.
25. ISO/IEC 14496-3:2019*, Information Technology — Coding of audio-visual objects — Part 3: Audio*, https://webstore.iec.ch/.
26. ISO/IEC 23003-1:2007, *Information technology — MPEG audio technologies — Part 1: MPEG Surround with AMD2:2008 and COR1:2009*, https://webstore.iec.ch/.
27. DVD Forum, *DVD Specifications for Read-Only Disc, Part 4, Audio Specifications, Version 1.0, Meridian Lossless Packing*, https://www.dvdfllc.co.jp/.
28. SCTE 127:2019, *Carriage of Vertical Blanking Interval (VBI) Data in North American Digital Television Bitstreams*, https://www.scte.org/standards/library/catalog/.
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32. IEC 61966-2-5:2007, *Multimedia systems and equipment — Colour measurement and management — Part 2-5: Colour management — Optional RGB colour space — opRGB*, https://webstore.iec.ch/.
33. IEC 61966-2-1:1999, *Multimedia systems and equipment — Colour measurement and management — Part 2-1: Colour management — Default RGB colour space — sRGB*, https://webstore.iec.ch/.
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35. SMPTE ST 2016-1:2009, *Format for Active Format Description and Bar Data*, https://www.smpte.org/standards/document-index/st.
36. ETSI TS 102 114 v1.6.1, *DTS Coherent Acoustics; Core and Extension with Additional Profiles*, https://www.etsi.org/standards.
37. ETSI TS 103 491 v1.2.1, *DTS-UHD Audio Format; Delivery of Channels, Objects and Ambisonic Sound Fields*, https://www.etsi.org/standards.
38. ANSI INCITS 4-1986[R2017], *Coded Character Sets — 7-Bit American National Standard Code for Information Interchange (7-Bit ASCII), Table 8*, https://www.incits.org/.
39. GB/T 22726-2008, *Specification for multichannel digital audio coding technology*, http://gbstandards.org/.

40. Recommendation ITU-R BT.2020-2, *Parameter values for ultra-high definition television systems for production and international programme exchange*, October 2015, https://www.itu.int/rec/R-REC-BT.2020.

41. SMPTE ST 2084:2014, *High Dynamic Range Electro-Optical Transfer Function of Mastering Reference Displays*, https://www.smpte.org/standards/document-index/st.

42. SMPTE ST 2086:2018, *Mastering Display Color Volume Metadata Supporting High Luminance and Wide Color Gamut Images*, https://www.smpte.org/standards/document-index/st.

43. ISO/IEC 62574:2020, *Audio, video and multimedia systems — General channel assignment of multichannel audio*, https://webstore.iec.ch/.

44. ISO/IEC 23008-3:2019, *Information technology — High efficiency coding and media delivery in heterogeneous environments — Part 3: 3D audio*, https://webstore.iec.ch/.

45. ETSI TS 103 190 v1.1.1, Digital Audio Compression (AC-4) Standard, April 2014, https://www.etsi.org/standards.

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48. Recommendation ITU-R BT.2100-2, *Image parameter values for high dynamic range television for use in production and international programme exchange*, July 2018, https://www.itu.int/rec/R-REC-BT.2100.

49. SMPTE ST 2113:2018, *Colorimetry of P3 Color Spaces,* https://www.smpte.org/standards/document-index/st.

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54. SMPTE ST 2094-10:2021, *Dynamic Metadata for Color Volume Transform — Application #1*, https://www.smpte.org/standards/document-index/st.

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57. *VESA DisplayID Standard*, Version 2.1, December 2021, https://vesa.org/vesa-standards/.

58. IEC 61937-15:2021, *Digital audio - Interface for non-linear PCM encoded audio bitstreams applying IEC 60958 - Part 15: Non-linear PCM bit streams according to Auro-Cx format*, https://webstore.iec.ch/.

59. ISO/IEC 23003-3:2020, *Information technology – MPEG audio technologies – Part 3: Unified speech and audio coding*, https://webstore.iec.ch/.

60. ETSI TS 103 572 v1.3.1, *HDR Signalling and Carriage of Dynamic Metadata for Colour Volume Transform; Application #1*, August 2021, https://www.etsi.org/standards.

61. - 99. [RESERVED]

2.1.2 Informative References

The following documents contain information that is useful in understanding this standard. At the time of publication, the editions indicated were valid.

2.1.2.1 Informative Document List

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101. SMPTE ST 296:2012, *1280 x 720 Progressive Image 4:2:2 and 4:4:4 Sample Structure — Analog and Digital Representation and Analog Interface*, https://www.smpte.org/standards/document-index/st.

102. SMPTE ST 125:1995, *Television — Component Video Signal 4:2:2 — Bit-Parallel Digital Interface*, https://www.smpte.org/standards/document-index/st.

103. SMPTE ST 2035:2009, *Audio Channel Assignments for Digital Television Recorders (DTRs),* https://www.smpte.org/standards/document-index/st.

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105. ANSI/CTA-708-E, *Digital Television (DTV) Closed Captioning*, August 2013, https://cta.tech/Standards/.

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112. ISO/IEC 13818-2:2013, *Information technology — Generic coding of moving pictures and associated audio information: Video*, https://webstore.iec.ch/.

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131. SMPTE RP 91:2014, *Specifications for 70-mm Projector Alignment and Screen Image Quality Test Film (R2007),* https://www.smpte.org/standards/document-index/rp.

132. SMPTE RP 202:2008, *Video Alignment for Compression Coding*, https://www.smpte.org/standards/document-index/rp.

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143. SMPTE RP 187:1995, *Center, Aspect Ratio and Blanking of Video Images*, https://www.smpte.org/standards/document-index/rp.

2.2 Definitions

For the purposes of CTA-861, the following definitions apply.

Active Format Description (AFD): A data structure that describes what portion of the Picture actually contains useful information (e.g., letterbox and pillarbox Bars are not considered useful information). It is a 4-bit field like that standardized in Annex B of ETSI TS 101 154 [107], but whose exact meaning may depend on whether the data is delivered per ATSC/SCTE or ETSI standard. See Section 6.4 for details. Note that the use of the term "active" in this definition is not consistent with the use of this term in other portions of CTA-861 and most of the other documents referenced by CTA-861.

Active Image: The useful portion of the image contained within a Picture. Active Image excludes letterbox and pillarbox Bars (see Annex N).

Active Line: A Video Line occurring during the Vactive period(s) containing both Active Pixels and Blank Pixels. Active Pixels and Blank Pixels fill the Hactive and Hblank portions of these lines, respectively (see Annex N).

Active Pixel: A Video Pixel that conveys Pixel Data (see Annex N).

Auxiliary Video Information (AVI): Additional information (defined in CTA-861) related to the video being sent from a Source to a Sink.

A/V: Audio and Video.

Bar Data: Data that enable computation of regions of the image that are outside of the Active Image within a Picture, for example areas of zero or uniform luminance (i.e., Bars).

Bar Pixel: An Active Pixel that conveys a portion of a Bar (see Annex N).

Bars: Region of the display screen that is being driven or scanned at either zero luminance or at a uniform luminance; or regions of a Picture that are intended to be driven (e.g., matrix addressed) or scanned (e.g., cathode ray tube (CRT)) at either zero luminance or at a uniform luminance. In other words, it is the portion of the Picture that does not contain useful information (see Annex N).

Basic Audio: Uncompressed, two channel, digital audio. Exact parameters are determined by the interface specification used with CTA-861 (e.g., 2 channel IEC 60958-3 [12] L-PCM, 32, 44.1, and 48 kHz sampling rates, 16 bits/sample).

Blank Pixel: A Video Pixel that carries data other than Pixel Data (see Annex N).

Blanking Line: A Video Line occurring during Vblank period(s) containing only Blank Pixels. Blank Pixels fill both Hactive and Hblank portions of these lines (see Annex N).

Byte: 8 bits of data.

CE Video Format: Any Video Format listed in Table 1, or listed in Table 11 and Table 12, except the 640x480p Video Format.

Channel, **Speaker**: There is a direct relation between a **Channel** and a **Speaker**. Audio feeding into a **Speaker** is sent via a **Channel.**

Color Component Sample: A value encoded using a Pixel Encoding that conveys a portion of the total color information about of a Video Pixel. A Color Component Sample might convey a red (R), green (G) or blue (B) color component, or might convey a luma component, or might convey a chroma component. A luma component, Y or I, represents luminance or intensity respectively, and a chroma component, C, represents a color difference.

Color Space: A tristimulus representation of colors typically based on the CIE 1931 color matching functions

Color Volume: Space of all colors and intensities that a device or signal can reproduce or convey

Colorimetry: Combination of color gamut, color primaries, dynamic range, and color representation (e.g., YC_BC_R or RGB). Note that Transfer Function, Component Depth, Quantization Range, and chroma sample location can often vary for a specific colorimetry, although defaults and/or constraints for some of these might exist for some colorimetries.

Component Depth: The number of bits used to represent a Color Component Sample. It is generally denoted as N.

Compressed Audio: All audio formats other than L-PCM and DSD Audio.

Content Pixel: An Active Pixel that conveys a portion of the Active Image (see Annex N).

CTA Extension: The E-EDID Standard [9] defines a VESA-assigned tag (0x02) that allows for an extension to be added with additional timing formats.

CTA Video Format: Any Video Format listed in Table 1, or listed in Table 11 and Table 12.

Digital Television (DTV): A device that receives, decodes, and presents audio and video material that has been transmitted in a digital compressed form. The device can be a single unit or it can be constructed from a number of individual components (e.g., a digital terrestrial STB and an analog television).

Direct Stream Digital (DSD): Audio format, also known as One Bit Audio.

Direct Stream Transfer (DST): A lossless compression scheme for the Direct Stream Digital (DSD) audio format.

Dual-Aspect Ratio DTV: A DTV that simultaneously supports both Picture Aspect Ratios of a Video Format Timing (e.g., 720x480p). Listing both formats in the EDID data structure at the same time signifies simultaneous support.

Dual-Aspect Ratio Timing: A Video Format Timing (e.g., 720x480p) that is available in both Picture Aspect Ratios (16:9 and 4:3) with no difference in the timing for the two formats.

Electro-Optical Transfer Function (EOTF): A mathematical function that describes the relationship between the luminance values input to a display device and the values output by the display.

Frame Rate: Rate, normally in Hz, of sequence of video frames

Full Range: R, G, B or Y Quantization Range that includes all code values. See Section 5.4.

Graphics Overlay: Content, possibly semi-transparent, that is superimposed over and/or around the original video, typically by a set-top-box or video playback device.

Harmonic Video Timing: Video Timing from a particular set of legacy timings based on exhaustive enumeration, sorting, and selection given criteria that includes "pixel clock shall be integer-divided from a GHz prime factor ($2^a*3^b*5^c*11^d$) clock."

High Channel Count Audio (HCCA): Channel based audio that can be presented in speaker configurations greater than 7.1. For CTA-861, HCCA is limited to 32 channels.

High Definition (HD): A CE Video Format that, inclusively, has 720 to 1080 active vertical lines (Vactive) lines per Video Frame.

High Definition Television (HDTV): A DTV capable of displaying a 1920x1080i or 1280x720p Video Format in 16:9 Picture Aspect Ratio. See Section 3.2.

HVT Video Format: Any CE Video Format based on HVT. HVT Video Formats are listed in Table 1.

IC_TC_P: A Constant Intensity video signal format, where the I component represents intensity, and the C_T and C_P components represent color. This signal format can be sub-sampled in the same manner as YC_BC_R.

InfoFrame: A data transfer structure for sending miscellaneous information from a Source to a Sink over a CTA-861 interface. Various InfoFrames are described in Section 6.

Interface Development Organization (IDO): The organization (e.g., HDMI LLC, HDMI Forum, VESA) responsible for the CTA-861 implementation that is present.

Interface VSIF: One or more VSIF(s) defined by the IDO.

IRE Unit: A percentage of reference white with respect to black (i.e., blanking level). Reference white is assigned a value of 100, blanking a value of 0.

IT Video Format: Any Video Format that is not a CE Video Format. Specifically, any Video Format not listed in Table 1, nor listed in Table 11 and Table 12, plus the 640x480p Video Format.

Limited Range: R, G, B or Y Quantization Range that excludes some code values at the extremes. See Section 5.4.

Multi-channel Audio: Digital audio with more than two channels, for example, L-PCM or AC-3.

Native Display Device Aspect Ratio: Ratio of maximum width to height dimension of the addressable portion of a physical display device screen, which is indicated in the EDID version 1, revision 3 block's "Max Horizontal Image Size" and "Max Vertical Image Size" fields.

Native Video Resolution: The exact number of horizontal pixels and vertical lines (or pixel mapping) that matches the physical structure of the display device.

Native Video Format: A Video Format with Native Video Resolution and scanning method that the display device accepts and displays without any internal scaling, de-interlacing, interlacing or frame rate conversion.

Object Based Audio (OBA): In Object Based Audio, each sound is expressed as an Object with a 3D position in space rather than a discrete number of channels or loudspeakers.

One Bit Audio: 1-bit Sigma-Delta (Delta-Sigma) modulated signal stream. Also known as DSD Audio.

opRGB: The optional RGB (opRGB) color space defined in IEC 61966-2-5 [32].

Opto-Electrical Transfer Function (OETF): A mathematical function that describes the relationship between scene light values received by a camera (adjusted for exposure) and values that represent the captured video signal.

$opYCC_{601}$: The luma-chroma-chroma (YCC) color space defined in Annex A of IEC 61966-2-5 [32]. The Rec. ITU-R BT.601 [6] color conversion matrix is used to transform RGB values to YCC values.

OVT Video Format: Any CE Video Format based on OVT. Standard OVT Video Formats are listed in Table 11 and Table 12.

Picture: An uncompressed video signal representing a rectangular array of pixels containing an image that may, in whole or part, be rendered onto a display. A Picture refers to the Pixel Data transferred in the uncompressed video signal during a single Video Frame. A Picture includes both the Active Image and Bars (see Annex N). The implementation of CTA-CEB16 [106] Total Image on the CTA-861 interface.

Picture Aspect Ratio: Ratio of width to height dimension of the Picture as delivered across the uncompressed digital interface, including any top, bottom, or side Bars. Only five Picture Aspect Ratios are specified for this interface: 4:3, 16:9, 64:27, 256:135, and 32:9 (see Annex N).

Picture Line: The complete set of contiguous Active Pixels along a single Active Line, where each Active Pixel contains a portion of either the Active Image or a Bar.

Picture Pixel: A single Active Pixel within a Picture Line containing a portion of either the Active Image or a Bar.

Pixel Data: Color Component Samples transmitted over the interface during a single Active Pixel. These samples may, but need not, completely define a single Video Pixel.

Pixel Encoding: Colorimetry in conjunction with a Component Depth, Transfer Function, Quantization Range, and chroma sample location (if applicable) used to represent the colors of transmitted or received pixels.

Preferred Picture Aspect Ratio: In a Dual-Aspect Ratio DTV, the preferred aspect ratio of a given Video Format Timing (e.g., 720x480p) is the aspect ratio of the first such timing listed in the EDID data structure (see Section 4.1). This would be the Picture Aspect Ratio that would be displayed if a DTV were to receive a Video Format Timing with no accompanying Picture Aspect Ratio information (i.e., no AVI sent from Source).

Preferred Video Format: The Video Format that a display manufacturer determines provides optimum image.

> Note: Source implementers are encouraged to review Section 7.2.2 for related guidance.

Quantization Range: The range of code values used to represent the color components of Active Pixels when transitioning between color extremes (e.g., black to white).

Resolution: Horizontal and vertical active pixel count of a video frame

RGB: A general representation of an analog or digital component video signal, where R represents the red color, G represents green, and B represents blue; and each component is sampled at a uniform rate (4:4:4). For the purpose of CTA-861, the signal is digital.

Sink: A device, which receives an uncompressed A/V signal.

Source: A device, which generates an uncompressed A/V signal.

Source Pass-through Mode: A mode supported by some media-based Sources, wherein decompressed video passes directly (in its original format) to a Sink without interlacing, deinterlacing, scaling, or frame rate conversion.

sRGB: The default RGB color space defined in IEC 61966-2-1 [33].

Standard Definition (SD): A CE Video Format that has less than 720 active vertical lines (Vactive) lines per Video Frame (e.g., 480 or 576 active vertical lines).

Standard Definition Television (SDTV): A DTV capable of displaying 720x480i[1] or 720x576i video in at least one of two Picture Aspect Ratios, 16:9 or 4:3.

$sYCC_{601}$: The luma-chroma-chroma (YCC) color space defined in Annex F of IEC 61966-2-1/Amendment 1 [34]. The Rec. ITU-R BT.601 [6] color conversion matrix is used to transform RGB values to YCC values. $sYCC_{601}$ color space can represent colors outside of the sRGB color gamut.

Transfer Function: Non-linear mathematical function that maps linear light levels to values representing a video signal or vice versa. Transfer functions include EOTF, inverse EOTF, OETF, and inverse OETF.

Uncompressed Audio: Linear Pulse Code Modulated (L-PCM) and DSD Audio.

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[1] Content is encoded as 704x480 or 720x483

Unique Active Pixel: A timing pattern, consisting of either fractional, single, or multiple contiguous Active Pixels, that effectively increases or lowers horizontal resolution (see Annex O). By splitting each Active Pixel into two Unique Active Pixels (e.g., using YC_BC_R 4:2:0 sampling), the effective horizontal resolution can be doubled. Alternately, Unique Active Pixels consisting of one or more contiguous Active Pixels having the same (systematically repeated) Pixel Data can be used to lower horizontal resolution. In this case, the number of Active Pixels in a Unique Active Pixel is equal to the pixel repetition factor (PR+1).

Unique Content Pixel: A timing pattern consisting of one or more contiguous Content Pixels having the same (systematically repeated) Pixel Data.

Video Field: The period from the leading (active) edge of one vertical sync (Vsync) pulse to the same edge of the next Vsync pulse or the timing pattern associated with that period.

Video Format: A Video Format is sufficiently defined such that when it is received at the DTV, the Sink has enough information to properly display an image. Although it is generally acceptable to define a Video Format by specifying only a Video Timing and a Picture Aspect Ratio, a more complete definition requires additional information including a Color Space, a Quantization Range, and a Component Depth (N).

Video Format Timing (or Video Timing): The resolution and frame rate associated with a Video Format. Note that a specific Video Format Timing may be associated with more than one Video Format (e.g., 720x480p formatted in the 4:3 Picture Aspect Ratio or a 720x480p formatted in the 16:9 Picture Aspect Ratio).

Video Frame: The period (beginning and ending where the active edges of horizontal and vertical sync align) for vertical total lines to elapse or the repetitive timing pattern associated with that period. Interlaced timings have two Video Fields per Video Frame, while progressive timings have only one. Therefore, in the case of progressive timings, the terms “video field” and “video frame” are synonymous.

Video Identification (ID) Code: An integer value used to identify a particular Video Format listed in Table 2. Table 1 uses Video Identification Codes to cross reference a particular Video Format with its exact Video Timing.

Video Line: The period, lasting Htotal pixel clock periods, beginning and ending with the active edge of a horizontal sync pulse (Hsync). The term may also refer to a timing pattern that occurs during this period consisting of Htotal contiguous Video Pixels.

Video Pixel: The period delimited by two sequential pixel clock active edges. The term may also refer to the portion of a timing pattern where the interface transfers one unit of data. The unit of data transferred may be related to a single pixel or other information (e.g., Audio) and may convey the same information as the preceding Video Pixel.

$xvYCC_{601}$: The extended gamut luma-chroma-chroma (YCC) color space defined in IEC 61966-2-4 [5]. The Rec. ITU-R BT.601 [6] color conversion matrix is used by IEC 61966-2-4 [5] to transform RGB values to YCC values. The extent of the gamut is device dependent.

xvYCC$_{709}$: The extended gamut luma-chroma-chroma (YCC) color space defined in IEC 61966-2-4 [5]. The Rec. ITU-R BT.709 [7] color conversion matrix is used by IEC 61966-2-4 [5] to transform RGB values to YCC values. The extent of the gamut is device dependent.

YC$_B$C$_R$: A general representation of a digital component video signal, where Y represents luminance, C_B represents the color blue, and C_R represents red; the color component may be sub-sampled at half the rate as luminance (4:2:2) or may be sampled at a uniform rate (4:4:4). For the purposes of CTA-861, it may be considered a digital representation of YP_BP_R.

2.3 Symbols and Abbreviations

AAC	Advanced Audio Coding
ADB	Audio Data Block
AFD	Active Format Description
AMOL96	Automated Measurement of Lineups 96 (bits/field)
ANSI	American National Standards Institute
AR	Aspect Ratio
ATRAC	Adaptive Transform Acoustic Coding
A/V	Audio/Video
AV/C	Audio/Video Control
AVI	Auxiliary Video Information
BD	Blu-Ray Disc
BDA	Blu-ray Disc Association
CD	Compact Disk
CDB	Colorimetry Data Block
CID	IEEE Company Identifier
CRT	Cathode Ray Tube
CTA	Consumer Technology Association
DAC	Digital to Analog Converter
DBS	Direct Broadcast Satellite
DCP	Digital Content Protection
DDWG	Digital Display Working Group
DMT	Display Monitor Timing
DRA	Digital Rise Audio
DSC	Digital Still Camera

DSD	Direct Stream Digital
DST	Direct Stream Transfer
DTD	Detailed Timing Descriptor
DTV	Digital Television
DVC	Digital Video Camera
DVD	Digital Versatile Disk
D-VHS	Digital VHS
DVI	Digital Visual Interface
E-DDC	Enhanced Display Data Channel
E-EDID	Enhanced Extended Display Identification Data Standard
ELB	End of Left Bar
EOTF	Electro-optical Transfer Function
ETB	End of Top Bar
EUI	Extended Unique Identifier
FFP	Fractional Fixed Point
HCCA	High Channel Count Audio
HD	High Definition
HDCP	High-bandwidth Digital Content Protection
HDD	Hard Disk Drive
HD DVD	High Definition DVD
HDMI	High-Definition Multimedia Interface
HDR	High Dynamic Range
HDR DMDB	HDR Dynamic Metadata Data Block
HDR SMDB	HDR Static Metadata Data Block
HDTV	High Definition Television
HLG	Hybrid Log-Gamma
HPD	Hot Plug Detect
HVT	Harmonic Video Timing
IDO	Interface Development Organization
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers

IFDB	InfoFrame Data Block
IRE	Institute of Radio Engineers
ISO	International Organization for Standardization
ITU	International Telecommunications Union
LCD	Liquid Crystal Display
LDI	LVDS Display Interface
L-PCM	Linear Pulse Code Modulation
LSB	Least Significant Byte
LVDS	Low Voltage Differential Signaling
MAT	MLP Audio Transport
MaxCLL	Maximum Content Light Level
MaxFALL	Maximum Frame-average Light Level
MPEG	Moving Picture Experts Group
MSB	Most Significant Byte
NABTS	North American Basic Teletext Specification
NVRDB	Native Video Resolution Data Block
OBA	Object Based Audio
OpenLDI	Open LVDS Display Interface
OUI	Organizationally Unique Identifier (Note: CID may be used as an alternative)
OVT	Optimized Video Timing
PES	Packetized Elementary Stream
PLP	Primary Listening Position
PMP	Portable Media Player
PQ	Perceptual Quantization
RCDB	Room Configuration Data Block
SACD	Super Audio CD
SADB	Speaker Allocation Data Block
SBB	Start of Bottom Bar
SD	Standard Definition
SDTV	Standard Definition Television
SLDB	Speaker Location Data Block

SMPTE	Society of Motion Picture & Television Engineers
SPM	Speaker Presence Mask
SRB	Start of Right Bar
STB	Set-Top Box
SVD	Short Video Descriptor
SVR	Short Video Reference
TVG2X	TVGuide 2X (bitrate)
VBI	Vertical Blanking Interval
VCDB	Video Capability Data Block
VCR	Video Cassette Recorder
VDB	Video Data Block
VESA	Video Electronics Standards Association
VESA DDDB	VESA Display Device Data Block
VESA DTCDB	VESA Display Transfer Characteristic Data Block
VFPDB	Video Format Preference Data Block
VIC	Video Identification (ID) Code
VSADB	Vendor-Specific Audio Data Block
VSDB	Vendor-Specific Data Block
VSIF	Vendor-Specific InfoFrame
VSVDB	Vendor-Specific Video Data Block
VTDB	DisplayID Video Timing Data Block (encompasses T7VTDB, T8VTDB, and T10VTDB)
WMA Pro	Windows Media Audio Professional
Y420CMDB	YC_BC_R 4:2:0 Capability Map Data Block
Y420VDB	YC_BC_R 4:2:0 Video Data Block

2.4 Compliance Notation

As used in this document, “shall” denotes mandatory provisions of the standard. “Should” denotes a provision that is recommended but not mandatory. “May” denotes a feature whose presence does not preclude compliance and implementation of which is optional. “Optional” denotes items that may or may not be present in a compliant device.

2.5 Hexadecimal Notation

The characters 0x followed by numbers or letters A through F designate those values as hexadecimal values.

2.6 Binary Notation

The characters 0b followed by numbers 0 and 1 designate those values as binary values.

2.7 HxV Video Timing Notation

Video Timings are sometimes expressed using HxV notation (e.g., “720x480”), where H and V are equal to the number of Active Pixels per Active Line and the number of Active Lines per Video Frame, respectively. The H value is sometimes surrounded by parenthesis and preceded by the number of Unique Active Pixels. Example: “720(1440)x480”. In this example, a Unique Active Pixel is formed by systematically repeating the preceding Active Pixel PR-number of times such that the effective horizontal resolution is lower than the value indicated in the parentheses by a factor of 1/(PR+1) (see Table 1, Note b for details).
Video Timings may also be expressed using the HxV @ F notation (e.g., “720x576i @ 50 Hz “), where a value F is appended to denote field frequency. The value F also refers to the Video Frame rate when the letter to the right of the value V is a ‘p’. North American Video Timings usually have a slash ‘/’ in the name (e.g., “720x480i @ 59.94/60 Hz”) to delimit dual vertical frequencies. Here, the first vertical frequency is adjusted by a factor of exactly 1000/1001 (for NTSC broadcast compatibility) relative to the second (see Section 4).

2.8 Bit Naming Conventions

The names of the individual bits of multi-bit data values are composed using a value’s mnemonic followed by a bit number. The significance of each bit is indicated by the bit number according to little-endian convention (i.e., bit number 0 is the least significant). For example, the quantization value is given the mnemonic ‘Q’, which is associated with two bits named ‘Q1’ and ‘Q0’. When the value Q = 2, bit Q1 = 1 and bit Q0 = 0.

Future bits are a special case. These bits begin with the mnemonic ‘F’ followed by a bit number. In this case, bit numbers indicate location — not significance. Future bits shall be set to zero and ignored.

2.9 ASCII Codes, Characters, and Strings

ASCII characters shall be encoded using either 7-bit or 8-bit codes as indicated. The least significant 7-bits shall encode characters according to ANSI INCITS [38], where ANSI INCITS bits b1 through b7 are mapped to bits 0 through 6, respectively. In the case of 8-bit codes, the most significant bit (bit 7) shall always be set to zero.

2.10 Operators and Functions

+	Addition
–	Subtraction (as a two-argument operator) or negation (as a unary prefix operator)
*****	Multiplication
/	Division where no truncation or rounding is intended
x % y	Modulus, remainder of x divided by y
x^y	Exponentiation, x to the power of y
>	Greater than
>=	Greater than or equal to
<	Less than
<=	Less than or equal to
==	Equal to
!=	Not equal to
&	Bit-wise "and"
|	Bit-wise "or"
^	Bit-wise "exclusive or"
~	Bit-wise "not"
x >> y	Arithmetic right shift of a two's complement integer representation of x by y binary digits
x << y	Arithmetic left shift of a two's complement integer representation of x by y binary digits
=	Assignment operator
++	Increment, x++ is equivalent to x = x + 1
––	Decrement, x–– is equivalent to x = x – 1
+=	Increment by amount specified, e.g., x += 3 is equivalent to x = x + 3
–=	Decrement by amount specified, e.g., x –= 3 is equivalent to x = x – 3

x ? y : z	If x is TRUE or not equal to 0, evaluates to the value of y; otherwise, evaluates to the value of z
Abs(x)	Absolute value, (x < 0) ? −x : x
Ceil(x)	The smallest integer greater than or equal to x
Floor(x)	The largest integer less than or equal to x
Min(x, y)	Minimum of two arguments, (x <= y) ? x : y
Max(x, y)	Maximum of two arguments, (x >= y) ? x : y
Sign(x)	Sign value, (x < 0) ? −1 : (x > 0) ? 1 : 0

3 OVERVIEW

CTA-861 describes requirements for video Sources and Sinks that include an uncompressed, baseband, digital video interface. These requirements apply to any baseband digital video interface that makes use of VESA E-EDID (structures for discovery of supported Video Formats) [9] and supports 24-bit RGB. The 60 Hz/59.94 Hz Video Timings are based on analog formats already standardized in CTA-770.2 [30] and CTA-770.3 [31]. A preferred physical/link interface is not specified in CTA-861. See the annexes on how to apply CTA-861 to the individual interfaces available at the time of this writing. Digital Visual Interface (DVI 1.0) [4] and OpenLDI 0.95 [8] can be used to enable minimal digital interface functionality. To take advantage of these enhancements, the physical interface also needs a way to transport CTA InfoFrames, digital audio, and YC_BC_R pixels from the Source to the Sink. The High-Definition Multimedia Interface (HDMI) [110] is capable of taking advantage of these enhancements.

Enhanced Extended Display Identification Data (E-EDID) was created by VESA to enable plug and play capabilities of Sinks. This data, which would be stored in the Sink, describes Video Formats that the Sink is capable of receiving and rendering. The information is supplied to the Source, over the interface, upon the request of the Source. The Source then chooses its output format, taking into account the format of the original video stream and the formats supported by the Sink. The Source (e.g., STB) is responsible for the format conversions necessary to supply video in an understandable form to the Sink.

CTA-861 includes the Sink's ability to describe other capabilities in the E-EDID — in addition to supported Video Formats (e.g., digital audio). In those cases, the same basic mechanism applies (i.e., the Source reads EDID data in the Sink to determine its capabilities and then the Source sends only audio and Video Formats the Sink is capable of receiving).

The physical/link standards in Annex B, Annex C, and Annex D do not support transport of closed captioning (CTA-608-C [104] and CTA-708-C [105]); therefore, the Source processes these elements. Specifically, if closed captioning is to be displayed, it is decoded by the Source, inserted into the video, and displayed as open captions. Similarly, if system Information, program information, events, service descriptors, etc. are displayed, related graphical information is inserted into the video by the Source. Control of closed captioning settings, programs, events, etc. is a feature of the Source, not supported by this interface and beyond the scope of CTA-861.

Furthermore, content advisory user menus, settings, and blocking are accommodated in the Source, and are beyond the scope of CTA-861.

3.1 General Video Format Requirements for Sources

A Source shall support at least one of the following Video Timings:

- 640x480p @ 59.94/60 Hz
- 720x480p @ 59.94/60 Hz
- 720x576p @ 50 Hz

A Source that accepts 60 Hz Video Formats, and that supports HDTV capability, should support 720x480p @ 59.94/60 Hz and 1280x720p @ 59.94/60 Hz or 1920x1080i @ 59.94/60 Hz Video Format Timings.

A Source that accepts 50 Hz Video Formats, and that supports HDTV capability, should support 720x576p @ 50 Hz and 1280x720p @ 50 Hz or 1920x1080i @ 50 Hz Video Format Timings.

3.2 General Video Format Requirements for Sinks

A Sink that accepts 60 Hz Video Formats shall support the 640x480p @ 59.94/60 Hz and 720x480p @ 59.94/60 Hz Video Format Timings.

A Sink that accepts 50 Hz Video Formats shall support the 640x480p @ 59.94/60 Hz and 720x576p @ 50 Hz Video Format Timings.

A Sink that accepts 60 Hz Video Formats, and that supports HDTV capability, shall support 1280x720p @ 59.94/60 Hz or 1920x1080i @ 59.94/60 Hz Video Format Timings.

A Sink that accepts 50 Hz Video Formats, and that supports HDTV capability, shall support 1280x720p @ 50 Hz or 1920x1080i @ 50 Hz Video Format Timings.

A Sink that supports 640x480p @ 59.94/60 Hz shall support video formatted in a 4:3 Picture Aspect Ratio, as described in Section 4.1, and coded with the default Component Depth, colorimetry, and Quantization Range for IT Video Timing as specified in Section 5.1, when receiving such timing.

A Sink that supports 720x480p @ 59.94/60 Hz or 720x576p @ 50 Hz should support video formatted in either 4:3 or 16:9 Picture Aspect Ratio, as described in Section 4.1, and coded with the default Component Depth, colorimetry, and Quantization Range for CE Video Timing as specified in Section 5.1, when receiving such timings.

4 VIDEO FORMATS AND WAVEFORM TIMINGS

CTA-861 interfaces transport uncompressed digital video using a variety of CE and IT Video Timings. This section describes the default IT 640x480 Video Timing as well as the standard CE Video Timings associated with a VIC. The balance of CE timings are documented in Annex U. The balance of IT timings are documented in the VESA DMT [124], GTF [120, 123], and VESA CVT [117, 121] standards.

Throughout CTA-861, the term “Video Format Timing” does not include aspect ratio, whereas the term “Video Format” does encompass the aspect ratio.

A Video Timing with a vertical frequency that is an integer multiple of 6.00 Hz (e.g., 24.00 or 120.00 Hz) is considered to be the same as a Video Timing with the equivalent detailed timing information but where the vertical frequency is adjusted by a factor of 1000/1001 (e.g., 24/1.001 or 120/1.001). That is, they are considered two versions of the same Video Timing but with slightly different pixel clock frequencies. Therefore, a DTV that declares it is capable of displaying a Video Timing with a vertical frequency that is either an integer multiple of 6 Hz or an integer multiple of 6 Hz adjusted by a factor of 1000/1001 shall be capable of displaying both versions of the Video Timing. 300 Hz, declared in a VFD (see Section 7.5.19.1) with the BFR50 and 6X flags, is excepted from this requirement. Frame Rates that are multiples of BRF60 are considered integer multiples of 6.00 Hz while those that are multiples of BRF50 are not. A Sink cannot declare 300 Hz with a BFR60 flag in a VFD, because no 5X flag exists.

Fractional Frame Rates expressed as decimal fractions, such as 23.976 Hz or 59.94 Hz, are approximations of the exact Frame Rate. The exact Frame Rate is calculated by applying a factor of 1000 / 1001 to the nearest adjacent integer frame rate (e.g., 23.976 Hz represents 24 Hz * 1000 / 1001).

The additional low-resolution progressive Video Format Timings (1440x240p, 2880x240p, 1440x288p, and 2880x288p) consist of one of several frame formats. These frame formats differ only by one or two scan lines in the vertical blanking interval. For that reason, they are treated as the same Video Format with a slight variation in the parameters (i.e., handled in a way similar to the 59.94 Hz / 60 Hz formats). For this reason, if a Sink declares support of one of these Video Formats of a specific Picture Aspect Ratio (through EDID), then it shall support all variations of that Video Format of the same Picture Aspect Ratio.

The mandatory and optional formats defined in CTA-861 shall comply with the timing parameters in Table 1.

In Table 1, note that the Vfront, Vsync, and Vback values are defined in terms of Video Lines (see Section 2.2 “Video Line”). The reader is advised that the signals Hsync, Vsync, Data Enable, and Clock are encoded in an interface-specific manner. For details, see the specifications for DVI [4], OpenLDI [8] or HDMI [110].

Standard-definition Video Timings generally use negative vertical and horizontal sync, while high-definition Video Timings use positive.

Lines are always numbered sequentially from 1 to Vtotal and match the line numbers found in the given reference standard. In the case of high-definition Video Timings, the leading-line of vertical sync in field 1 is always line 1. In the case of standard-definition Video Timings, line 1 may coincide with the leading-line of vertical sync in field 1 or a line slightly before it. CTA-861 Video Timings are sometimes based on legacy 60 Hz standard-definition television standards that have slightly larger Vactive values (e.g., 483-lines vs. 480-lines). Such Video Timings begin line numbering before the leading-line of vertical sync in field 1 — in order to keep line-numbers in alignment with the legacy standard. The “Ln” column in Table 1 provides the line number of the leading-line of vertical sync in field 1 for each Video Timing code. See Annex L for examples.

For progressive Video Timings and Field 1 of interlaced Video Timings, the leading (active) edge of Hsync and Vsync transitions shall be perfectly aligned plus or minus zero pixel clocks. In Field 2 of interlaced Video Timings, the alignment between the leading (active) edge of Hsync and Vsync transitions shall be precisely a half-line (Htotal / 2) plus or minus zero pixel clocks.

Table 1 - Video Format Timings — Detailed Timing Information

Field Rate[d]	VIC	Figure	Hactive	Vactive	I / P	Htotal	Hblank[d]	Hfront	Hsync	Hback	Hpol[v]	Vtotal	Vblank[d]	Vfront	Vsync	Vback	Vpol[v]	H Freq[d] (kHz)	V Freq (Hz)	Pixel Freq[d] (MHz)	Ln	Reference Standard	Notes
Low	60, 65	2	1280	720	Prog	3300	2020	1760	40	220	P	750	30	5	5	20	P	18.000	24.000[c]	59.400	1	SMPTE ST 296 [101]	d, f, x
	61, 66	2	1280	720	Prog	3960	2680	2420	40	220	P	750	30	5	5	20	P	18.750	25.000	74.250	1		d, f
	62, 67	2	1280	720	Prog	3300	2020	1760	40	220	P	750	30	5	5	20	P	22.500	30.000[c]	74.250	1		d, f
	108, 109	2	1280	720	Prog	2500	1220	960	40	220	P	750	30	5	5	20	P	36.000	48.000[c]	90.000	1		d, f, x
	79	2	1680	720	Prog	3300	1620	1360	40	220	P	750	30	5	5	20	P	18.000	24.000[c]	59.400	1		y
	80	2	1680	720	Prog	3168	1488	1228	40	220	P	750	30	5	5	20	P	18.750	25.000	59.400	1		y
	81	2	1680	720	Prog	2640	960	700	40	220	P	750	30	5	5	20	P	22.500	30.000[c]	59.400	1		y
	110	2	1680	720	Prog	2750	1070	810	40	220	P	750	30	5	5	20	P	36.000	48.000[c]	99.000	1		y
	32, 72	2	1920	1080	Prog	2750	830	638	44	148	P	1125	45	4	5	36	P	27.000	24.000[c]	74.250	1	SMPTE ST 274 [2]	r
	33, 73	2	1920	1080	Prog	2640	720	528	44	148	P	1125	45	4	5	36	P	28.125	25.000	74.250	1		r
	34, 74	2	1920	1080	Prog	2200	280	88	44	148	P	1125	45	4	5	36	P	33.750	30.000[c]	74.250	1		r
	111, 112	2	1920	1080	Prog	2750	830	638	44	148	P	1125	45	4	5	36	P	54.000	48.000[c]	148.500	1		r
	86	2	2560	1080	Prog	3750	1190	998	44	148	P	1100	20	4	5	11	P	26.400	24.000[c]	99.000	1		z
	87	2	2560	1080	Prog	3200	640	448	44	148	P	1125	45	4	5	36	P	28.125	25.000	90.000	1		z
	88	2	2560	1080	Prog	3520	960	768	44	148	P	1125	45	4	5	36	P	33.750	30.000[c]	118.800	1		z
	113	2	2560	1080	Prog	3750	1190	998	44	148	P	1100	20	4	5	11	P	52.800	48.000[c]	198.000	1		z
	93, 103	2	3840	2160	Prog	5500	1660	1276	88	296	P	2250	90	8	10	72	P	54.000	24.000[c]	297.000	1		d, f, aa
	94, 104	2	3840	2160	Prog	5280	1440	1056	88	296	P	2250	90	8	10	72	P	56.250	25.000	297.000	1		d, f, aa
	95, 105	2	3840	2160	Prog	4400	560	176	88	296	P	2250	90	8	10	72	P	67.500	30.000[c]	297.000	1		d, f, aa
	114, 116	2	3840	2160	Prog	5500	1660	1276	88	296	P	2250	90	8	10	72	P	108.000	48.000[c]	594.000	1		d, f, aa
	98	2	4096	2160	Prog	5500	1404	1020	88	296	P	2250	90	8	10	72	P	54.000	24.000[c]	297.000	1		d, f, bb
	99	2	4096	2160	Prog	5280	1184	968	88	128	P	2250	90	8	10	72	P	56.250	25.000	297.000	1		d, f, bb
	100	2	4096	2160	Prog	4400	304	88	88	128	P	2250	90	8	10	72	P	67.500	30.000[c]	297.000	1		d, f, bb
	115	2	4096	2160	Prog	5500	1404	1020	88	296	P	2250	90	8	10	72	P	108.000	48.000[c]	594.000	1		d, f, bb
	121	2	5120	2160	Prog	7500	2380	1996	88	296	P	2200	40	8	10	22	P	52.800	24.000[c]	396.000	1		cc
	122	2	5120	2160	Prog	7200	2080	1696	88	296	P	2200	40	8	10	22	P	55.000	25.000	396.000	1		cc
	123	2	5120	2160	Prog	6000	880	664	88	128	P	2200	40	8	10	22	P	66.000	30.000[c]	396.000	1		cc
	124	2	5120	2160	Prog	6250	1130	746	88	296	P	2475	315	8	10	297	P	118.800	48.000[c]	742.500	1		cc

Table 1 - Video Format Timings — Detailed Timing Information (continued)

Field Rate[d]	VIC	Figure	Hactive	Vactive	I / P	Htotal	Hblank[d]	Hfront	Hsync	Hback	Hpol[v]	Vtotal	Vblank[d]	Vfront	Vsync	Vback	Vpol[v]	H Freq[d] (kHz)	V Freq (Hz)	Pixel Freq[d] (MHz)	Ln	Reference Standard	Notes
Low (cont.)	194, 202	2	7680	4320	Prog	11000	3320	2552	176	592	P	4500	180	16	20	144	P	108.000	24.000[c]	1188.000	1	SMPTE ST 274 [2]	
	195, 203	2	7680	4320	Prog	10800	3120	2352	176	592	P	4400	80	16	20	44	P	110.000	25.000	1188.000	1		
	196, 204	2	7680	4320	Prog	9000	1320	552	176	592	P	4400	80	16	20	44	P	132.000	30.000[c]	1188.000	1		
	197, 205	2	7680	4320	Prog	11000	3320	2552	176	592	P	4500	180	16	20	144	P	216.000	48.000[c]	2376.000	1		
	210	2	10240	4320	Prog	12500	2260	1492	176	592	P	4950	630	16	20	594	P	118.800	24.000[c]	1485.000	1		dd
	211	2	10240	4320	Prog	13500	3260	2492	176	592	P	4400	80	16	20	44	P	110.000	25.000	1485.000	1		dd
	212	2	10240	4320	Prog	11000	760	288	176	296	P	4500	180	16	20	144	P	135.000	30.000[c]	1485.000	1		dd
	213	2	10240	4320	Prog	12500	2260	1492	176	592	P	4950	630	16	20	594	P	237.600	48.000[c]	2970.000	1		dd
50 Hz	17, 18	1	720	576	Prog	864	144	12	64	68	N	625	49	5	5	39	N	31.250	50.000	27.000	1	Rec. ITU-R BT.1358 [116]	
	21, 22	3	1440[b]	576	Int	1728[b]	288	24	126	138	N	625	24.5[a]	2	3	19	N	15.625	50.000	27.000	1	Rec. ITU-R BT.656 [114]	j, s
	23, 24	1	1440[b]	288	Prog	1728[b]	288	24	126	138	N	312	24	2	3	19	N	15.625	50.080	27.000	1	Rec. ITU-R BT.1358 [116]	k, r, s, w
												313	25	3					49.920				
												314	26	4					49.761				
	29, 30	1	1440[b]	576	Prog	1728[b]	288	24	128	136	N	625	49	5	5	39	N	31.250	50.000	54.000	1		m, n, r
	25, 26	3	2880[b]	576	Int	3456[b]	576	48	252	276	N	625	24.5[a]	2	3	19	N	15.625	50.000	54.000	1	Rec. ITU-R BT.656 [114]	l, q, r, u
	27, 28	1	2880[b]	288	Prog	3456[b]	576	48	252	276	N	312	24	2	3	19	N	15.625	50.080	54.000	1		k, l, p, q, u, w
												313	25	3					49.920				
												314	26	4					49.761				
	37, 38	1	2880[b]	576	Prog	3456[b]	576	48	256	272	N	625	49	5	5	39	N	31.250	50.000	108.000	1	Rec. ITU-R BT.1358 [116]	m, o
	19, 68	2	1280	720	Prog	1980	700	440	40	220	P	750	30	5	5	20	P	37.500	50.000	74.250	1	SMPTE ST 296 [101]	d, f
	82	2	1680	720	Prog	2200	520	260	40	220	P	750	30	5	5	20	P	37.500	50.000	82.500	1		d, f, y
	39	5	1920	1080	Int	2304	384	32	168	184	P	1250	85	23	5	57	N	31.250	50.000	72.000	1	AS 4933.1-2005 [126]	i
	20	4	1920	1080	Int	2640	720	528	44	148	P	1125	22.5[a]	2	5	15	P	28.125	50.000	74.250	1	SMPTE ST 274 [2]	d, f
	31, 75	2	1920	1080	Prog	2640	720	528	44	148	P	1125	45	4	5	36	P	56.250	50.000	148.500	1		r
	89	2	2560	1080	Prog	3300	740	548	44	148	P	1125	45	4	5	36	P	56.250	50.000	185.625	1		r, z
	96, 106	2	3840	2160	Prog	5280	1440	1056	88	296	P	2250	90	8	10	72	P	112.500	50.000	594.000	1		d, f, aa
	101	2	4096	2160	Prog	5280	1184	968	88	128	P	2250	90	8	10	72	P	112.500	50.000	594.000	1		d, f, bb
	125	2	5120	2160	Prog	6600	1480	1096	88	296	P	2250	90	8	10	72	P	112.500	50.000	742.500	1		cc
	198, 206	2	7680	4320	Prog	10800	3120	2352	176	592	P	4400	80	16	20	44	P	220.000	50.000	2376.000	1		
	214	2	10240	4320	Prog	13500	3260	2492	176	592	P	4400	80	16	20	44	P	220.000	50.000	2970.000	1		dd

Table 1 - Video Format Timings — Detailed Timing Information (continued)

Field Rate[d]	VIC	Figure	Hactive	Vactive	I / P	Htotal	Hblank[d]	Hfront	Hsync	Hback	Hpol[v]	Vtotal	Vblank[d]	Vfront	Vsync	Vback	Vpol[v]	H Freq[d] (kHz)	V Freq (Hz)	Pixel Freq[d] (MHz)	Ln	Reference Standard	Notes
60 Hz[c]	1	1	640	480	Prog	800	160	16	96	48	N	525	45	10	2	33	N	31.469	59.940[c]	25.175	1	VESA DMT [124]	d, h
	2, 3	1	720	480	Prog	858	138	16	62	60	N	525	45	9	6	30	N	31.469	59.940[c]	27.000	7	CTA-770.2 [30]	f
	6, 7	3	1440[b]	480	Int	1716[b]	276	38	124	114	N	525	22.5[a]	4	3	15	N	15.734	59.940[c]	27.000	4		f, s
	8, 9	1	1440[b]	240	Prog	1716[b]	276	38	124	114	N	262	22	4	3	15	N	15.734	60.054[c]	27.000	4		k, r, s, u, w
												263	23	5					59.826[c]				
	14, 15	1	1440[b]	480	Prog	1716[b]	276	32	124	120	N	525	45	9	6	30	N	31.469	59.940[c]	54.000	7		m, n, q, r
	10, 11	3	2880[b]	480	Int	3432[b]	552	76	248	228	N	525	22.5[a]	4	3	15	N	15.734	59.940[c]	54.000	4		l, q, u
	12, 13	1	2880[b]	240	Prog	3432[b]	552	76	248	228	N	262	22	4	3	15	N	15.734	60.054[c]	54.000	4		k, l, q, u, w
												263	23	5					59.826[c]				
	35, 36	1	2880[b]	480	Prog	3432[b]	552	64	248	240	N	525	45	9	6	30	N	31.469	59.940[c]	108.000	7		m, o
	4, 69	2	1280	720	Prog	1650	370	110	40	220	P	750	30	5	5	20	P	45.000	60.000[c]	74.250	1	CTA-770.3 [31]	d, f
	83	2	1680	720	Prog	2200	520	260	40	220	P	750	30	5	5	20	P	45.000	60.000[c]	99.000	1		d, f, y
	5	4	1920	1080	Int	2200	280	88	44	148	P	1125	22.5[a]	2	5	15	P	33.750	60.000[c]	74.250	1		d, f
	16, 76	2	1920	1080	Prog	2200	280	88	44	148	P	1125	45	4	5	36	P	67.500	60.000[c]	148.500	1	SMPTE ST 274 [2]	r
	90	2	2560	1080	Prog	3000	440	248	44	148	P	1100	20	4	5	11	P	66.000	60.000[c]	198.000	1		r, z
	97, 107	2	3840	2160	Prog	4400	560	176	88	296	P	2250	90	8	10	72	P	135.000	60.000[c]	594.000	1		d, f, aa
	102	2	4096	2160	Prog	4400	304	88	88	128	P	2250	90	8	10	72	P	135.000	60.000[c]	594.000	1		d, f, bb
	126	2	5120	2160	Prog	5500	380	164	88	128	P	2250	90	8	10	72	P	135.000	60.000[c]	742.500	1		cc
	199, 207	2	7680	4320	Prog	9000	1320	552	176	592	P	4400	80	16	20	44	P	264.000	60.000[c]	2376.000	1		
	215	2	10240	4320	Prog	11000	760	288	176	296	P	4500	180	16	20	144	P	270.000	60.000[c]	2970.000	1		dd
100 Hz	42, 43	1	720	576	Prog	864	144	12	64	68	N	625	49	5	5	39	N	62.500	100.000	54.000	1	Rec. ITU-R BT.1358 [116]	
	44, 45	3	1440[b]	576	Int	1728[b]	288	24	126	138	N	625	24.5[a]	2	3	19	N	31.250	100.000	54.000	1	Rec. ITU-R BT.656 [114]	t
	41, 70	2	1280	720	Prog	1980	700	440	40	220	P	750	30	5	5	20	P	75.000	100.000	148.500	1	SMPTE ST 296 [101]	
	84	2	1680	720	Prog	2000	320	60	40	220	P	825	105	5	5	95	P	82.500	100.000	165.000	1		y
	40	4	1920	1080	Int	2640	720	528	44	148	P	1125	22.5[a]	2	5	15	P	56.250	100.000	148.500	1	SMPTE ST 274 [2]	
	64, 77	2	1920	1080	Prog	2640	720	528	44	148	P	1125	45	4	5	36	P	112.500	100.000	297.000	1		
	91	2	2560	1080	Prog	2970	410	218	44	148	P	1250	170	4	5	161	P	125.000	100.000	371.250	1		z
	117, 119	2	3840	2160	Prog	5280	1440	1056	88	296	P	2250	90	8	10	72	P	225.000	100.000	1188.000	1		
	218	2	4096	2160	Prog	5280	1184	800	88	296	P	2250	90	8	10	72	P	225.000	100.000	1188.000	1		
	127	2	5120	2160	Prog	6600	1480	1096	88	296	P	2250	90	8	10	72	P	225.000	100.000	1485.000	1		cc
	200, 208	2	7680	4320	Prog	10560	2880	2112	176	592	P	4500	180	16	20	144	P	450.000	100.000	4752.000	1		
	216	2	10240	4320	Prog	13200	2960	2192	176	592	P	4500	180	16	20	144	P	450.000	100.000	5940.000	1		dd

Table 1 - Video Format Timings — Detailed Timing Information (continued)

Field Rate[d]	VIC	Figure	Hactive	Vactive	I / P	Htotal	Hblank[d]	Hfront	Hsync	Hback	Hpol[v]	Vtotal	Vblank[d]	Vfront	Vsync	Vback	Vpol[v]	H Freq[d] (kHz)	V Freq (Hz)	Pixel Freq[d] (MHz)	Ln	Reference Standard	Notes
120 Hz[c]	48, 49	1	720	480	Prog	858	138	16	62	60	N	525	45	9	6	30	N	62.937	119.880[c]	54.000	7	CTA-770.2 [30]	
	50, 51	3	1440[b]	480	Int	1716[b]	276	38	124	114	N	525	22.5[a]	4	3	15	N	31.469	119.880[c]	54.000	4		t
	47, 71	2	1280	720	Prog	1650	370	110	40	220	P	750	30	5	5	20	P	90.000	120.000[c]	148.500	1	SMPTE ST 296 [101]	
	85	2	1680	720	Prog	2000	320	60	40	220	P	825	105	5	5	95	P	99.000	120.000[c]	198.000	1		y
	46	4	1920	1080	Int	2200	280	88	44	148	P	1125	22.5[a]	2	5	15	P	67.500	120.000[c]	148.500	1	SMPTE ST 274 [2]	
	63, 78	2	1920	1080	Prog	2200	280	88	44	148	P	1125	45	4	5	36	P	135.000	120.000[c]	297.000	1		
	92	2	2560	1080	Prog	3300	740	548	44	148	P	1250	170	4	5	161	P	150.000	120.000[c]	495.000	1		z
	118, 120	2	3840	2160	Prog	4400	560	176	88	296	P	2250	90	8	10	72	P	270.000	120.000[c]	1188.000	1		
	219	2	4096	2160	Prog	4400	304	88	88	128	P	2250	90	8	10	72	P	270.000	120.000[c]	1188.000	1		
	193	2	5120	2160	Prog	5500	380	164	88	128	P	2250	90	8	10	72	P	270.000	120.000[c]	1485.000	1		cc
	201, 209	2	7680	4320	Prog	8800	1120	352	176	592	P	4500	180	16	20	144	P	540.000	120.000[c]	4752.000	1		
	217	2	10240	4320	Prog	11000	760	288	176	296	P	4500	180	16	20	144	P	540.000	120.000[c]	5940.000	1		dd
200 Hz	52, 53	1	720	576	Prog	864	144	12	64	68	N	625	49	5	5	39	N	125.000	200.000	108.000	1	Rec. ITU-R BT.1358 [116]	
	54, 55	3	1440[b]	576	Int	1728[b]	288	24	126	138	N	625	24.5[a]	2	3	19	N	62.500	200.000	108.000	1	Rec. ITU-R BT.656 [114]	t
240 Hz[c]	56, 57	1	720	480	Prog	858	138	16	62	60	N	525	45	9	6	30	N	125.874	239.760[c]	108.000	7	CTA-770.2 [30]	
	58, 59	3	1440[b]	480	Int	1716[b]	276	38	124	114	N	525	22.5[a]	4	3	15	N	62.937	239.760[c]	108.000	4		t

Notes:

a. Fractional Vblank values indicate that the number of Blanking Lines varies (see timing diagram for more details).

b. The pixels for the 720(1440)x480i@59.94/60 Hz, 720(1440)x240p@59.94/60 Hz, 720(1440)x576i@50 Hz, and 720(1440)x288p@50 Hz Video Formats are double clocked to meet minimum speed requirements of the interface, thus H active is shown as 1440, instead of 720. At higher field rates, these formats continue to be double clocked — even though double clocking is unnecessary. Each pixel of the 1440xN 480p and 576p formats, as well as the 2880xN 480i, 240p, 480p, 576i, 288p, and 576p formats, is repeated a variable number of times. The repeat value is communicated using the AVI InfoFrames (see Section 6.4).

c. The vertical frequencies of the 240p, 480p, and 480i Video Formats are tuned by a factor of exactly 1000/1001 for NTSC video compatibility, while the 576p, 576i, and the HDTV Video Formats are typically not adjusted. The VESA DMT standard [124] specifies a ± 0.5% pixel clock frequency tolerance. Therefore, the nominally 25.175 MHz pixel clock frequency value given for Video Identification Code 1 may be adjusted to 25.2 MHz to obtain an exact 60 Hz vertical frequency.

d. Data in this column is provided for informational purposes only.

Table 1 - Video Format Timings — Detailed Timing Information (continued)

e. The reference standard uses tri-level sync, while CTA-861 uses bi-level. Bi-level sync timing is accomplished using the second half of the reference standard's tri-level sync, defining the actual sync time to be the rising edge of that pulse.

f. The reference standard uses a composite sync while CTA-861 uses separate sync signals, thus eliminating the need for serrations during vertical sync.

g. VESA defines blanking as not including the border while CTA-861 includes the border within the blanking interval.

h. Uses default IT color space, RGB components, and Full Range 8-bit color coding and quantization (see Section 5.1).

i. Is specifically designed for use with 31.25 kHz constant horizontal rate cathode-ray tube televisions and has a total of 1250 vertical lines — instead of the normal 1125 found in SMPTE ST 274 based timings. It has a frame, which is split into two unequal fields of 624.5 and 625.5 lines. The Video Format is specifically designed for use with special 31.25 kHz constant horizontal rate cathode-ray tube televisions and should be used with caution. Timing is similar to the 1250/50/2:1 system that described in Australian AS 4933.1-2005 standard [126].

j. Same as the reference standard except for horizontal and vertical synchronization pulse durations, which are specified in Rec. ITU-R BT.711-1 [115] and Rec. ITU-R BT.1700 [113]. Thus, the clock is 27 MHz.

k. There are two or three Video Frame timings that differ only in the number of Blanking Lines in the vertical blanking interval of the frame. All are considered variations of the same Video Timing.

l. Represents a superset of game console timings with variable repetition factors. Typically has Bars on the left and right sides that are 160/n pixels wide, where n is the repetition factor. Encompasses all of the following cases:
   - 2880/10 = 288 pixels/line
   - 2880/8 = 360 pixels/line
   - 2880/7 = 411 pixels/line
   - 2880/5 = 576 pixels/line
   - 2880/4 = 720 pixels/line

m. Represents a superset of timings with specific repetition factors that provide either additional bandwidth for carrying audio data or increased horizontal video resolution.

n. Is a superset of Video Formats encompassing all of the following cases:
   - 1440/2 = 720 pixels/line
   - 1440/1 = 1440 pixels/line

o. Is a superset of Video Formats encompassing all of the following cases:
   - 2880/4 = 720 pixels/line
   - 2880/2 = 1440 pixels/line
   - 2880/1 = 2880 pixels/line

p. The exact Video Timing depends upon the pixel repetition factor specified in the AVI InfoFrame.

q. If this Video Timing is advertised in the EDID, the Sink shall have an interface capable of signaling pixel repetition via AVI InfoFrames (e.g., HDMI) and shall accept all listed pixel repetition factors.

r. It is likely that non-HDMI Sources may not recognize this Video Format in a Detailed Timing Descriptor.

s. Assumes the pixels are double-clocked to meet minimum clock speed requirements for the interface.

t. Assumes the pixels are double-clocked.

Table 1 - Video Format Timings — Detailed Timing Information (continued)

u. This is a “gaming” format. Progressive timing is obtained by removing the second field of the reference standard’s interlaced timing.

v. Hpol and Vpol stand for horizontal and vertical sync pulse polarity, respectively. The value ‘N’ signifies negative polarity, where the signal stays mostly high (at a logic ‘1’) and only pulses low (to a logic ‘0’) during the sync pulse. Likewise, the value ‘P’ signifies positive polarity, where the signal stays mostly low (at a logic ‘0’) and only pulses high (to a logic ‘1’) during the sync pulse.

w. Vfront varies as a function of Vtotal.

x. This Video Timing has been modified from the one listed in the reference standard. The reference standard specifies an odd Htotal greater than 4096, which is incompatible with legacy silicon. CTA-861 instead uses the referenced standard’s 720p30 Video Timing with a reduced pixel clock frequency to obtain 24 Hz.

y. This Video Timing has been modified from the one listed in the reference standard. The pixel rate and Hfront have been modified to provide Hactive of 1680 pixels needed to achieve a 64:27 Picture Aspect Ratio with approximately square (64:63) pixels.

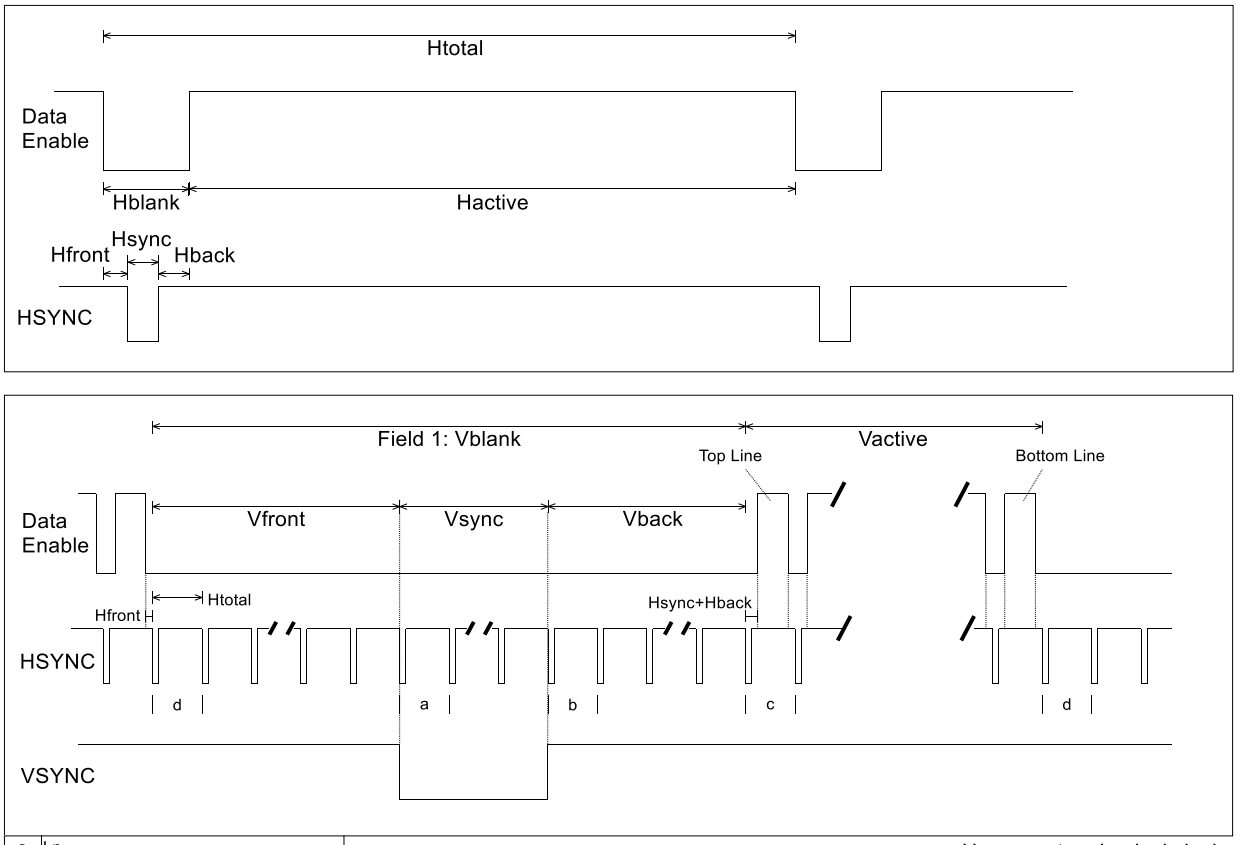
z. This Video Timing has been modified from the one listed in the reference standard. The pixel rate and Hfront have been modified to provide Hactive of 2560 pixels needed to achieve a 64:27 Picture Aspect Ratio with perfectly square pixels.

aa. Using UHDTV1 image sample structures and frame rates from SMPTE ST 2036-1 [136].

bb. Using UHDTV1 image sample structures and frame rates from SMPTE ST 2048-1 [137].

cc. This Video Timing has been modified from the one listed in the reference standard. The pixel rate and Hfront have been modified to provide Hactive of 5120 pixels needed to achieve a 64:27 Picture Aspect Ratio with perfectly square pixels.

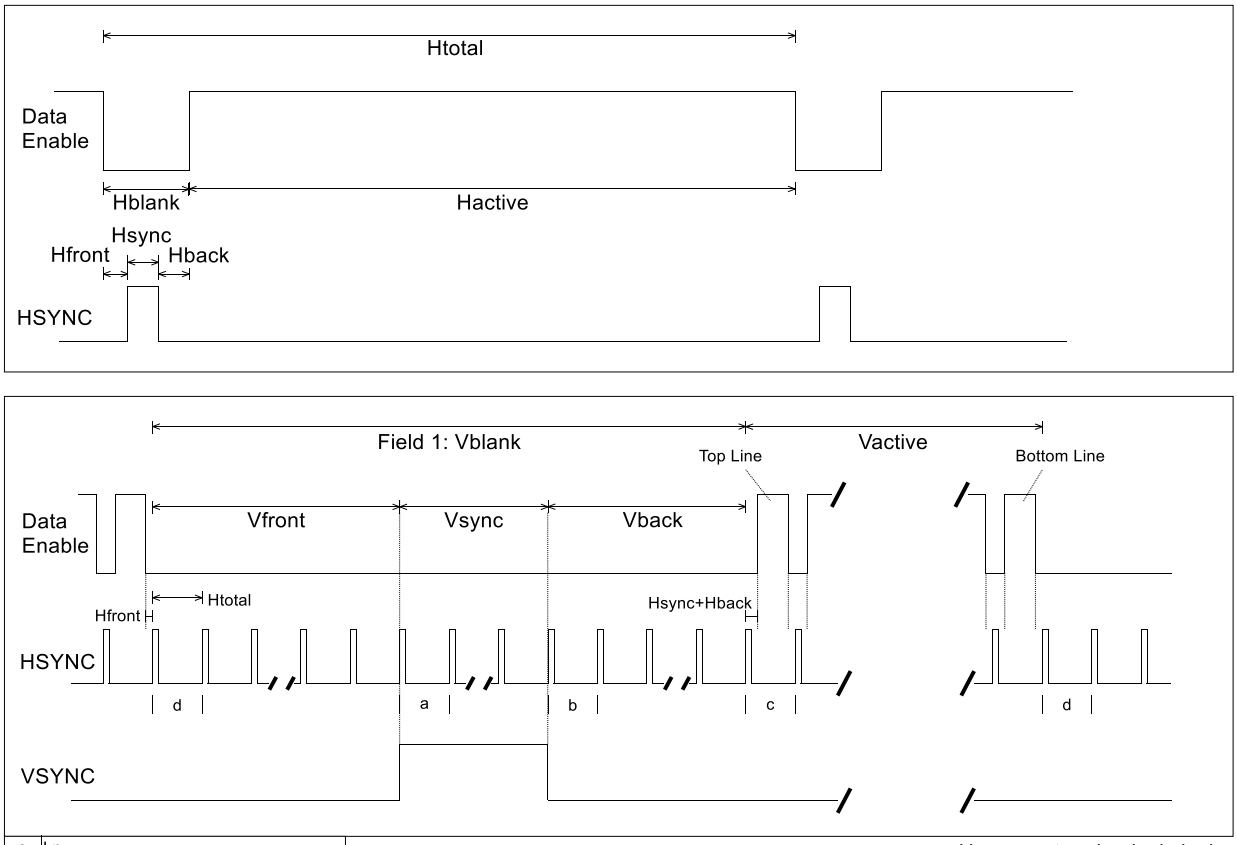
dd. This Video Timing has been modified from the one listed in the reference standard. The pixel rate and Hfront have been modified to provide Hactive of 10240 pixels needed to achieve a 64:27 Picture Aspect Ratio with perfectly square pixels.



a	Ln
b	Ln+Vsync
c	Ln+Vsync+Vback
d	Ln+Vsync+Vback+Vactive

H parameters in pixel clocks
V parameters in lines
Letters a, b, c, … represent line numbers

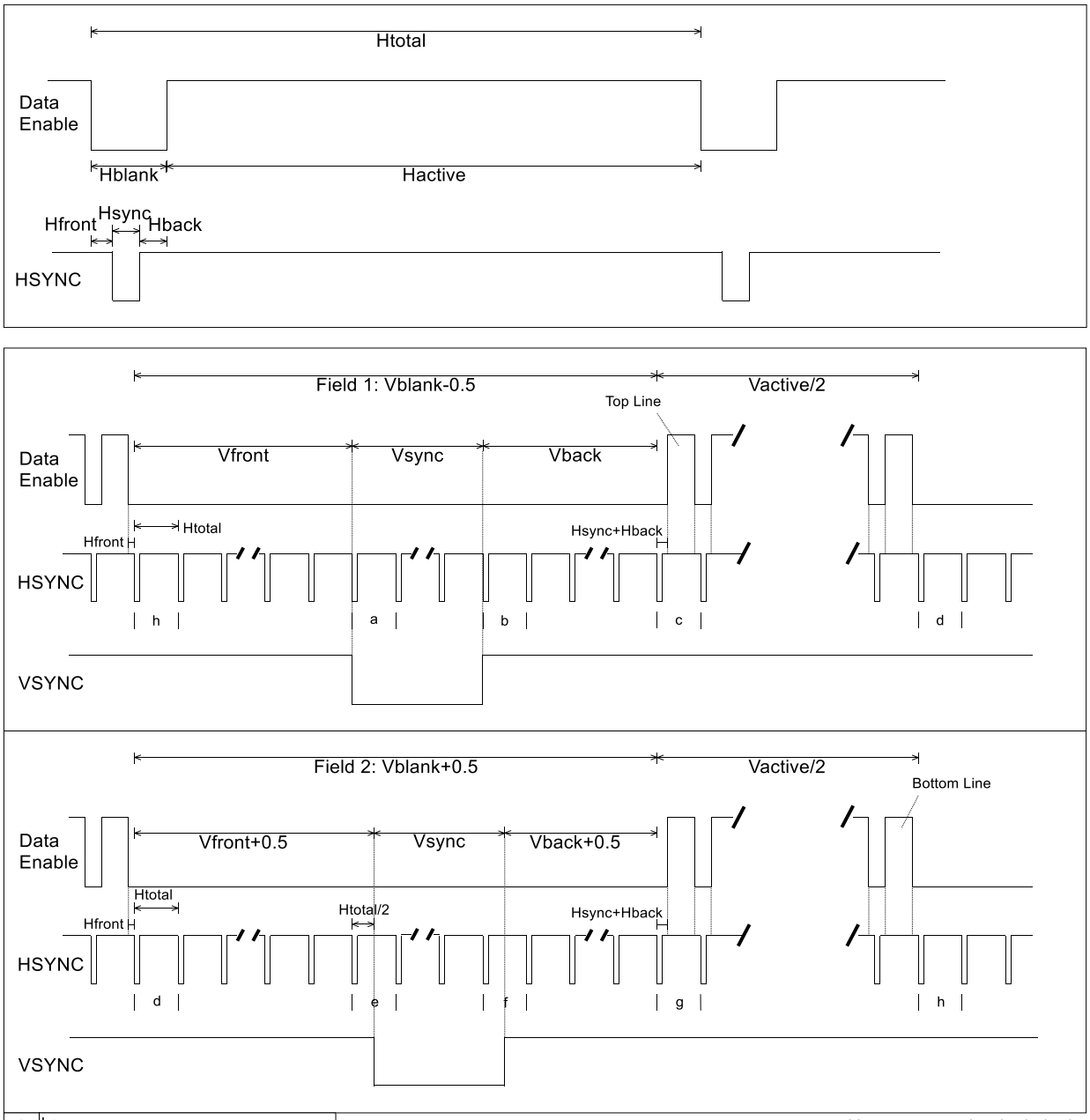
Figure 1 - General Progressive Video Format Timing (Negative Sync)



a	Ln
b	Ln+Vsync
c	Ln+Vsync+Vback
d	Ln+Vsync+Vback+Vactive

H parameters in pixel clocks
V parameters in lines
Letters a, b, c, … represent line numbers

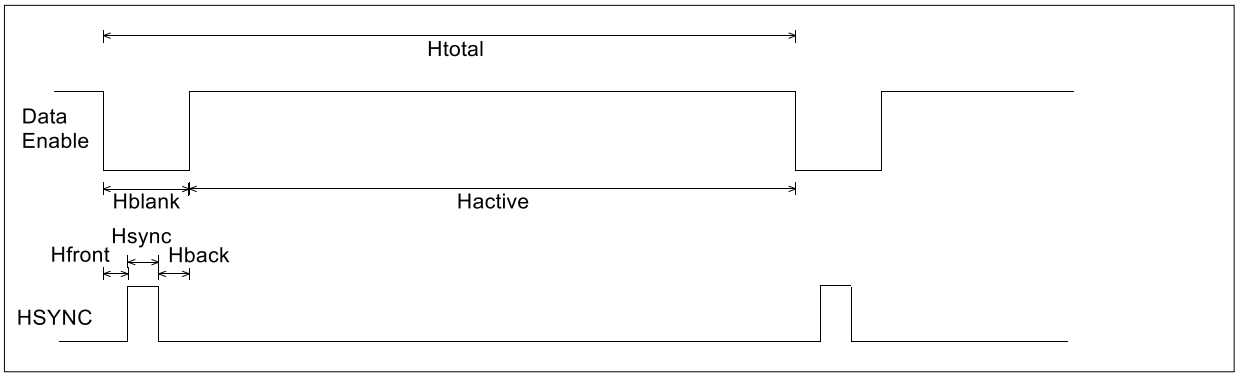
Figure 2 - General Progressive Video Format Timing (Positive Sync)

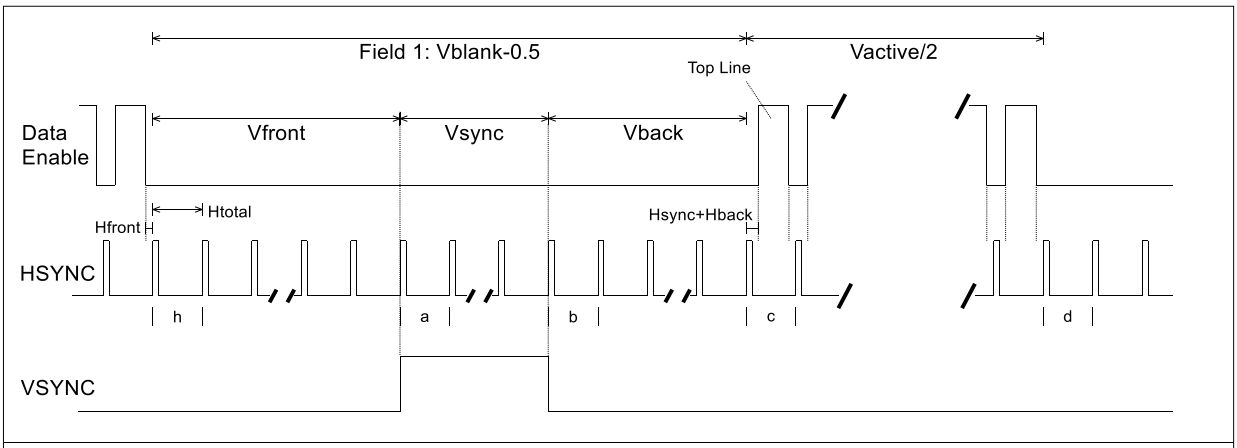


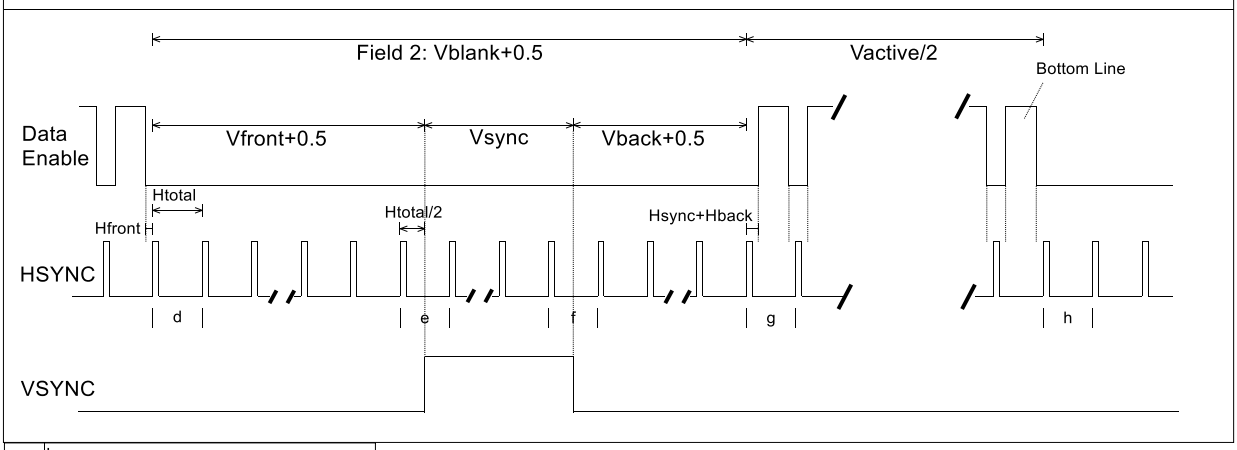
a	Ln
b	Ln+Vsync
c	Ln+Vsync+Vback
d	Ln+Vsync+Vback+Vactive/2
e	Ln+Vsync+Vback+Vactive/2+Vfront
f	Ln+2*Vsync+Vback+Vactive/2+Vfront
g	Ln+2*Vsync+2*Vback+Vactive/2+Vfront+1
h	Ln+2*Vsync+2*Vback+Vactive+Vfront+1

H parameters in pixel clocks
V parameters in lines
Letters a, b, c, … represent line numbers.

Figure 3 - General Interlaced Video Format Timing (Negative Sync)



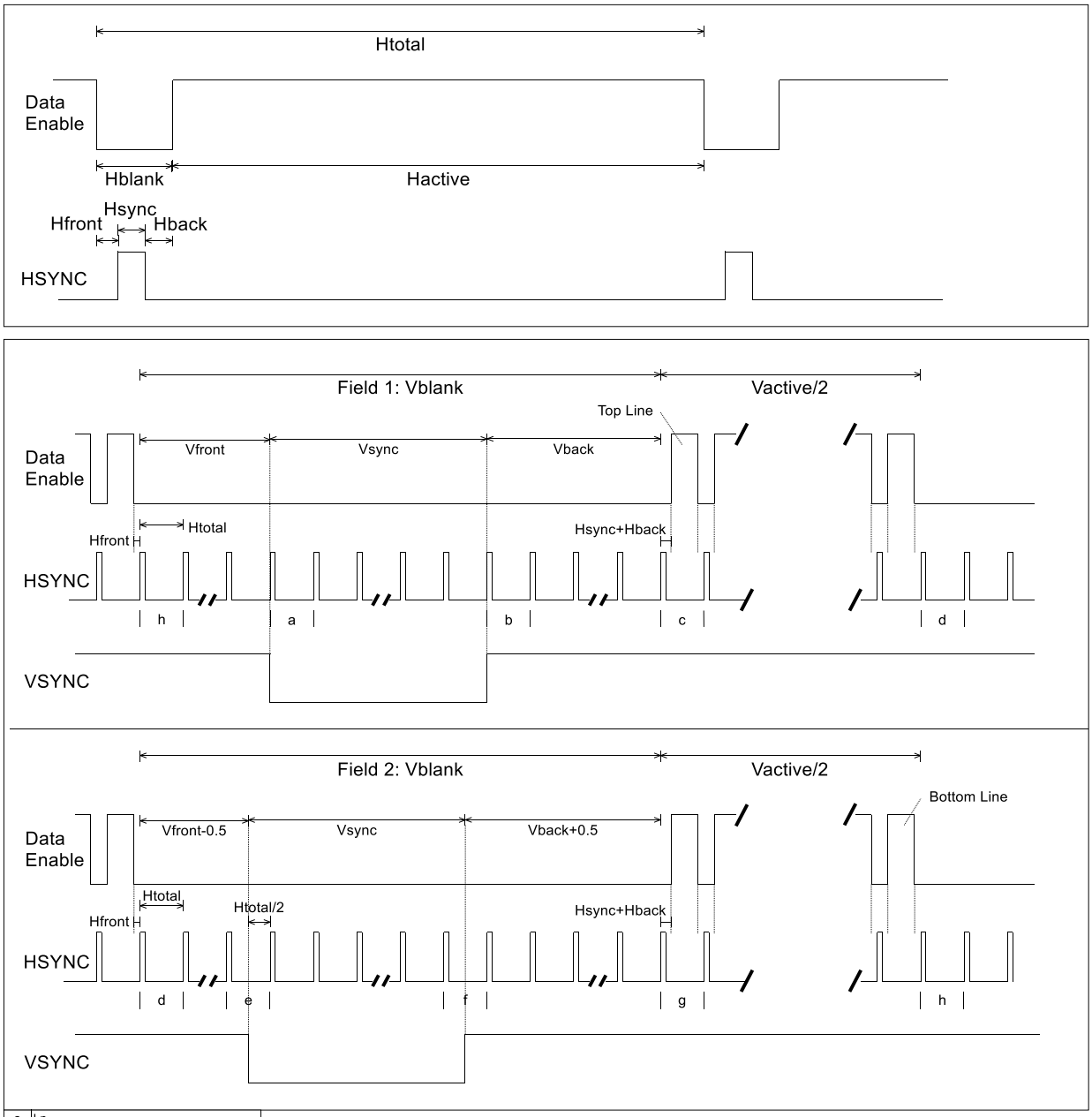




a	Ln
b	Ln+Vsync
c	Ln+Vsync+Vback
d	Ln+Vsync+Vback+Vactive/2
e	Ln+Vsync+Vback+Vactive/2+Vfront
f	Ln+2*Vsync+Vback+Vactive/2+Vfront
g	Ln+2*Vsync+2*Vback+Vactive/2+Vfront+1
h	Ln+2*Vsync+2*Vback+Vactive+Vfront+1

H parameters in pixel clocks
V parameters in lines.
Letters a, b, c, … represent line numbers.

Figure 4 - General Interlaced Video Format Timing (Positive Sync)



a	Ln
b	Ln+Vsync
c	Ln+Vsync+Vback
d	Ln+Vsync+Vback+Vactive/2
e	Ln+Vsync+Vback+Vactive/2+Vfront-1
f	Ln+2*Vsync+Vback+Vactive/2+Vfront-1
g	Ln+2*Vsync+2*Vback+Vactive/2+Vfront
h	Ln+2*Vsync+2*Vback+Vactive+Vfront

H parameters in pixel clocks
V parameters in lines
Letters a, b, c, … represent line numbers

Figure 5 - Special Interlaced Video Format Timing (Even Vtotal)

4.1 Aspect Ratio

A DTV should always indicate its native (physical) aspect ratio in the EDID version 1, revision 3 block's "Max Horizontal Image Size" and "Max Vertical Image Size" fields even if the maximum image size is unknown or variable. Typically, the ratio of these two fields is 4:3, 16:9, or 64:27 — though this may not be true for some displays with non-standard aspect ratios. The Source should use these fields to determine the Native Display Device Aspect Ratio.

Video formats with the same timing, but different Picture Aspect Ratios (e.g., 4:3 and 16:9) are considered different formats that can be independently supported and discovered. These are referred to as Dual-Aspect Ratio Timings.

For any Dual-Aspect Ratio Video Timing, the Preferred Picture Aspect Ratio for that timing is indicated by the first listing of that timing in the EDID. When receiving a signal not accompanied by an aspect ratio indication (because no AVI InfoFrame is transmitted) a DTV shall assume that the aspect ratio is the Preferred Picture Aspect Ratio for the transmitted Video Timing.

If a Dual-Aspect Ratio DTV is receiving a Video Format Timing for which it has declared support for both Picture Aspect Ratios in EDID and the Source has indicated the Picture Aspect Ratio by including the AVI in the video stream, then the DTV shall display the Picture in the aspect ratio that has been indicated by the Source in the AVI. If the Source does not support transmission of the AVI and the Source supports both of the Dual-Aspect Ratio Video Timing Formats for a particular Video Timing defined by the Sink, then the Source shall provide the video to the DTV in the Preferred Picture Aspect Ratio.

For a display device to simultaneously support both formats, the Source needs a way to let the display device know the Picture Aspect Ratio in which the video should be displayed. A DTV shall list only one Picture Aspect Ratio of any Dual-Aspect Ratio Timing unless it is capable of receiving and decoding the AVI InfoFrame defined in Section 6.

However, it is possible for a DTV that has no support for the AVI InfoFrame to still support both aspect ratios of such formats as a user programmable option. In that case, the EDID Detailed Timing Descriptor could be modified during operation to reflect the selected Picture Aspect Ratio and the change could be signaled to the Source (e.g., with Hot Plug Detect on DVI or HDMI). The effects on the EDID data structure are explained in Section 7.2. See Table 2 for Video ID Code and Aspect Ratios.

Table 2 - Video Formats — Video ID Code and Aspect Ratios

VIC	Formats	Field Rate[e]	Picture Aspect Ratio (H:V)[a]	Pixel Aspect Ratio (H:V)
1	640x480p	59.94Hz/60Hz	4:3	1:1
2	720x480p	59.94Hz/60Hz	4:3	8:9
3	720x480p	59.94Hz/60Hz	16:9	32:27
4	1280x720p	59.94Hz/60Hz	16:9	1:1
5	1920x1080i	59.94Hz/60Hz	16:9	1:1
6	720(1440)x480i	59.94Hz/60Hz	4:3	8:9
7	720(1440)x480i	59.94Hz/60Hz	16:9	32:27
8	720(1440)x240p	59.94Hz/60Hz	4:3	4:9
9	720(1440)x240p	59.94Hz/60Hz	16:9	16:27
10	2880x480i	59.94Hz/60Hz	4:3	2:9 – 20:9[b]
11	2880x480i	59.94Hz/60Hz	16:9	8:27 -80:27[b]
12	2880x240p	59.94Hz/60Hz	4:3	1:9 – 10:9[b]
13	2880x240p	59.94Hz/60Hz	16:9	4:27 – 40:27[b]
14	1440x480p	59.94Hz/60Hz	4:3	4:9 or 8:9[c]
15	1440x480p	59.94Hz/60Hz	16:9	16:27 or 32:27[c]
16	1920x1080p	59.94Hz/60Hz	16:9	1:1
17	720x576p	50Hz	4:3	16:15
18	720x576p	50Hz	16:9	64:45
19	1280x720p	50Hz	16:9	1:1
20	1920x1080i	50Hz	16:9	1:1
21	720(1440)x576i	50Hz	4:3	16:15
22	720(1440)x576i	50Hz	16:9	64:45
23	720(1440)x288p	50Hz	4:3	8:15
24	720(1440)x288p	50Hz	16:9	32:45
25	2880x576i	50Hz	4:3	2:15 – 20:15[b]
26	2880x576i	50Hz	16:9	16:45-160:45[b]
27	2880x288p	50Hz	4:3	1:15 – 10:15[b]
28	2880x288p	50Hz	16:9	8:45 – 80:45[b]
29	1440x576p	50Hz	4:3	8:15 or 16:15[c]
30	1440x576p	50Hz	16:9	32:45 or 64:45[c]
31	1920x1080p	50Hz	16:9	1:1
32	1920x1080p	23.98Hz/24Hz	16:9	1:1
33	1920x1080p	25Hz	16:9	1:1
34	1920x1080p	29.97Hz/30Hz	16:9	1:1
35	2880x480p	59.94Hz/60Hz	4:3	2:9, 4:9, or 8:9[d]
36	2880x480p	59.94Hz/60Hz	16:9	8:27, 16:27, or 32:27[d]
37	2880x576p	50Hz	4:3	4:15, 8:15, or 16:15[d]
38	2880x576p	50Hz	16:9	16:45, 32:45, or 64:45[d]
39	1920x1080i (1250 total)	50Hz	16:9	1:1
40	1920x1080i	100Hz	16:9	1:1
41	1280x720p	100Hz	16:9	1:1
42	720x576p	100Hz	4:3	16:15
43	720x576p	100Hz	16:9	64:45
44	720(1440)x576i	100Hz	4:3	16:15
45	720(1440)x576i	100Hz	16:9	64:45

Table 2 - Video Formats — Video ID Code and Aspect Ratios (continued)

VIC	Formats	Field Rate[e]	Picture Aspect Ratio (H:V)[a]	Pixel Aspect Ratio (H:V)
46	1920x1080i	119.88Hz/120Hz	16:9	1:1
47	1280x720p	119.88Hz/120Hz	16:9	1:1
48	720x480p	119.88Hz/120Hz	4:3	8:9
49	720x480p	119.88Hz/120Hz	16:9	32:27
50	720(1440)x480i	119.88Hz/120Hz	4:3	8:9
51	720(1440)x480i	119.88Hz/120Hz	16:9	32:27
52	720x576p	200Hz	4:3	16:15
53	720x576p	200Hz	16:9	64:45
54	720(1440)x576i	200Hz	4:3	16:15
55	720(1440)x576i	200Hz	16:9	64:45
56	720x480p	239.76Hz/240Hz	4:3	8:9
57	720x480p	239.76Hz/240Hz	16:9	32:27
58	720(1440)x480i	239.76Hz/240Hz	4:3	8:9
59	720(1440)x480i	239.76Hz/240Hz	16:9	32:27
60	1280x720p	23.98Hz/24Hz	16:9	1:1
61	1280x720p	25Hz	16:9	1:1
62	1280x720p	29.97Hz/30Hz	16:9	1:1
63	1920x1080p	119.88Hz/120Hz	16:9	1:1
64	1920x1080p	100Hz	16:9	1:1
65	1280x720p	23.98Hz/24Hz	64:27[f]	4:3
66	1280x720p	25Hz	64:27[f]	4:3
67	1280x720p	29.97Hz/30Hz	64:27[f]	4:3
68	1280x720p	50Hz	64:27[f]	4:3
69	1280x720p	59.94Hz/60Hz	64:27[f]	4:3
70	1280x720p	100Hz	64:27[f]	4:3
71	1280x720p	119.88Hz/120Hz	64:27[f]	4:3
72	1920x1080p	23.98Hz/24Hz	64:27[f]	4:3
73	1920x1080p	25Hz	64:27[f]	4:3
74	1920x1080p	29.97Hz/30Hz	64:27[f]	4:3
75	1920x1080p	50Hz	64:27[f]	4:3
76	1920x1080p	59.94Hz/60Hz	64:27[f]	4:3
77	1920x1080p	100Hz	64:27[f]	4:3
78	1920x1080p	119.88Hz/120Hz	64:27[f]	4:3
79	1680x720p	23.98Hz/24Hz	64:27[f]	64:63
80	1680x720p	25Hz	64:27[f]	64:63
81	1680x720p	29.97Hz/30Hz	64:27[f]	64:63
82	1680x720p	50Hz	64:27[f]	64:63
83	1680x720p	59.94Hz/60Hz	64:27[f]	64:63
84	1680x720p	100Hz	64:27[f]	64:63
85	1680x720p	119.88Hz/120Hz	64:27[f]	64:63
86	2560x1080p	23.98Hz/24Hz	64:27[f]	1:1
87	2560x1080p	25Hz	64:27[f]	1:1
88	2560x1080p	29.97Hz/30Hz	64:27[f]	1:1
89	2560x1080p	50Hz	64:27[f]	1:1

Table 2 - Video Formats — Video ID Code and Aspect Ratios (continued)

VIC	Formats	Field Rate[e]	Picture Aspect Ratio (H:V)[a]	Pixel Aspect Ratio (H:V)
90	2560x1080p	59.94Hz/60Hz	64:27[f]	1:1
91	2560x1080p	100Hz	64:27[f]	1:1
92	2560x1080p	119.88Hz/120Hz	64:27[f]	1:1
93	3840x2160p	23.98Hz/24Hz	16:9	1:1
94	3840x2160p	25Hz	16:9	1:1
95	3840x2160p	29.97Hz/30Hz	16:9	1:1
96	3840x2160p	50Hz	16:9	1:1
97	3840x2160p	59.94Hz/60Hz	16:9	1:1
98	4096x2160p	23.98Hz/24Hz	256:135	1:1
99	4096x2160p	25Hz	256:135	1:1
100	4096x2160p	29.97Hz/30Hz	256:135	1:1
101	4096x2160p	50Hz	256:135	1:1
102	4096x2160p	59.94Hz/60Hz	256:135	1:1
103	3840x2160p	23.98Hz/24Hz	64:27[f]	4:3
104	3840x2160p	25Hz	64:27[f]	4:3
105	3840x2160p	29.97Hz/30Hz	64:27[f]	4:3
106	3840x2160p	50Hz	64:27[f]	4:3
107	3840x2160p	59.94Hz/60Hz	64:27[f]	4:3
108	1280x720p	47.95Hz/48Hz	16:9	1:1
109	1280x720p	47.95Hz/48Hz	64:27[f]	4:3
110	1680x720p	47.95Hz/48Hz	64:27[f]	64:63
111	1920x1080p	47.95Hz/48Hz	16:9	1:1
112	1920x1080p	47.95Hz/48Hz	64:27[f]	4:3
113	2560x1080p	47.95Hz/48Hz	64:27[f]	1:1
114	3840x2160p	47.95Hz/48Hz	16:9	1:1
115	4096x2160p	47.95Hz/48Hz	256:135	1:1
116	3840x2160p	47.95Hz/48Hz	64:27[f]	4:3
117	3840x2160p	100Hz	16:9	1:1
118	3840x2160p	119.88Hz/120Hz	16:9	1:1
119	3840x2160p	100Hz	64:27[f]	4:3
120	3840x2160p	119.88Hz/120Hz	64:27[f]	4:3
121	5120x2160p	23.98Hz/24Hz	64:27[f]	1:1
122	5120x2160p	25Hz	64:27[f]	1:1
123	5120x2160p	29.97Hz/30Hz	64:27[f]	1:1
124	5120x2160p	47.95Hz/48Hz	64:27[f]	1:1
125	5120x2160p	50Hz	64:27[f]	1:1
126	5120x2160p	59.94Hz/60Hz	64:27[f]	1:1
127	5120x2160p	100Hz	64:27[f]	1:1
128-192	Forbidden			
193	5120x2160p	119.88Hz/120Hz	64:27[f]	1:1
194	7680x4320p	23.98Hz/24Hz	16:9	1:1
195	7680x4320p	25Hz	16:9	1:1
196	7680x4320p	29.97Hz/30Hz	16:9	1:1

Table 2 - Video Formats — Video ID Code and Aspect Ratios (continued)

VIC	Formats	Field Rate[e]	Picture Aspect Ratio (H:V)[a]	Pixel Aspect Ratio (H:V)
197	7680x4320p	47.95Hz/48Hz	16:9	1:1
198	7680x4320p	50Hz	16:9	1:1
199	7680x4320p	59.94Hz/60Hz	16:9	1:1
200	7680x4320p	100Hz	16:9	1:1
201	7680x4320p	119.88Hz/120Hz	16:9	1:1
202	7680x4320p	23.98Hz/24Hz	64:27[f]	4:3
203	7680x4320p	25Hz	64:27[f]	4:3
204	7680x4320p	29.97Hz/30Hz	64:27[f]	4:3
205	7680x4320p	47.95Hz/48Hz	64:27[f]	4:3
206	7680x4320p	50Hz	64:27[f]	4:3
207	7680x4320p	59.94Hz/60Hz	64:27[f]	4:3
208	7680x4320p	100Hz	64:27[f]	4:3
209	7680x4320p	119.88Hz/120Hz	64:27[f]	4:3
210	10240x4320p	23.98Hz/24Hz	64:27[f]	1:1
211	10240x4320p	25Hz	64:27[f]	1:1
212	10240x4320p	29.97Hz/30Hz	64:27[f]	1:1
213	10240x4320p	47.95Hz/48Hz	64:27[f]	1:1
214	10240x4320p	50Hz	64:27[f]	1:1
215	10240x4320p	59.94Hz/60Hz	64:27[f]	1:1
216	10240x4320p	100Hz	64:27[f]	1:1
217	10240x4320p	119.88Hz/120Hz	64:27[f]	1:1
218	4096x2160p	100Hz	256:135	1:1
219	4096x2160p	119.88Hz/120Hz	256:135	1:1
220-255	Reserved for the Future			
0	No Video Identification Code Available (Used with AVI InfoFrame only)			

Notes:

a. Picture Aspect Ratio — For example, with the 720x480 (16:9) data format and a 4:3 display, the Source could (1) use pan and scan information to crop the data to 540 horizontal pixels and then resample up to the required 720 pixels for output to the display or (2) vertically resample to 360 lines and create Bars of 60 lines above and below it to send this "letterbox" with the required 480 lines for output. Other Picture scaling methods are possible in either Source or Sink. For example, Picture Aspect Ratio scaling (Picture expand, shrink, etc.) can be accomplished in the Source, including, possibly, added black/gray lines in the pixel portion of the video. The exception to this is the 640x480 format, which is always sent as 4x3 data, and is rendered according to the characteristics of the Sink.

b. The pixel repeat value can vary from 9 to 0 (see the PR field in Section 6.4.1) resulting in 10 variations of Pixel Aspect Ratio.

c. The pixel repeat value can be set to 0 or 1 (see the PR field in Section 6.4.1) resulting in 2 variations of Pixel Aspect Ratio.

d. The pixel repeat value can be set to 0, 1 or 3 (see the PR field in Section 6.4.1) resulting in 3 variations of Pixel Aspect Ratio.

e. In the case of interlaced formats, the frame rate is 1/2 the field rate.

f. This Picture Aspect Ratio continues the progression $(4:3)^N$, where N = 1, 2, and 3, and is near other wide cinematic values in use such as 2.2:1 (SMPTE RP 91 [131]), 21:9, and 2.39:1 (SMPTE RP 40 [130]).

4.2 Frame Rate Relationships

Some Video Formats have a high frame rate that is an integer (2X, 4X, or 5X) multiple of the frame rate of a base (1X) Video Format. While receiving certain low frame rate Video Formats, some Sources may calculate extra interpolated frames, and output a related high frame rate Video Format — in order to optimize display performance. Table 3 lists the VICs of base Video Formats along with the VICs of their higher frame rate counterparts.

Table 3 - Frame Rate Relationships — Base to High Frame Rate VICs

Base (1X)	2X	4X	5X
2	48	56	
3	49	57	
4	47		
5	46		
6	50	58	
7	51	59	
17	42	52	
18	43	53	
19	41		
20	40		
21	44	54	
22	45	55	
32			63
33	31	64	
34	16		
60			47
61	19	41	
62	4	47	
65			71
66	68	70	
67	69	71	
72			78
73	75	77	
74	76	78	
79			85
80	82	84	
81	83	85	
86			92
87	89	91	
88	90	92	
94	96	117	
95	97	118	
99	101	218	
100	102	219	
104	106		
105	107		

5 COLOR ENCODING, SAMPLING, AND CONVERSION

5.1 Default Encoding Parameters

When present, encoding parameters specified in the AVI InfoFrame, or other interface-specific controls (e.g., HDMI General Control Packets), shall take precedence over any default parameters defined in this sub-section.

The default Component Depth shall be 8-bits (N = 8). Other elements of the default parameter set vary as a function of Video Format Timing type (IT or CE) and (in the case of CE Video Format Timings) vertical Active Line count (Vactive).

Table 4 defines default Quantization Ranges for various colorimetries and Video Formats[2]. Methods may be provided to override these default Quantization Ranges for any Video Format (see Sections 6.4.3 and 7.5.6).

Table 4 - Default Quantization Ranges

Colorimetry	CE Video Format	IT Video Format
RGB	Limited	Full
SMPTE ST 170	Limited	Limited
Rec. ITU-R BT.709	Limited	Limited
$xvYCC_{601}$	Limited [a]	Limited [a]
$xvYCC_{709}$	Limited [a]	Limited [a]
$sYCC_{601}$	Limited	Full
$opYCC_{601}$	Limited	Full
opRGB	Limited	Full
Rec. ITU-R BT.2020 $Y'_CC'_{BC}C'_{RC}$	Limited	Limited
Rec. ITU-R BT.2020 R'G'B'	Limited	Full
Rec. ITU-R BT.2020 $Y'C'_BC'_R$	Limited	Limited
SMPTE ST 2113 P3 R'G'B'	Limited	Full
Rec. ITU-R BT.2100 IC_TC_P	Limited	Limited
sRGB	Limited	Full
defaultRGB	Limited [b]	Full [b]

Notes:

a. The quantization range of the $xvYCC_{601}$ and $xvYCC_{709}$ colorimetries is always Limited Range, but values outside that range are allowed to represent an extended gamut (see IEC 61966-2-4 [5]).

b. Uses the default RGB color space defined in the Basic Display Parameters and Features Block of the EDID.

[2] Historically video for Consumer Electronics applications uses Limited Range due to both legacy considerations (analog TV overshoot/undershoot) and the prevalence of interfaces such as SDI that use embedded synchronization packets that prevent the use of Full Range Quantization.

When transmitting IT Video Formats, the default color space shall be RGB using Full Range quantization levels. The RGB color space used for the transmission of IT Video Formats should[3] be the RGB color space the Sink declares in the Basic Display Parameters and Features Block of its EDID (see Sections A.2.6 and A.2.7 for further information). Most Sources default to RGB color space when transmitting IT Video Formats.

If a Source determines that a Sink is incapable of receiving AVI InfoFrames or is incapable of receiving YC_BC_R Pixel Data, then it shall, by default, encode the CE Video Format Pixel Data in RGB color space using Limited Range levels. If a Sink is incapable of receiving AVI InfoFrames, incapable of receiving YC_BC_R Pixel Data, or does not receive an AVI InfoFrame, then it should, by default, assume CE Video Format Pixel Data is encoded in RGB color space using Limited Range levels and IT Video Format Pixel Data is encoded in RGB color space using Full Range levels. In all cases described above, the RGB color space used should[3] be the RGB color space the Sink declares in the Basic Display Parameters and Features Block of its EDID.

If a Source determines that a Sink is capable of receiving AVI InfoFrames and is capable of receiving YC_BC_R Pixel Data, then it shall, by default, encode CE Video Format Pixel Data in a color space determined by the vertical Active Line count (Vactive) using Limited Range levels. By default, an SD Video Format shall be encoded according to SMPTE ST 170 [1] color space, and non-SD Video Formats shall be encoded according to Rec. ITU-R BT.709 [7] color space.

5.2 Color Component Samples

Color is communicated using a set of three components. This interface shall be capable of supporting RGB (red, green, and blue), with encoding parameters based on the format.

The interface may optionally support color formats that allow sub-sampling of chroma, referred to as YC_BC_R[4] or YCC in this specification. The colorimetries used by these formats include SMPTE ST 170, Rec. ITU-R BT.709, xvYCC, sYCC, opYCC, $Y'_CC'_{BC}C'_{RC}$, $Y'C'_BC'_R$, and IC_TC_P. When any of these colorimetries are used, the chroma sampling format (4:4:4, 4:2:2, or 4:2:0) shall be signaled using the Y field in the AVI InfoFrame (see Section 6.4.2).

IC_TC_P is signaled and transmitted equivalent to the YC_BC_R colorimetries. All YCC-related fields and flags shall apply, including, but not limited to, signaling with the Y and YQ fields of the AVI InfoFrame (see Section 6.4.3), the QY flag in the VCDB (see Section 7.5.6) and the Y420VDB and Y420CMDB (see Sections 7.5.10 and 7.5.11).

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[3] In practice, Sources often ignore the default RGB color space that the Sink declares in the Basic Display Parameters and Features Block of its EDID and instead will use the sRGB[33] color space. It is advisable that Sinks declare explicit support for sRGB and defaultRGB colorimetry in the Colorimetry Data Block, and that Sources explicitly signal in the AVI InfoFrame whether sRGB or defaultRGB colorimetry is transmitted.

[4] Typically, YC_BC_R notation is used for digital domains; and YP_BP_R is used for analog domains. RGB signals have the same notation in the digital and analog domains.

Pixel Data with IC_TC_P colorimetry shall use the following component mapping: The I component shall be carried in the Y Color Component Sample, C_T shall be carried in the C_B Color Component Sample, and C_P shall be carried in the C_R Color Component Sample.

5.2.1 RGB Conversion Matrices

A transformation from YC_BC_R or IC_TC_P to RGB generally occurs within the DTV after it receives a YC_BC_R or IC_TC_P encoded Picture. A transformation between YC_BC_R Color Component Samples and RGB Color Component Samples can be accomplished by applying one of four conversion matrices: Rec. ITU-R BT.601 [6], Rec. ITU-R BT.709 [7], Rec. ITU-R BT.2020 [40] non-constant luminance, or Rec. ITU-R BT.2020 [40] constant luminance. A transformation between IC_TC_P Color Component Samples and RGB Color Component Samples can be accomplished by applying one of two conversion matrices: Rec. ITU-R BT.2100 [48] constant intensity IC_TC_P color conversion matrix for PQ or HLG. The specific conversion matrix required depends on the Colorimetry and Extended Colorimetry fields in the AVI InfoFrame. The conversion matrix is either specified explicitly (e.g., the Colorimetry field is set to Rec. ITU-R BT.601 or Rec. ITU-R BT.709) or it is denoted in the subscript of the short name of the selected YC_BC_R colorimetry. For example, the Rec. ITU-R BT.601 conversion matrix applied to $sYCC_{601}$ or $opYCC_{601}$ Color Component Samples results in RGB Color Component Samples (both positive and negative).

5.2.2 Sample Lattice

In order to improve color reproduction, the sample lattice for RGB, YC_BC_R 4:2:2, and IC_TC_P 4:2:2 Pixel Data should conform to the Rec. ITU-R BT.709 [7] sampling lattice. The sample lattice for these Pixel Data encodings are described below for convenience:

- R, G, B, Y, and I components are orthogonal, line- and picture-repetitive. R, G, and B components are co-sited with each other.
- C_B, C_R, C_T, and C_P are orthogonal, line- and picture-repetitive co-sited with each other and with alternate Y or I samples (starting with the first active Y or I sample in a line).

The sample lattice for YC_BC_R 4:4:4 and IC_TC_P 4:4:4 Pixel Data should be the same as the sample lattice for RGB Pixel Data.

5.3 Transfer Characteristic

The transfer characteristics for all supported colorimetries are shown in Table 5. The first column for the SDR Gamma Transfer Function applies when no Dynamic Range and Mastering InfoFrame is being used to specify a Transfer Function; the remaining columns apply when using a Dynamic Range and Mastering InfoFrame (see Section 6.9 for details on Gamma - HDR, PQ, and HLG Transfer Functions). In Table 5, P refers to the peak encoded luminance, and γ is the assumed gamma value of the display.

Table 5 - Colorimetry Transfer Characteristics

	Transfer Function			
Colorimetry	**Gamma - SDR (P = 100 cd/m² unless specified below)**	**Gamma - HDR (P = max_display_-mastering_luminance)**	**PQ**	**HLG**
RGB	Rec. ITU-R BT.709 [7] Item 1.2	Rec. ITU-R BT.709 [7] Item 1.2	Rec. ITU-R BT.2100 [48] Table 4	Rec. ITU-R BT.2100 [48] Table 5
opRGB	γ = 2.19921875, P = 160 cd/m²	γ = 2.19921875	Rec. ITU-R BT.2100 [48] Table 4	Rec. ITU-R BT.2100 [48] Table 5
Rec. ITU-R BT.2020 R'G'B'	Rec. ITU-R BT.2020 [40] Table 4	Rec. ITU-R BT.2020 [40] Table 4	Rec. ITU-R BT.2100 [48] Table 4	Rec. ITU-R BT.2100 [48] Table 5
SMPTE ST 2113 P3D65 R'G'B'	γ = 2.6, P = 48 cd/m²	γ = 2.6	Rec. ITU-R BT.2100 [48] Table 4	Rec. ITU-R BT.2100 [48] Table 5
SMPTE ST 2113 P3DCI R'G'B'	γ = 2.6, P = 48 cd/m²	γ = 2.6	Rec. ITU-R BT.2100 [48] Table 4	Rec. ITU-R BT.2100 [48] Table 5
SMPTE ST 170 or Rec. ITU-R BT.709	Rec. ITU-R BT.709 [7] Item 1.2	Rec. ITU-R BT.709 [7] Item 1.2	Rec. ITU-R BT.2100 [48] Table 4	Rec. ITU-R BT.2100 [48] Table 5
$xvYCC_{601}$	IEC 61966-2-4 [5] Section 4.2	IEC 61966-2-4 [5] Section 4.2	Reserved	Reserved
$xvYCC_{709}$	IEC 61966-2-4 [5] Section 4.3	IEC 61966-2-4 [5] Section 4.3	Reserved	Reserved
$sYCC_{601}$	IEC 61966-2-1/Amd 1 [34] Section F.2	IEC 61966-2-1/Amd 1 [34] Section F.2	Reserved	Reserved
$opYCC_{601}$	γ = 2.19921875, P = 160 cd/m²	γ = 2.19921875	Reserved	Reserved
Rec. ITU-R BT.2020 $Y'_C C'_{BC} C'_{RC}$	Rec. ITU-R BT.2020 [40] Table 4 (CL)	Rec. ITU-R BT.2020 [40] Table 4 (CL)	Reserved	Reserved
Rec. ITU-R BT.2020 $Y'C'_B C'_R$	Rec. ITU-R BT.2020 [40] Table 4 (NCL)	Rec. ITU-R BT.2020 [40] Table 4 (NCL)	Rec. ITU-R BT.2100 [48] Table 4	Rec. ITU-R BT.2100 [48] Table 5
Rec. ITU-R BT.2100 $IC_T C_P$	Reserved	Reserved	Rec. ITU-R BT.2100 [48] Table 7 (PQ)	Rec. ITU-R BT.2100 [48] Table 7 (HLG)
sRGB	IEC 61966-2-1 [33] Section 5.2	IEC 61966-2-1 [33] Section 5.2	Rec. ITU-R BT.2100 [48] Table 4	Rec. ITU-R BT.2100 [48] Table 5
defaultRGB	Gamma as defined in the Basic Display Parameters and Features Block of the EDID for the default colorimetry	Gamma as defined in the Basic Display Parameters and Features Block of the EDID for the default colorimetry	Rec. ITU-R BT.2100 [48] Table 4	Rec. ITU-R BT.2100 [48] Table 5

5.4 Color Coding and Quantization

Component Depth: The coding shall be N-bit, where N = 8, 10, 12, or 16 bits/component — except in the case of the default 640x480 Video Timing 1, where the value of N shall be 8.

Rounding: code = Floor (X + 0.5), where X is the result of a floating point calculation. After rounding, code shall be limited according to the range given below.

Range: Limited Range R, G, B, Y, and I signals shall have $(219 * 2^{(N-8)}) + 1$ quantization levels. Limited Range C_B, C_R, C_T, and C_P signals shall have $(224 * 2^{(N-8)}) + 1$ quantization levels. Full Range R, G, B, Y, I, C_B, C_R, C_T, and C_P signals shall have 2^N quantization levels.

Levels: Limited Range R, G, B, Y, and I signals shall have black level corresponding to code $16 * 2^{(N-8)}$ and peak white level corresponding to code $235 * 2^{(N-8)}$; Limited Range C_B, C_R, C_T, and C_P signals shall have a zero level corresponding to digital code $2^{(N-1)}$ and range spanning codes $16 * 2^{(N-8)}$ to $240 * 2^{(N-8)}$. Full Range R, G, B, Y, and I signals shall have a black level corresponding to code 0 and peak white level corresponding to code $(2^N) - 1$. Full Range C_B, C_R, C_T, and C_P signals shall have a zero level corresponding to digital code $2^{(N-1)}$ and range spanning codes 0 to $(2^N) - 1$.

Overshoot/Undershoot Regions: If the N-bit digital video signal is converted to an analog signal in the Sink, it is recommended that for RGB or Y or I, the black level (i.e., sync level and blanking level) be aligned with the video portion of the signal at black and white digital levels $16 * 2^{(N-8)}$ and $235 * 2^{(N-8)}$, respectively, such that the Limited Range digital signal swing corresponds to the nominal analog video swing (e.g., 0 to 700 mV per Sections 9.4, 10.5, and 10.6 of SMPTE ST 274 [2]). This means that zero analog level (0.0 IRE Units) should be associated with digital level $16 * 2^{(N-8)}$. Digital levels in an undershoot region 1 to $(16 * 2^{(N-8)}) - 1$ and overshoot region $(235 * 2^{(N-8)}) + 1$ to $(2^N) - 2$ are recommended to be passed through the digital to analog converter; however, limited range of the analog signal should be aligned with the range $16 * 2^{(N-8)}$ to $235 * 2^{(N-8)}$ since it is expected that this range contains essential video. For the 640x480p format, it is recommended that the full 0 – 255 range be displayed for this format.

Forbidden Values: For Limited Range R, G, B, Y, I, C_B, C_R, C_T, C_P signals, codes outside the range $2^{(N-8)}$ to $(255 * 2^{(N-8)}) - 1$ are reserved and shall not be considered video.

6 AUXILIARY INFORMATION CARRIED FROM SOURCE TO SINK

Various types of auxiliary data can be carried from the Source to the Sink using InfoFrames. This section describes the InfoFrames that have been defined so far.

The actual mechanism for carrying these InfoFrames may vary depending on the digital interface being used[5].

Sources and Sinks shall not rely on the Revision Number in the CTA Extension of the Sink's EDID to determine whether a Sink can accept InfoFrames. Sinks shall declare InfoFrame capability by including an interface related (e.g., HDMI) VSDB in their EDID CTA Extension. Sources shall only assume InfoFrame capability, when an appropriate (e.g., HDMI) VSDB is found.

> Note: Previous versions of CTA-861 relied on a Revision Number in the included CTA Extension to indicate whether the Sink could accept InfoFrames. Due to a significant number of DVI (not InfoFrame-capable) Sinks having the Revision Number set to 3, indicating support of InfoFrames, and not being capable of doing so, it is necessary to deprecate this requirement.

DVI does not support the transmission of any InfoFrames, independent of CTA Extension version number. Sinks with a VSDB indicating support for reception of InfoFrames shall accept any of the InfoFrames defined here.

The assigned type codes for InfoFrames are shown in Table 6. The first byte of the InfoFrame designates the type of InfoFrame while the second byte indicates the version of that particular InfoFrame. All future versions of a specific InfoFrame shall be backward compatible with previous versions. They may contain additional information, but old and new devices should be able to access and interpret the information previously received.

Extended InfoFrames are distinct from InfoFrames and are described in Section 6.10.

[5] Neither DVI 1.0 [4] nor OpenLDI 0.95 [8] contain a mechanism for transporting InfoFrames. These physical interfaces can be used to implement this standard with reduced functionality. HDMI, which is backward compatible with DVI 1.0 and contains mechanisms for transferring InfoFrames, digital audio, and YC_BC_R Pixel Data, is available and can be used to implement the full capabilities of CTA-861.

Table 6 - List of InfoFrame Type Codes

InfoFrame Type Code	Type of InfoFrame
0x00	Reserved
0x01	Vendor-Specific (defined in Section 6.1)
0x02	Auxiliary Video Information (defined in Section 6.4)
0x03	Source Product Description (defined in Section 6.5)
0x04	Audio (defined in Section 6.6)
0x05	MPEG Source (defined in Section 6.7)
0x06	NTSC VBI (defined in Section 6.8)
0x07	Dynamic Range and Mastering (defined in Section 6.9)
0x08-0x1F	Reserved for future use
0x20-0xFF	Forbidden

The InfoFrame Length Field is contained in the third byte of each InfoFrame. This length field is the total number of bytes in the InfoFrame Payload. It does not include the Type, Version, or Length fields. In the case of the Vendor-Specific InfoFrame, the length includes the IEEE Registration ID, also called OUI, as well as any additional bytes defined by the vendor to be in the InfoFrame (see Table 7). The IEEE Company ID (CID) can be used in place of the OUI. If the InfoFrame Length field is not set correctly, Sinks might not be able to parse the InfoFrame correctly.

6.1 Vendor-Specific InfoFrames

The content of the Vendor-Specific InfoFrame is defined in Table 7 and Table 8. This InfoFrame can be used by product manufacturers or organizations that have an assigned 24-bit IEEE Registration Identifier to transport information not defined elsewhere. The Vendor-Specific Payload would be defined by the organization to which the 24-bit IEEE number refers. The 24-bit IEEE number is sent the least significant byte first. It is recommended that the Vendor-Specific Payload contain a “length field” to facilitate extensibility, but this is not required.

Any organization or vendor that wishes to define a Vendor-Specific InfoFrame shall obtain a registration ID (also known as vendor ID, organizationally unique ID, or company ID) from the IEEE Registration Authority [128].

Table 7 - Vendor-Specific InfoFrame (Version 1)

InfoFrame Type Code		InfoFrame Type = 0x01
InfoFrame Version Number		Version = 0x01
InfoFrame Length (L_V)		Total number of bytes in InfoFrame Payload including IEEE Registration ID
Data Byte 1	24 bit IEEE Registration Identifier	IEEE OUI third two hex digits
Data Byte 2		IEEE OUI second two hex digits
Data Byte 3		IEEE OUI first two hex digits
Data Byte 4		Vendor-Specific Payload
...		
Data Byte L_V		

Table 8 - Vendor-Specific InfoFrame (Version 2)

InfoFrame Type Code		InfoFrame Type = 0x01	
InfoFrame Version Number (bits 0 – 6) and VSIF Change Bit (bit 7)		0 or 1	Version = 0x02
InfoFrame Length (L_V)		Total number of bytes in InfoFrame Payload including IEEE Registration ID	
Data Byte 1	24 bit IEEE Registration Identifier	IEEE OUI third two hex digits	
Data Byte 2		IEEE OUI second two hex digits	
Data Byte 3		IEEE OUI first two hex digits	
Data Byte 4		Vendor-Specific Payload	
...			
Data Byte L_V			

6.1.1 Multiple VSIF Handling

The following requirements ensure the highest level of interoperability with legacy devices that may or may not support 1 or more VSIFs and devices supporting the use of VSIFs.

a) Source devices transmitting one or more Vendor-Specific InfoFrames (VSIFs) shall transmit all such VSIFs during a single vertical blanking interval (VBI). In the case of interlaced formats, the VSIFs shall be transmitted immediately prior to the first of the two fields. It is permissible to also send the VSIF data immediately prior to the second of two fields.
b) When transmitting Vendor-Specific InfoFrames, Source devices shall transmit all of them at least once per two Video Fields.
c) If a Source device transmits Interface VSIF(s) as well as other VSIF(s), it shall transmit the Interface VSIF(s) last.

d) The information carried in one or more VSIFs transmitted in the same VBI shall apply to the Video Field/Frame directly following the VBI.
e) Sink devices should apply the VSIF information from a given VSIF type to the subsequent Field/Frames until a change is transmitted or until a VSIF has not been received for two sequential Fields.
f) InfoFrame information transmitted in non-VSIF(s) and the Interface VSIF(s) should supersede any contradictory information transmitted by ordinary VSIFs.
g) If two (or more) VSIFs in the same VBI carry conflicting information and both are being acted upon by the Sink, the information contained within the VSIF received last should take precedence.
h) Sink devices shall read the IEEE Registration Identifier of all VSIFs received in order to properly identify and process such VSIFs.
i) Sink devices that support the processing of multiple VSIFs should support at least four (4) VSIFs that arrive as frequently as every VBI.
j) Source devices shall not transmit identical VSIFs during a single VBI.
    - Vendors may choose to define more than one VSIF, however simultaneous use of multiple VSIFs with same OUI is discouraged as this wastes scarce resources in the Sink.
k) Source and Sink devices should handle any changes to the contents of the VSIFs as a group.
    - When a Source transmits a sole VSIF, it shall be version 1 (see Table 7).
    - When a set of multiple VSIFs are transmitted and the Sink's EDID contains an InfoFrame Data Block (see Section 7.5.9), the first VSIF shall be version 2 (see Table 8) and the VSIFs that follow it shall be version 1 (see Table 7). In this case, the first VSIF shall indicate changes in the set of multiple VSIFs via its VSIF Change Bit[6], which shall toggle from 0-to-1 or from 1-to-0 each time the balance of the information in the first VSIF, or contents of any other VSIF in the set, changes.
    - When a set of multiple VSIFs are transmitted and the Sink's EDID does not contain an InfoFrame Data Block, then all VSIFs (including the first) in the set of multiple VSIFs shall be version 1 (see Table 7). In this case, the first VSIF, in the series of VSIFs,

[6] Some Sinks have a single VSIF buffer and the ability to generate an interrupt in response to VSIF data changes. When the Source transmits a sole VSIF, VSIF interrupts can be interpreted as a change in this sole VSIF and VSIF processing in the Sink is infrequent (i.e., only occurs when the Source changes the content of the VSIF) and reasonable. However, when a single VSIF buffer is required to process multiple VSIFs, interrupts occur much more often (i.e., once for each VSIF received – since each VSIF received is different from the VSIF received before it) and no longer correlate with changes to individual VSIF data. Instead, interrupts signal the arrival of the next VSIF in a set of VSIFs. Without special measures, Sink VSIF processing might otherwise dominate available processor bandwidth as software would be forced to check for changes in a set of VSIFs on a frame-by-frame basis. To prevent this interrupt-overload, designs are used which filter only the first VSIF (or actually the first few bytes of the first VSIF), and provide an interrupt if something changes in these bytes. It is with this in mind that a toggle bit is added – to make it easy for Sink software to check when data, in a set of VSIFs, has changed – since the Source will toggle the toggle bit in the first VSIF to trigger the interrupt. For the Source device, which is creating a set of VSIFs, it is relatively easy to toggle a bit when there is a change in any of the VSIFs.

should[7] have its contents changed by at least one bit whenever the contents of any other VSIF, in the set of multiple VSIFs, changes.

- The Source shall transmit the VSIFs in the same order until any of the VSIFs change.

l) Sink devices that support additional VSIF processing capabilities shall report those capabilities using an InfoFrame Data Block. Source devices shall inspect the InfoFrame Data Block and send the appropriate VSIF(s) per the Sink device's capabilities and instructions from the associated product manufacturer or organization. Repeaters should report their VSIF processing capability and only send those VSIF(s) consistent with the reported capabilities of the downstream Sink device.

6.2 Auxiliary Video Information (AVI) InfoFrame

This section has been removed (see Section 6.4).

6.3 Format of Version 1 AVI InfoFrame

Previous versions of CTA-861 defined a Version 1 AVI InfoFrame. Support for version 1 is no longer included in CTA-861. The format of the Version 2 AVI InfoFrame is backward compatible with Version 1. All of the fields that were contained in the Version 1 AVI InfoFrame are also contained in the Version 2 AVI InfoFrame (see Section 6.4). Their purpose and use remain unchanged.

6.4 Format of Version 2, 3, and 4 AVI InfoFrames

The Auxiliary Video Information (AVI) InfoFrame contains information that describes the Pixel Data carried in the next Video Field. It also contains information about the composition of the Picture. Also see Section 7 for EDID requirements that relate to the processing of AVI InfoFrames in both Source and Sinks.

If the Source supports the transmission of the Auxiliary Video Information (AVI) and if it determines that the Sink is capable of receiving that information, it shall send the AVI to the Sink once per Video Field. The data from the AVI applies to the next full frame of video data.

A Sink capable of receiving a Video Format with Video Identification Code greater than 7 or capable of receiving Dual-Aspect Ratio Timing shall be able to receive and decode the AVI InfoFrame described in this Section. As required in Section 7.1, a Sink declares the capability of receiving Video Formats generated at different Picture Aspect Ratios by listing both Video Formats in its EDID data structure. Simultaneous support of timings available in two different aspect ratios shall be indicated by listing both formats in the EDID data structure at the same time.

[7] Otherwise, some legacy Sink devices may not know to look for (and act upon) changes in other VSIFs.

If, for some reason, an indication is received that conflicts with the Video Format being received (e.g., the Source indicates 4:3 but sends the 1920x1080i format), then the Sink shall ignore the conflicting information in the AVI.

If a Sink is capable of receiving YC_BC_R Pixel Data, then, as defined in Section 7.1, it is required to include the Version 3 CTA Extension in its EDID with at least one of the YC_BC_R chroma sampling format bits set. When a Sink's EDID indicates that it is capable of receiving YC_BC_R Pixel Data the Sink shall be capable of receiving AVI InfoFrames. If no AVI InfoFrame is sent from the Source, then, as defined in Section 5.1, the Sink is required to assume that all CE Video Formats are encoded in RGB color space using Limited Range levels with an 8-bit Component Depth.

The information on "Active Format Aspect Ratio," Bar widths, overscan/underscan, non-uniform picture scaling, and colorimetry is information that can be used by the Sink to improve the picture quality. Use of this information by the Sink is optional. If this information is present at the Source and valid[8], and if the Sink is capable of receiving the AVI, the Source shall send the information.

Sources shall not use AVI InfoFrame version 1.

All fields of the Version 3 AVI InfoFrame are described in Table 9. All fields of the Version 2 AVI InfoFrame are the same as the Version 3, except for the Y, VIC, and Version fields. The Y and VIC fields of a Version 2 AVI InfoFrame do not include the most-significant bits Y2 and VIC7 shown. The Y2 and VIC7 bits are simply set to zero in a Version 2 AVI InfoFrame and might not be decoded by some Sinks.

Table 9 - Auxiliary Video Information (AVI) InfoFrame Format (Versions 2 and 3)

InfoFrame Type Code	InfoFrame Type = 0x02							
InfoFrame Version Number	Version = 0x02 or [0x03]							
Length of AVI InfoFrame	Length of AVI InfoFrame (13)							
Data Byte 1	[Y2]	Y1	Y0	A0	B1	B0	S1	S0
Data Byte 2	C1	C0	M1	M0	R3	R2	R1	R0
Data Byte 3	ITC	EC2	EC1	EC0	Q1	Q0	SC1	SC0
Data Byte 4	[VIC7]	VIC6	VIC5	VIC4	VIC3	VIC2	VIC1	VIC0
Data Byte 5	YQ1	YQ0	CN1	CN0	PR3	PR2	PR1	PR0
Data Byte 6	ETB07-ETB00 (Line Number of End of Top Bar — lower 8 bits)							
Data Byte 7	ETB15-ETB08 (Line Number of End of Top Bar — upper 8 bits)							
Data Byte 8	SBB07-SBB00 (Line Number of Start of Bottom Bar — lower 8 bits)							
Data Byte 9	SBB15-SBB08 (Line Number of Start of Bottom Bar — upper 8 bits)							
Data Byte 10	ELB07-ELB00 (Pixel Number of End of Left Bar — lower 8 bits)							
Data Byte 11	ELB15-ELB08 (Pixel Number of End of Left Bar — upper 8 bits)							
Data Byte 12	SRB07-SRB00 (Pixel Number of Start of Right Bar — lower 8 bits)							
Data Byte 13	SRB15-SRB08 (Pixel Number of Start of Right Bar — upper 8 bits)							

[8] The data may not be valid if, for example, the stream was converted from an analog signal with no reliable aspect ratio or format information.

All fields of the Version 4 AVI InfoFrame are described in Table 10. All fields of the Version 4 AVI InfoFrame are the same as Version 3 AVI InfoFrame, except for the InfoFrame Version Number, Length of AVI InfoFrame, and additional Data Bytes 14 and 15.

Table 10 - Auxiliary Video Information (AVI) InfoFrame Format (Version 4)

InfoFrame Type Code	InfoFrame Type = 0x02							
InfoFrame Version Number	Version = 0x04							
Length of AVI InfoFrame	Length of AVI InfoFrame (14 or 15)							
Data Byte 1	Y2	Y1	Y0	A0	B1	B0	S1	S0
Data Byte 2	C1	C0	M1	M0	R3	R2	R1	R0
Data Byte 3	ITC	EC2	EC1	EC0	Q1	Q0	SC1	SC0
Data Byte 4	VIC7	VIC6	VIC5	VIC4	VIC3	VIC2	VIC1	VIC0
Data Byte 5	YQ1	YQ0	CN1	CN0	PR3	PR2	PR1	PR0
Data Byte 6	ETB07-ETB00 (Line Number of End of Top Bar — lower 8 bits)							
Data Byte 7	ETB15-ETB08 (Line Number of End of Top Bar — upper 8 bits)							
Data Byte 8	SBB07-SBB00 (Line Number of Start of Bottom Bar — lower 8 bits)							
Data Byte 9	SBB15-SBB08 (Line Number of Start of Bottom Bar — upper 8 bits)							
Data Byte 10	ELB07-ELB00 (Pixel Number of End of Left Bar — lower 8 bits)							
Data Byte 11	ELB15-ELB08 (Pixel Number of End of Left Bar — upper 8 bits)							
Data Byte 12	SRB07-SRB00 (Pixel Number of Start of Right Bar — lower 8 bits)							
Data Byte 13	SRB15-SRB08 (Pixel Number of Start of Right Bar — upper 8 bits)							
Data Byte 14	ACE3	ACE2	ACE1	ACE0	FR3	FR2	FR1	FR0
Data Byte 15 (if Length >= 15)	F157=0	FR4	RID5	RID4	RID3	RID2	RID1	RID0

Sinks that support AVI InfoFrames shall include support for AVI InfoFrame Version 2.

Sinks implicitly advertise support for Version 3 AVI InfoFrames by declaring support for VIC >= 128 or Y = 7 or both.

Sinks implicitly advertise support for Version 4 AVI InfoFrames with Length = 14 by declaring support for any standard indicated by ACE (Additional Colorimetry Extension listed in Table 19) in their Colorimetry Data Block (Table 82). Sinks that support AVI InfoFrame Version 4 with Length = 14 shall also support AVI InfoFrame Version 3.

Sinks implicitly advertise support for Version 4 AVI InfoFrames with Length = 15 through Length = 27 by declaring support for video timings through a VFDB (Video Format Data Block, see Section 7.5.19) in their EDID. Sinks that support AVI InfoFrame Version 4 with Length = 15 through Length = 27 shall also support AVI InfoFrame Version 4 with Length = 14.

Sources shall not send AVI InfoFrame Version 3 to Sinks that do not indicate support for AVI InfoFrame Version 3 as described above.

Sources shall not send AVI InfoFrame Version 4 with Length = 14 to Sinks that do not indicate support for AVI InfoFrame Version 4 as described above.

Sources shall not send AVI InfoFrame Version 4 with Length = 15 through Length = 27 to Sinks that do not indicate support for AVI InfoFrame Version 4 with Length = 15 through Length = 27 as described above.

Sources shall not send a Version 4 InfoFrame with Length = 15, RID = 0, and FR = 0, and shall send a Version 4 (or lower) InfoFrame with Length = 14 (or less) instead.

Although the current AVI InfoFrame syntax defines lengths up to 15 Data Bytes, Sinks that include a VFDB shall be capable of receiving and handling an AVI InfoFrame of longer lengths up to 27 Data Bytes. Any bytes past the length specified in the syntax are considered Reserved for future use and shall be cleared by the Source and ignored by the Sink.

When Y = 7, the IDO defines the C, EC, and ACE fields. In this case the Source shall set the AVI InfoFrame Version field to no less than 3. AVI InfoFrame Version 3 indicates that Byte 14 is not transmitted, while AVI InfoFrame Version 4 indicates the presence of Byte 14. A Length field value of 15 in an AVI InfoFrame Version 4 indicates the presence of Byte 15. When Y < 7, Sources shall use the following algorithm for AVI InfoFrame Version selection:

```
If (RID > 0 or FR > 0)
        Sources shall use AVI InfoFrame Version 4 with Length = 15.
Elseif (C = 3 and EC = 7)
        Sources shall use AVI InfoFrame Version 4 with Length = 14.
Elseif (VIC >= 193)
        Sources shall use AVI InfoFrame Version 3.
Else
        Sources shall use AVI InfoFrame Version 2.
End if
```

6.4.1 Video Format, Picture Aspect Ratio and Pixel Repetition

Bits 0 through 7 of Data Byte 4 contain a Video Identification Code (VIC), bits 0 through 5 of Data Byte 15 contain a Resolution ID (RID), and bits 0 through 4 of Data Byte 14 and bit 6 of Data Byte 15 contain a Frame Rate code (FR) to identify the Video Format. In most cases, the Video Format can be uniquely determined from the Video Format Timing itself.

However, if the Source is sending one of the Video Formats defined in Table 1 of CTA-861, then it shall set the VIC field to the proper code. If the Frame Rate represented by the VIC (indicated by the Field Rate column in Table 2) can be expressed by the FR field, the FR field shall be set to the proper code, or to 0. For Video Timings with a vertical frequency that is an integer multiple of 6.00 Hz (except 300 Hz), or an integer multiple of 6.00 Hz adjusted by a factor of 1000/1001, the Source may use FR to indicate the version of the Timing (see Section 4). If the Source is sending a Video Format that can be indicated by RID and FR, and is not listed in Table 13, then it shall set the RID and FR fields to the proper codes.

If a Video Format not listed in Table 1 of CTA-861 is sent, then the Video Identification Code shall be set to 0. If any of the following is true, then the RID shall be set to 0:

- If a Video Format not listed in Table 11 is sent;

- if a Video Format with Frame Rates not listed in Table 12 is sent; or
- if a Video Format listed in Table 13 is sent.

If a Video Format is indicated with RID > 0, the Frame Rate shall be indicated with FR. A Video Format with a Frame Rate that cannot be indicated by FR shall not be indicated by RID.

If the Frame Rate represented by the VIC can be expressed by the FR field, the FR field shall be set to the proper code, or to 0.

If a modified[9] version of a Video Format listed in CTA-861 is sent (e.g., modified for some 3D modes or YC_BC_R 4:2:0), then the VIC, RID, and FR fields shall be set in accordance with the IDO's specification.

If no CTA-861 Video Format was identified (VIC = 0 and RID = 0), or the Video Format Timing does not match that of the identified CTA-861 Video Format, then the Sink should infer the default Quantization Range by determining whether the timing is a CE Video Format based on the Video Format Timing itself or not.

The VICs associated with each Video Format defined in Table 1 are shown in Table 2. These same codes are used in the Short Video Descriptors used in the Version 3 CTA Extension, which is described in Section 7.3.3. The RID codes and their association with Video Resolutions and Aspect Ratios are shown in Table 11. The values for FR are shown in Table 12. These same codes are used in the Video Format Descriptors used in the Version 3 CTA Extension, which is described in Section 7.3.3.

The list of VICs should not be expanded in future versions of CTA-861, and any VICs added in a future version of CTA-861 shall describe Video Formats that cannot be expressed with a current or future VFD, and shall describe Video Timings other than OVT timings.

[9] Video timing might be modified from that given in Table 1 or derived from the OVT algorithm (see Annex U), when Y = 3 (Y2 = 0, Y1 = 1, Y0 = 1) in Data Byte 1 (i.e., YC_BC_R 4:2:0), or when Y = 7 (Y2 = 1, Y1 = 1, Y0 = 1) in Data Byte 1 (i.e., IDO-defined), or when an overriding IDO-defined 3D mode is present (see Annex O). In the case of YC_BC_R 4:2:0, the pixel rate is reduced by half, along with the Htotal, Hactive, Hblank, Hfront, Hsync, and Hback timing parameters. In the case of 3D, the pixel frequency might be doubled and Vsync pulses skipped.

Table 11 - Resolution Identification (RID)

RID	Resolution[a]	Aspect Ratio		Common Name
		Picture	Pixel	
0	No Resolution Identification Available (Used with AVI InfoFrame only)			
1	1280x720p	16:9	1:1	HD, 720p
2	1280x720p	64:27	4:3	HD, 720p, 21:9 anamorphic
3	1680x720p	64:27	64:63[b]	21:9 "1.5K"
4	1920x1080p	16:9	1:1	Full HD, 1080p
5	1920x1080p	64:27	4:3	Full HD, 1080p, 21:9 anamorphic
6	2560x1080p	64:27	1:1	21:9 "2.5K"
7	3840x1080p	32:9	1:1	32:9 "4K"
8	2560x1440p	16:9	1:1	QHD, 1440p
9	3440x1440p	64:27	128:129[c]	WQHD
10	5120x1440p	32:9	1:1	32:9 "5K"
11	3840x2160p	16:9	1:1	UHD "4K", 2160p
12	3840x2160p	64:27	4:3	UHD "4K", 2160p, 21:9 anamorphic
13	5120x2160p	64:27	1:1	21:9 "5K"
14	7680x2160p	32:9	1:1	32:9 "8K"
15	5120x2880p	16:9	1:1	2880p
16	5120x2880p	64:27	4:3	2880p, 21:9 anamorphic
17	6880x2880p	64:27	128:129[c]	21:9 "6K"
18	10240x2880p	32:9	1:1	32:9 "10K"
19	7680x4320p	16:9	1:1	UHD "8K", 4320p
20	7680x4320p	64:27	4:3	UHD "8K", 4320p, 21:9 anamorphic
21	10240x4320p	64:27	1:1	21:9 "10K"
22	15360x4320p	32:9	1:1	32:9 "15K"
23	11520x6480p	16:9	1:1	UHD "12K", 6480p
24	11520x6480p	64:27	4:3	UHD "12K", 6480p, 21:9 anamorphic
25	15360x6480p	64:27	1:1	21:9 "15K"
26	15360x8640p	16:9	1:1	UHD "16K", 8640p
27	15360x8640p	64:27	4:3	UHD "16K", 8640p, 21:9 anamorphic
28	20480x8640p	64:27	1:1	21:9 "20K"
29 ... 62	reserved – future use			
63	reserved – future extension			

Notes:

a. Shows the horizontal resolution (Active Pixel per Line), followed by an "x", then the vertical resolution (Active Lines), and a "p" indicating Progressive Video Resolutions.

b. The Pixel Aspect Ratio 64:63 is used in lieu of a square Pixel Aspect Ratio 1:1 with this resolution.

c. The Pixel Aspect Ratio 128:129 is used in lieu of a square Pixel Aspect Ratio 1:1 with this resolution.

Table 12 - AVI InfoFrame Video Format Frame Rate

FR4	FR3	FR2	FR1	FR0	Frame Rate (Hz)	Rate Index	Rate Group
0	0	0	0	0	No Data	0	0
0	0	0	0	1	23.98 (24/1.001)	1	1
0	0	0	1	0	24		
0	0	0	1	1	25	2	
0	0	1	0	0	29.97 (30/1.001)	3	
0	0	1	0	1	30		
0	0	1	1	0	47.95 (48/1.001)	4	2
0	0	1	1	1	48		
0	1	0	0	0	50	5	
0	1	0	0	1	59.94 (60/1.001)	6	
0	1	0	1	0	60		
0	1	0	1	1	100	7	3
0	1	1	0	0	119.88 (120/1.001)	8	
0	1	1	0	1	120		
0	1	1	1	0	143.86 (144/1.001)	9	4
0	1	1	1	1	144		
1	0	0	0	0	200	10	5
1	0	0	0	1	239.76 (240/1.001)	11	
1	0	0	1	0	240		
1	0	0	1	1	300	12	6
1	0	1	0	0	359.64 (360/1.001)	13	
1	0	1	0	1	360		
1	0	1	1	0	400	14	7
1	0	1	1	1	479.52 (480/1.001)	15	
1	1	0	0	0	480		
1	1	0	0	1	reserved		
1	1	0	1	X	reserved		
1	1	1	X	X	reserved		
1	1	X	X	X	reserved		
1	X	X	X	X	reserved		

Table 13 - RID To VIC Mapping

RID	Frame Rate (Hz)							
	23.98 & 24	25	29.97 & 30	47.95 & 48	50	59.94 & 60	100	119.88 & 120
1	60	61	62	108	19	4	41	47
2	65	66	67	109	68	69	70	71
3	79	80	81	110	82	83	84	85
4	32	33	34	111	31	16	64	63
5	72	73	74	112	75	76	77	78
6	86	87	88	113	89	90	91	92
11	93	94	95	114	96	97	117	118
12	103	104	105	116	106	107	119	120
13	121	122	123	124	125	126	127	193
19	194	195	196	197	198	199	200	201
20	202	203	204	205	206	207	208	209
21	210	211	212	213	214	215	216	217

The following pseudocode illustrates how Sink devices shall interpret VIC and RID codes sent by a Source:

```
If VIC >= 1 and VIC <= 64 then
        Video Format is defined according to Table 1 and Table 2, with 7-bit VIC
Elseif VIC >= 65 and VIC <= 127 then
        Video Format is defined according to Table 1 and Table 2, with 8-bit VIC
Elseif VIC == 128 then
        Reserved
Elseif VIC >= 129 and VIC <= 192 then
        Forbidden
Elseif VIC >= 193 and VIC <= 253 then
        Video Format is defined according to Table 1 and Table 2, with 8-bit VIC
Elseif VIC == 254 then
        Reserved
Elseif VIC == 255 then
        Reserved
Elseif RID > 0 and FR > 0 then
        Video Format is defined according to Table 11 and Table 12
Else
        Video Format not documented in CTA-861 (not a "CE Video Format" or
        "640x480p")
End if
```

Data Byte 2, bits M1 and M0, contain the Picture Aspect Ratio. When VIC = 0, the AVI M field may be used to signal a Picture Aspect Ratio. When M = 0 (M1 = 0, M0 = 0) and VIC = 0, the Picture is considered to be formatted according to the Preferred Picture Aspect Ratio. If the AVI VIC is not 0, or the AVI RID is not 0, then the Sink shall ignore this field and instead use the

Picture Aspect Ratio entry in Table 2 associated with the non-zero AVI VIC, or the Picture Aspect Ratio entry in Table 11 associated with the non-zero AVI RID, and the Source shall set M to the matching Picture Aspect Ratio of the AVI VIC or RID, or to 0 if there is no match.

The values for M are shown in Table 14.

Table 14 - AVI InfoFrame Picture Aspect Ratio Field, Data Byte 2

M1	M0	Picture Aspect Ratio
0	0	No Data
0	1	4:3
1	0	16:9
1	1	Reserved

Data Byte 5, bits PR2 through PR0, contain the pixel repetition field (PR). The first transmitted Active Pixel of an Active Line shall be unique. When PR is zero, the second through the last transmitted Active Pixel shall each be unique. When PR is greater than zero, Unique Active Pixels are transmitted less often as the Source shall repeat each Unique Active Pixel PR-times. Unique Active Pixels are always vertically aligned and horizontally spaced at PR+1 Active Pixel (clock) intervals. The values for PR are shown in Table 15.

Table 15 - AVI InfoFrame Pixel Repetition Field, Data Byte 5

PR3	PR2	PR1	PR0	Pixel Repetition Factor
0	0	0	0	No Repetition (i.e., pixel data sent once)
0	0	0	1	Pixel Data sent 2 times (i.e., repeated once)
0	0	1	0	Pixel Data sent 3 times
0	0	1	1	Pixel Data sent 4 times
0	1	0	0	Pixel Data sent 5 times
0	1	0	1	Pixel Data sent 6 times
0	1	1	0	Pixel Data sent 7 times
0	1	1	1	Pixel Data sent 8 times
1	0	0	0	Pixel Data sent 9 times
1	0	0	1	Pixel Data sent 10 times
0x0A-0x0F				Reserved

A Source shall correctly set the PR field whenever it sends an AVI InfoFrame to a Sink — no matter what Video Timing Format is being transmitted. A list of allowable PR values for each CE Video Format is shown in Table 16. Note that this characteristic is independent of Picture Aspect Ratio. When a Source outputs a Video Timing Format with non-repeated pixels, PR shall be set to 0. When a Source outputs a double-clocked Video Timing Format, PR shall be set to 1. When a Source outputs Video Timing Formats 10 through 15, 25 through 30, or 35 through 38, it shall send an AVI InfoFrame indicating the specific PR being used and the Sink shall properly interpret it — decimating or repeating pixels depending on the signal process.

Video Timing Formats with Video Identification Codes 10 through 15, 25 through 30, and 35 through 38 support variable horizontal resolution. These Video Formats maintain a fixed 1440-

or 2880-pixel Hactive and use pixel repetition to, in effect, provide different horizontal resolutions.

Video formats with Video Identification Codes 10 through 13 and 25 through 28 keep Hactive fixed at 2880 pixels and allow PR to be varied over a 9-to-0 range thereby providing effective resolutions of 288, 320, 360, 411, 480, 576, 720, 960, 1440, and 2880 Unique Active Pixels per Video Line, respectively. In addition, gaming formats typically utilize optional left and right Bars, which ensure that all of the pixels in a game are visible on overscanned displays and further reduce the number of Unique Content Pixels to 256, 284, 320, 366, 427, 512, 640, 853, 1280, and 2560 pixels, respectively. When Hactive is not an integer multiple of PR+1, the Source shall adjust the Bars on each side so that the width of the left Bar is an integer multiple of PR+1 and the right Bar begins with a Unique Active Pixel. Table 17 gives recommended Bar placement for each value of PR in the form of AVI Bar Data.

Video formats with Video Identification Codes 14, 15, 29, and 30 allow PR to be set to 1 or 0 thereby providing effective resolutions of 720 or 1440 Unique Active Pixels per Video Line, respectively.

Video formats with Video Identification Codes 35 through 38 allow PR to be set to 3, 1, or 0 thereby providing effective resolutions of 720, 1440, or 2880 Unique Active Pixels per Video Line, respectively.

Table 16 - Valid Pixel Repeat Values for Each Video Format Timing

VIC	Video Description	Valid Pixel Repeat Values	AVI w/PR Required
1	640x480p @ 59.94/60Hz	No Repetition	No
2, 3	720x480p @ 59.94/60Hz	No Repetition	No
4, 69	1280x720p @ 59.94/60Hz	No Repetition	No
5	1920x1080i @ 59.94/60Hz	No Repetition	No
6, 7	720(1440)x480i @ 59.94/60Hz	Pixel Data sent 2 times	No
8, 9	720(1440)x240p @ 59.94/60Hz	Pixel Data sent 2 times	No
10, 11	2880x480i @ 59.94/60Hz	Pixel Data sent 1 to 10 times	Yes
12, 13	2880x240p @ 59.94/60Hz	Pixel Data sent 1 to 10 times	Yes
14, 15	1440x480p @ 59.94/60Hz	Pixel Data sent 1 to 2 times	Yes
16, 76	1920x1080p @ 59.94/60Hz	No Repetition	No
17, 18	720x576p @ 50Hz	No Repetition	No
19, 68	1280x720p @ 50Hz	No Repetition	No
20	1920x1080i @ 50Hz	No Repetition	No
21, 22	720(1440)x576i @ 50Hz	Pixel Data sent 2 times	No
23, 24	720(1440)x288p @ 50Hz	Pixel Data sent 2 times	No
25, 26	2880x576i @ 50Hz	Pixel Data sent 1 to 10 times	Yes
27, 28	2880x288p @ 50Hz	Pixel Data sent 1 to 10 times	Yes
29, 30	1440x576p @ 50Hz	Pixel Data sent 1 or 2 times	Yes
31, 75	1920x1080p @ 50Hz	No Repetition	No
32, 72	1920x1080p @ 23.98/24Hz	No Repetition	No
33, 73	1920x1080p @ 25Hz	No Repetition	No
34, 74	1920x1080p @ 29.98/30Hz	No Repetition	No
35, 36	2880x480p @ 59.94/60Hz	Pixel Data sent 1, 2 or 4 times	Yes
37, 38	2880x576p @ 50Hz	Pixel Data sent 1, 2 or 4 times	Yes
39	1920x1080i (1250) @ 50Hz	No Repetition	No
40	1920x1080i @ 100Hz	No Repetition	No
41, 70	1280x720p @ 100Hz	No Repetition	No
42, 43	720x576p @ 100Hz	No Repetition	No
44, 45	720(1440)x576i @ 100Hz	Pixel Data sent 2 times	No
46	1920x1080i @ 119.88/120Hz	No Repetition	No
47, 71	1280x720p @ 119.88/120Hz	No Repetition	No
48, 49	720x480p @ 119.88/120Hz	No Repetition	No
50, 51	720(1440)x480i @ 119.88/120Hz	Pixel Data sent 2 times	No
52, 53	720x576p @ 200Hz	No Repetition	No
54, 55	720(1440)x576i @ 200Hz	Pixel Data sent 2 times	No
56, 57	720x480p @ 239.76/240Hz	No Repetition	No
58, 59	720(1440)x480i @ 239.76/240Hz	Pixel Data sent 2 times	No
60, 65	1280x720p @ 23.98/24Hz	No Repetition	No
61, 66	1280x720p @ 25Hz	No Repetition	No
62, 67	1280x720p @ 29.97/30Hz	No Repetition	No
63, 78	1920x1080p @ 119.88/120Hz	No Repetition	No
64, 77	1920x1080p @ 100Hz	No Repetition	No
79	1680x720p @ 23.98/24Hz	No Repetition	No
80	1680x720p @ 25Hz	No Repetition	No
81	1680x720p @ 29.97/30Hz	No Repetition	No
82	1680x720p @ 50Hz	No Repetition	No
83	1680x720p @ 59.94/60Hz	No Repetition	No
84	1680x720p @ 100Hz	No Repetition	No
85	1680x720p @ 119.88/120Hz	No Repetition	No

Table 16 - Valid Pixel Repeat Values for Each Video Format Timing (continued)

VIC	Video Description	Valid Pixel Repeat Values	AVI w/PR Required
86	2560x1080p @ 23.98/24Hz	No Repetition	No
87	2560x1080p @ 25Hz	No Repetition	No
88	2560x1080p @ 29.97/30Hz	No Repetition	No
89	2560x1080p @ 50Hz	No Repetition	No
90	2560x1080p @ 59.94/60Hz	No Repetition	No
91	2560x1080p @ 100Hz	No Repetition	No
92	2560x1080p @ 119.88/120Hz	No Repetition	No
93, 103	3840x2160p @ 23.98/24Hz	No Repetition	No
94, 104	3840x2160p @ 25Hz	No Repetition	No
95, 105	3840x2160p @ 29.97/30Hz	No Repetition	No
96, 106	3840x2160p @ 50Hz	No Repetition	No
97, 107	3840x2160p @ 59.94/60Hz	No Repetition	No
98	4096x2160p @ 23.98/24Hz	No Repetition	No
99	4096x2160p @ 25Hz	No Repetition	No
100	4096x2160p @ 29.97/30Hz	No Repetition	No
101	4096x2160p @ 50Hz	No Repetition	No
102	4096x2160p @ 59.94/60Hz	No Repetition	No
108, 109	1280x720p @ 47.95/48Hz	No Repetition	No
110	1680x720p @ 47.95/48Hz	No Repetition	No
111, 112	1920x1080p @ 47.95/48Hz	No Repetition	No
113	2560x1080p @ 47.95/48Hz	No Repetition	No
114, 116	3840x2160p @ 47.95/48Hz	No Repetition	No
115	4096x2160p @ 47.95/48Hz	No Repetition	No
117, 119	3840x2160p @ 100Hz	No Repetition	No
118, 120	3840x2160p @ 119.88/120Hz	No Repetition	No
121	5120x2160p @ 23.98/24Hz	No Repetition	No
122	5120x2160p @ 25Hz	No Repetition	No
123	5120x2160p @ 29.97/30Hz	No Repetition	No
124	5120x2160p @ 47.95/48Hz	No Repetition	No
125	5120x2160p @ 50Hz	No Repetition	No
126	5120x2160p @ 59.94/60Hz	No Repetition	No
127	5120x2160p @ 100Hz	No Repetition	No
193	5120x2160p @ 119.88/120Hz	No Repetition	No
194, 202	7680x4320p @ 23.98/24Hz	No Repetition	No
195, 203	7680x4320p @ 25Hz	No Repetition	No
196, 204	7680x4320p @ 29.97/30Hz	No Repetition	No
197, 205	7680x4320p @ 47.95/48Hz	No Repetition	No
198, 206	7680x4320p @ 50Hz	No Repetition	No
199, 207	7680x4320p @ 59.94/60Hz	No Repetition	No
200, 208	7680x4320p @ 100Hz	No Repetition	No
201, 209	7680x4320p @ 119.88/120Hz	No Repetition	No
210	10240x4320p @ 23.98/24Hz	No Repetition	No
211	10240x4320p @ 25Hz	No Repetition	No
212	10240x4320p @ 29.97/30Hz	No Repetition	No
213	10240x4320p @ 47.95/48Hz	No Repetition	No
214	10240x4320p @ 50Hz	No Repetition	No
215	10240x4320p @ 59.94/60Hz	No Repetition	No
216	10240x4320p @ 100Hz	No Repetition	No
217	10240x4320p @ 119.88/120Hz	No Repetition	No
218	4096x2160p @ 100Hz	No Repetition	No
219	4096x2160p @ 119.88/120Hz	No Repetition	No

Table 17 - Typical Gaming Format AVI InfoFrame Parameters

Unique Active Pixel Spacing (in Video Pixels)	Unique Active Pixels	Unique Content Pixels	AVI PR	AVI B	AVI ELB	AVI SRB
1	2880	2560	0	1	160	2721
2	1440	1280	1	1	160	2721
3	960	853	2	1	162	2722
4	720	640	3	1	160	2721
5	576	512	4	1	160	2721
6	480	427	5	1	162	2725
7	411	366	6	1	161	2724
8	360	320	7	1	160	2721
9	320	284	8	1	162	2719
10	288	256	9	1	160	2721

6.4.2 Color Component Sample Format and Colorimetry

Data Bytes 1, 2, 3, and 14 contain bits that indicate the Color Component Sample format and chroma sampling, and describe colorimetry.

Data Byte 1, bits Y2, Y1, and Y0 set the Color Component Sample format and chroma sampling format of the next Picture. The Source shall first determine if the Sink is capable of receiving YC_BC_R Pixel Data in the defined chroma sampling format prior to sending such data. See Sections 7.1 and 7.3.3 for a description of a Sink's support of YC_BC_R chroma sampling formats.

Table 18 - AVI InfoFrame RGB or YC_BC_R Field, Data Byte 1

Y2	Y1	Y0	RGB or YC_BC_R
0	0	0	RGB (default)
0	0	1	YC_BC_R 4:2:2
0	1	0	YC_BC_R 4:4:4
0	1	1	YC_BC_R 4:2:0
1	0	0	Reserved
1	0	1	Reserved
1	1	0	Reserved
1	1	1	IDO-Defined

Data Byte 2, bits C1 and C0, are used in conjunction with Data Byte 3, EC2 through EC0, and Data Byte 14, ACE3 through ACE0, to override the default color spaces and explicitly indicate the colorimetry of the next Picture. The extended colorimetry bits, EC2, EC1, and EC0, describe optional colorimetry encoding that may be applicable to some implementations.

If bits C0 and C1 are zero, the colorimetry shall correspond to the default colorimetry defined in Section 5.1. A Source shall be prohibited from setting C = 1 (C1 = 0, C0 = 1) or C = 2 (C1 = 1, C0 = 0) when Y = 0 (Y2 = 0, Y1 = 0, Y0 = 0) in Data Byte 1. When C = 3, colorimetry is indicated in bits EC0 through EC2. When EC = 7, colorimetry is indicated in bits ACE0 through ACE3. The EC field shall be set to 0 by the Source unless C = 3. The EC field shall be ignored by the Sink unless

C = 3. The ACE field (if present) shall be set to 0 by the Source unless EC = 7. The ACE field (if present) shall be ignored by the Sink unless EC = 7.

Table 19 - AVI InfoFrame Colorimetry Fields

C1	C0	Colorimetry
0	0	No Data
0	1	SMPTE ST 170 [1]
1	0	Rec. ITU-R BT.709 [7]
1	1	Extended Colorimetry Information Valid (colorimetry indicated in bits EC0, EC1, and EC2)

EC2	EC1	EC0	Extended Colorimetry
0	0	0	$xvYCC_{601}$[10] [5]
0	0	1	$xvYCC_{709}$[10] [5]
0	1	0	$sYCC_{601}$ [34]
0	1	1	$opYCC_{601}$ [32]
1	0	0	opRGB [32]
1	0	1	Rec. ITU-R BT.2020 $Y'_{C}C'_{BC}C'_{RC}$ [40]
1	1	0	Rec. ITU-R BT.2020 R'G'B' or $Y'C'_{B}C'_{R}$ [40]
1	1	1	Additional Colorimetry Extension Information Valid (colorimetry indicated in bits ACE0, ACE1, ACE2, and ACE3)

ACE3	ACE2	ACE1	ACE0	Additional Colorimetry Extension
0	0	0	0	SMPTE ST 2113 P3D65 R'G'B' [49]
0	0	0	1	SMPTE ST 2113 P3DCI R'G'B' [49]
0	0	1	0	Rec. ITU-R BT.2100 $IC_{T}C_{P}$ [48]
0	0	1	1	sRGB [33]
0	1	0	0	defaultRGB; colorimetry based on the default chromaticity information of the display as defined in the Basic Display Parameters and Features Block of the EDID.
0x05-0x0F				reserved

[10] The YCC Quantization Range YQ bits shall be ignored by the Sink for these color spaces.

The Sink shall interpret the Y, C, EC, and ACE fields of Data Bytes 1, 2, 3, and 14 according to Table 20. A value of 'X' in the table indicates the Sink shall ignore the bit. A value of 'D' in the table indicates the Sink shall refer to the IDO's standard. A reserved value in the table means the colorimetry assumed by the Sink is indeterminate. Sources shall be prohibited from setting reserved colorimetry.

Table 20 - Picture Colorimetry Indicated by the RGB or YC_BC_R (Y), Colorimetry (C), Extended Colorimetry (EC) and Additional Colorimetry Extension (ACE) Field Settings

	RGB or YC_BC_R (from Data Byte 1)			Colorimetry (from Data Byte 2)		Extended Colorimetry (from Data Byte 3)			Additional Colorimetry Extension (from Data Byte 14)				Colorimetry of Next Transmitted Picture	Notes
	Y2	Y1	Y0	C1	C0	EC2	EC1	EC0	ACE3	ACE2	ACE1	ACE0		
RGB 4:4:4	0	0	0	0	0	X	X	X	X	X	X	X	RGB	a
	0	0	0	0	1	X	X	X	X	X	X	X	Reserved	
	0	0	0	1	0	X	X	X	X	X	X	X	Reserved	
	0	0	0	1	1	0	X	X	X	X	X	X	Reserved	
	0	0	0	1	1	1	0	0	X	X	X	X	opRGB [32]	b
	0	0	0	1	1	1	0	1	X	X	X	X	Reserved	
	0	0	0	1	1	1	1	0	X	X	X	X	Rec. ITU-R BT.2020 R'G'B' [40]	b, e
	0	0	0	1	1	1	1	1	0	0	0	0	SMPTE ST 2113 P3D65 R'G'B' [49]	
	0	0	0	1	1	1	1	1	0	0	0	1	SMPTE ST 2113 P3DCI R'G'B' [49]	
	0	0	0	1	1	1	1	1	0	0	1	0	Reserved	
	0	0	0	1	1	1	1	1	0	0	1	1	sRGB [33]	b
	0	0	0	1	1	1	1	1	0	1	0	0	defaultRGB	b, h
	0	0	0	1	1	1	1	1	0	1	0	1	Reserved	
	0	0	0	1	1	1	1	1	0	1	1	X	Reserved	
	0	0	0	1	1	1	1	1	1	X	X	X	Reserved	
YC_BC_R 4:2:2	0	0	1	0	0	X	X	X	X	X	X	X	SMPTE ST 170 [1] or Rec. ITU-R BT.709 [7]	c, d
	0	0	1	0	1	X	X	X	X	X	X	X	SMPTE ST 170 [1]	d
	0	0	1	1	0	X	X	X	X	X	X	X	Rec. ITU-R BT.709 [7]	d
	0	0	1	1	1	0	0	0	X	X	X	X	$xvYCC_{601}$ [5]	b, d
	0	0	1	1	1	0	0	1	X	X	X	X	$xvYCC_{709}$ [5]	b, d
	0	0	1	1	1	0	1	0	X	X	X	X	$sYCC_{601}$ [34]	b, d
	0	0	1	1	1	0	1	1	X	X	X	X	$opYCC_{601}$ [32]	b, d
	0	0	1	1	1	1	0	0	X	X	X	X	Reserved	
	0	0	1	1	1	1	0	1	X	X	X	X	Rec. ITU-R BT.2020 $Y'_CC'_{BC}C'_{RC}$ [40]	b, e
	0	0	1	1	1	1	1	0	X	X	X	X	Rec. ITU-R BT.2020 $Y'C'_BC'_R$ [40]	b, e
	0	0	1	1	1	1	1	1	0	0	0	X	Reserved	
	0	0	1	1	1	1	1	1	0	0	1	0	Rec. ITU-R BT.2100 IC_TC_P [48]	g
	0	0	1	1	1	1	1	1	0	0	1	1	Reserved	
	0	0	1	1	1	1	1	1	0	1	X	X	Reserved	
	0	0	1	1	1	1	1	1	1	X	X	X	Reserved	

Table 20 - Picture Colorimetry Indicated by the RGB or YC_BC_R (Y), Colorimetry (C), Extended Colorimetry (EC) and Additional Colorimetry Extension (ACE) Field Settings (continued)

	RGB or YC_BC_R (from Data Byte 1)			Colorimetry (from Data Byte 2)		Extended Colorimetry (from Data Byte 3)			Additional Colorimetry Extension (from Data Byte 14)				Colorimetry of Next Transmitted Picture	Notes
	Y2	Y1	Y0	C1	C0	EC2	EC1	EC0	ACE3	ACE2	ACE1	ACE0		
YC_BC_R 4:4:4	0	1	0	0	0	X	X	X	X	X	X	X	SMPTE ST 170 [1] or Rec. ITU-R BT.709 [7]	c, d
	0	1	0	0	1	X	X	X	X	X	X	X	SMPTE ST 170 [1]	d
	0	1	0	1	0	X	X	X	X	X	X	X	Rec. ITU-R BT.709 [7]	d
	0	1	0	1	1	0	0	0	X	X	X	X	$xvYCC_{601}$ [5]	b, d
	0	1	0	1	1	0	0	1	X	X	X	X	$xvYCC_{709}$ [5]	b, d
	0	1	0	1	1	0	1	0	X	X	X	X	$sYCC_{601}$ [34]	b, d
	0	1	0	1	1	0	1	1	X	X	X	X	$opYCC_{601}$ [32]	b, d
	0	1	0	1	1	1	0	0	X	X	X	X	Reserved	
	0	1	0	1	1	1	0	1	X	X	X	X	Rec. ITU-R BT.2020 $Y'_CC'_{BC}C'_{RC}$ [40]	b, e, f
	0	1	0	1	1	1	1	0	X	X	X	X	Rec. ITU-R BT.2020 $Y'C'_BC'_R$ [40]	b, e
	0	1	0	1	1	1	1	1	0	0	0	X	Reserved	
	0	1	0	1	1	1	1	1	0	0	1	0	Rec. ITU-R BT.2100 IC_TC_P [48]	g
	0	1	0	1	1	1	1	1	0	0	1	1	Reserved	
	0	1	0	1	1	1	1	1	0	1	X	X	Reserved	
	0	1	0	1	1	1	1	1	1	X	X	X	Reserved	
YC_BC_R 4:2:0	0	1	1	0	0	X	X	X	X	X	X	X	SMPTE ST 170 [1] or Rec. ITU-R BT.709 [7]	c, d
	0	1	1	0	1	X	X	X	X	X	X	X	SMPTE ST 170 [1]	d
	0	1	1	1	0	X	X	X	X	X	X	X	Rec. ITU-R BT.709 [7]	d
	0	1	1	1	1	0	0	0	X	X	X	X	$xvYCC_{601}$ [5]	b, d
	0	1	1	1	1	0	0	1	X	X	X	X	$xvYCC_{709}$ [5]	b, d
	0	1	1	1	1	0	1	0	X	X	X	X	$sYCC_{601}$ [34]	b, d
	0	1	1	1	1	0	1	1	X	X	X	X	$opYCC_{601}$ [32]	b, d
	0	1	1	1	1	1	0	0	X	X	X	X	Reserved	
	0	1	1	1	1	1	0	1	X	X	X	X	Rec. ITU-R BT.2020 $Y'_CC'_{BC}C'_{RC}$ [40]	b, e
	0	1	1	1	1	1	1	0	X	X	X	X	Rec. ITU-R BT.2020 $Y'C'_BC'_R$ [40]	b, e
	0	1	1	1	1	1	1	1	0	0	0	X	Reserved	
	0	1	1	1	1	1	1	1	0	0	1	0	Rec. ITU-R BT.2100 IC_TC_P [48]	g
	0	1	1	1	1	1	1	1	0	0	1	1	Reserved	
	0	1	1	1	1	1	1	1	0	1	X	X	Reserved	
	0	1	1	1	1	1	1	1	1	X	X	X	Reserved	

Table 20 - Picture Colorimetry Indicated by the RGB or YC_BC_R (Y), Colorimetry (C), Extended Colorimetry (EC) and Additional Colorimetry Extension (ACE) Field Settings (continued)

	RGB or YC_BC_R (from Data Byte 1)			Colorimetry (from Data Byte 2)		Extended Colorimetry (from Data Byte 3)			Additional Colorimetry Extension (from Data Byte 14)				Colorimetry of Next Transmitted Picture	Notes
	Y2	Y1	Y0	C1	C0	EC2	EC1	EC0	ACE3	ACE2	ACE1	ACE0		
Reserved	1	0	X	X	X	X	X	X	X	X	X	X	Reserved	
Reserved	1	1	0	X	X	X	X	X	X	X	X	X	Reserved	
IDO-Defined	1	1	1	D	D	D	D	D	D	D	D	D	Defined by IDO	

Notes:

a. A DTV declares it is capable of displaying Pictures encoded in sRGB color space (as defined in IEC 61996-2-1 [33]) by setting bit 2 in the Feature Support byte (0x18) of the Basic Display Parameters and Features Block in its EDID. A Sink that declares it is not capable of displaying Pictures encoded in sRGB color space declares its colorimetry via the values set in bytes 0x17 and 0x19 through 0x22 of the Basic Display Parameters and Features Block in its EDID. See Section 5.1 for further information.

b. A DTV declares it is capable of displaying Pictures encoded in this colorimetry by setting the associated bit in Bytes 3 and 4 of the Colorimetry Data Block in its EDID. See Section 7.5.5 for further information.

c. The Picture colorimetry is dependent on the value of Vactive for the Video Identification Code set in the AVI InfoFrame. See Section 5 for further information.

d. A DTV declares it is capable of displaying Pictures encoded in this colorimetry by setting bit 4 and/or bit 5 in Byte 3 of the CTA Extension Version 3 block in its EDID. See Section 7.3.3.

e. Rec. ITU-R BT.2020 [40] colorimetry is only defined for Component Depths of 10 and 12-bits/component and shall not be used at 8-bits/component.

f. In the case of 4:4:4 sampling, applying the constant luminance ($Y'_CC'_{BC}C'_{RC}$) transform of Rec. ITU-R BT.2020 [40] might be of little benefit.

g. Rec. ITU-R BT.2100 [48] colorimetry is only defined for Component Depths of 10 and 12-bits/component and shall not be used at 8-bits/component. If IC_TC_P colorimetry is used, the Source shall transmit a Dynamic Range and Mastering InfoFrame (see Section 6.9) and set the Transfer Function to be either 2 (PQ) or 3 (HLG), matching the encoding of the Picture.

h. Colorimetry based on the default chromaticity information of the display as defined in the Basic Display Parameters and Features Block of the EDID.

6.4.3 Quantization Range

Displays conforming to CTA-861 accept both Limited and Full Range Quantization Range Pixel Data when receiving Pictures encoded in an RGB color space. The RGB quantization bits in Data Byte 3, bits Q1 and Q0, allow the Source to override the default RGB Quantization Range and to explicitly indicate the RGB Quantization Range of the next Picture. A Source shall always explicitly signal the RGB Quantization Range as either Limited Range (Q = 1) or Full Range (Q = 2). The value Q = 0 (Q1 = 0, Q0 = 0) indicates that the Quantization Range corresponds to the default RGB Quantization Range defined in Section 5.1, but the use of this value is deprecated and it shall not be used in new designs for Sources.

If the Sink does not support Selectable RGB Quantization Range (QS = 0 in the Video Capabilities Data Block, see Section 7.5.6), then the Source should send a Q value corresponding to the default RGB Quantization Range specified for the colorimetry transmitted (see Table 4).

When the Source is transmitting any non-RGB colorimetry (Y is not 0), then the Q-field is unused and shall be set to 0. The Sink shall ignore the Q field when any non-RGB colorimetry is received.

Table 21 - AVI InfoFrame RGB Quantization Range Field, Data Byte 3

Q1	Q0	RGB Quantization Range
0	0	Default (depends on Video Format)
0	1	Limited Range
1	0	Full Range
1	1	Reserved

Displays conforming to CTA-861 may accept both Limited and Full Range Quantization Range Pixel Data when receiving Pictures encoded in an YCC color space. The YCC quantization bits in Data Byte 5, bits YQ1 and YQ0, allow the Source to override the default YCC Quantization Range and to explicitly indicate the YCC Quantization Range of the next Picture. Table 22 illustrates the meaning assigned to these YCC Quantization Range bits YQ1 and YQ0 in Data Byte 5. A Source shall always explicitly signal the YCC Quantization Range as either Limited Range (YQ = 0) or Full Range (YQ = 1).

If the Sink does not support Selectable YCC Quantization Range (QY = 0 in the Video Capabilities Data Block, see Section 7.5.6), then the Source should send a YQ value corresponding to the default YCC Quantization Range specified for the colorimetry transmitted (see Table 4).

When the Source is transmitting any RGB colorimetry (Y is 0), the YQ field is unused and shall be set to 0. The Sink shall ignore the YQ field when any RGB colorimetry is received.

Table 22 - AVI InfoFrame YCC Quantization Range Field, Data Byte 5

YQ1	YQ0	YCC Quantization Range	
		Y = YCbCr	Y = RGB
0	0	Limited Range	Required Setting
0	1	Full Range	Reserved
1	0	Reserved	Reserved
1	1	Reserved	Reserved

6.4.4 Active Format and Bar Data

Data Bytes 1, 2, and 6 through 13 contain bits that indicate the active format and/or Bar Data information. If the Bar Data and the active format information do not agree, then the Bar Data shall take precedence.

Data Byte 1, bit A0, indicates whether Active Format Data is present in Data Byte 2 bits R3 through R0. A Source device shall set A0 = 1 when any of the AFD bits are set.

Data Byte 1, bits B1 and B0, indicate the presence and type of Bar Data transmitted in Data Bytes 6 through 13. The contents of Data Bytes 6 through 13 are described later in this Section. The presence of both vertical and horizontal Bar Data is not standardized.

Table 23 - AVI InfoFrame Active Format Information and Bar Data Present Fields, Data Byte 1

A0	Active Format Information Present
0	No Active Format Information
1	Active Format (R3…R0) Information present

B1	B0	Bar Data Present
0	0	Bar Data not present
0	1	Vert. Bar Info present
1	0	Horiz. Bar Info present
1	1	Vert. and Horiz. Bar Info present

Data Byte 2, bits R3 through R0, may be used to communicate common "Active Format" aspect ratio information from a Source to a display device. Table 25 illustrates the terminology and gives examples of common aspect ratio information. It illustrates some of the possibilities given three standard aspect ratios (4:3, 16:9, and 64:27) within the Picture (i.e., CTA-CEB16 Total Image). For the "greater than 16:9" case, an example aspect ratio of 64:27 was chosen.

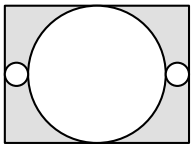
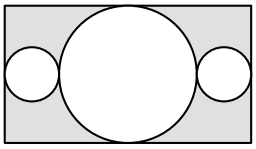
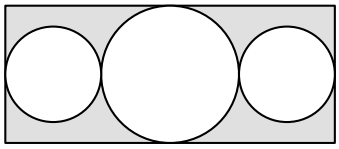
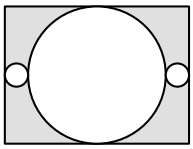
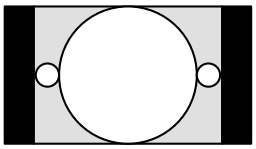
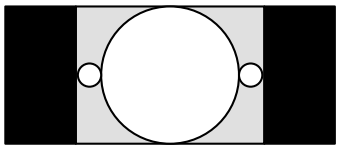
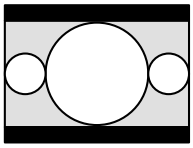
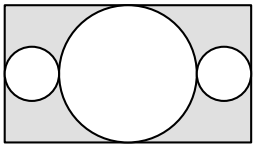
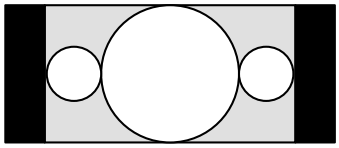
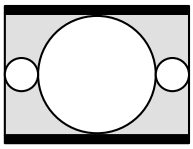
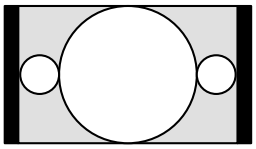
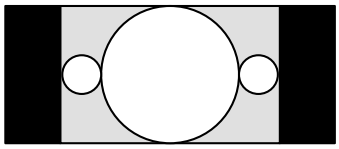
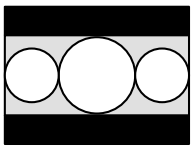
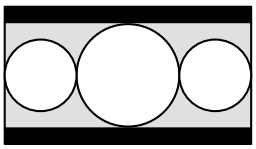
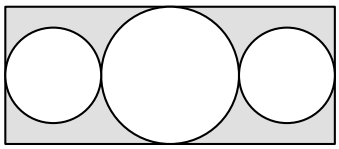
Table 24 - AVI InfoFrame Active Portion Aspect Ratio Field, Data Byte 2

R3	R2	R1	R0	Active Portion Aspect Ratio
1	0	0	0	Same as Picture Aspect Ratio
1	0	0	1	4:3 (Center)
1	0	1	0	16:9 (Center)
1	0	1	1	14:9 (Center)
other values				Varies. See Annex H.

"Active Format" codes shall be transmitted when received with content. Originating devices supplying such codes may provide codes in accordance with the Active Format Description[11] (AFD) in SMPTE ST 2016-1 [35] and Annex B of ETSI TS 101 154 [107]. All active format codes defined in [35] and [107] are reproduced in informative Annex H.

[11] Note that the use of the word "active" in the "Active Format Description" differs from how it is used in other places of this standard and documents referenced by this standard. The word, "active" usually refers to all Active Pixels. In this case of AFD, the word, "active" refers only to the area containing Content Pixels and its format relative to areas potentially containing Bar Pixels.

Table 25 - Common Active Formats

Active Format Code	Description		
	4:3 Total Images	16:9 Total Images	64:27 Total Images
0b1000 Same as Picture Aspect Ratio			
	4:3 full frame image	16:9 full frame image	64:27 full frame image
0b1001 4:3 (Center)			
	4:3 full frame image	4:3 pillarbox image	4:3 pillarbox image
0b1010 16:9 (Center)			
	16:9 letterbox image	16:9 full frame image	16:9 pillarbox image
0b1011 14:9 (Center)			
	14:9 letterbox image	14:9 pillarbox image	14:9 pillarbox image
0b0100 > 16:9 (Center)			
	“>16:9” letterbox image	“>16:9” letterbox image	“>16:9” full frame image

See CTA-CEB16-B [106] for more information about AFD processing.

Data Bytes 6 through 13 provide Bar Data encoded according to the pixel and line numbering scheme of CTA-861 as shown in Table 27, which is not compatible with specifications such as SMPTE ST 2016-1.

Line counts used to describe a Coded Frame typically conform to SMPTE ST 2016-1 [35], Table 2, which is informatively reproduced in Table 26.

Table 26 - Video Format Information (Informative)

Format	Applicable Standard	Coding Range Pixels x lines	Coded Pixels[a]	Coded Lines[b]		
				Field 1	Field 2	Frame
480 Interlaced	SMPTE ST 125 [102]	720 x 480	0 – 719	23 – 262[c]	286 – 525[c]	-
480 Progressive	SMPTE ST 293 [100]	720 x 480	0 – 719	-	-	45 – 524
576 Interlaced	Rec. ITU-R BT.656 [114]	720 x 576	0 – 719	23 – 310	336 – 623	-
576 Progressive	Rec. ITU-R BT.1358 [116]	720 x 576	0 – 719	-	-	45 – 620
720 Progressive	SMPTE ST 296 [101]	1280 x 720	0 – 1279	-	-	26 – 745
1080 Interlaced	SMPTE ST 274 [2]	1920 x 1080	0 – 1919	21 – 560	584 – 1123	-
1080 Progressive	SMPTE ST 274 [2]	1920 x 1080	0 – 1919	-	-	42 – 1121

a. SMPTE specifications number Coded Pixels 0–719, while CTA-861 numbers Hactive pixels 1–720.

b. SMPTE Coded Line numbering scheme is timing-dependent, while CTA-861 Picture Line numbering scheme is not.

c. CTA-861 and SMPTE ST 2016-1 480i Vactive centers are different. CTA-861's Vactive (22–261/285–524) follows archived SMPTE RP 187 [143] recommended practice, while SMPTE ST 2016-1's Vactive (23–262/286–525) is shifted down one line and follows current SMPTE RP 202-2008 [132] recommended practice. See SMPTE RP 202-2008 [132] Table 1, Note 2 and Section 6.1 for further details.

The equivalent pixel and line numbering scheme of CTA-861 is shown in Table 27.

Table 27 - CTA-861 Picture Pixel and Line Numbers

VIC	Format	Coding Range Pixels x lines	Picture Pixel Number	Picture Line Number		
				Field 1	Field 2	Frame
6, 7	480 Interlaced	720 x 480	1 – 720	Odd 1 – 479	Even 2 – 480	-
2, 3	480 Progressive	720 x 480	1 – 720	-	-	1 – 480
21, 22	576 Interlaced	720 x 576	1 – 720	Odd 1 – 575	Even 2 – 576	-
17, 18	576 Progressive	720 x 576	1 – 720	-	-	1 – 576
4, 19, 60 – 62, 108 – 109	720 Progressive	1280 x 720	1 – 1280	-	-	1 – 720
5, 20	1080 Interlaced	1920 x 1080	1 – 1920	Odd 1 – 1079	Even 2 – 1080	-
16, 32 – 34, 111 – 112	1080 Progressive	1920 x 1080	1 – 1920	-	-	1 – 1080
79 – 85, 110	720 Progressive	1680 x 720	1 – 1680	-	-	1 – 720
86 – 92, 113	1080 Progressive	2560 x 1080	1 – 2560	-	-	1 – 1080
93 – 97,103 – 107, 114, 116 – 120	2160 Progressive	3840 x 2160	1 – 3840	-	-	1 – 2160
98 – 102, 115, 218 – 219	2160 Progressive	4096 x 2160	1 – 4096	-	-	1 – 2160
121 – 127, 193	2160 Progressive	5120 x 2160	1 – 5120	-	-	1 – 2160
194 – 201	4320 Progressive	7680 x 4320	1 – 7680	-	-	1 – 4320
202 – 217	4320 Progressive	10240 x 4320	1 – 10240	-	-	1 – 4320

Note that the pixel and line numbering schemes used to encode AFD Bar Data for program material conforming to SMPTE ST 2016-1 are incompatible with those specified in CTA-861. With respect to pixel numbering, SMPTE ST 2016-1 begins with zero, while CTA-861 begins with one. With respect to line numbering, SMPTE ST 2016-1 and CTA-861 timing-based line numbering schemes have been harmonized. However, where SMPTE ST 2016-1 uses timing-based line-numbering for encoding its AFD Bar Data, CTA-861 uses a separate Picture-based line-numbering scheme to encode its InfoFrame AFD Bar Data. Timing-based line numbering begins with one and at a prescribed line in the blanking interval 1–Ln lines relative to the leading line of Vsync. CTA-861's Picture-based line-numbering also begins with one, but always begins at the leading line of Vactive (i.e., the topmost line of the Picture).

Sources that receive Bar Data from external media (e.g., media carrying Bar Data in accordance with SMPTE ST 2016-1) and output it to Sink via AVI InfoFrame Bar Data, should convert the Bar Data according to CTA-861 standard line number and pixel number conventions (given below) prior to outputting. The equation for converting SMPTE ST 2016-1 Coded Pixel numbers (P_{SMPTE}) to equivalent CTA-861 Picture Pixel numbers (P_{CTA}) is shown in Table 28. The equations for converting interlaced and progressive format SMPTE ST 2016-1 Coded Line numbers (L_{SMPTE}) to equivalent CTA-861 Picture Line numbers (L_{CTA}) are shown in Table 29 and Table 30, respectively. The equations in Table 28, Table 29, and Table 30 only apply when a Bar is present. CTA-861 utilizes special values (given below) when a Bar is omitted. The variables Ln, Vsync, Vback, Vfront, Vactive, in the equations come from Table 1 of CTA-861.

Table 28 - Bar Data Pixel Number Normalization Equation

Format	SMPTE ST 2016-1 Coded Pixel Number to CTA-861 Picture Pixel Number Conversion Equation
All formats	$P_{CTA} = P_{SMPTE} + 1$

Table 29 - Interlaced Bar Data Line Number Normalization Equations

Format	SMPTE ST 2016-1 Coded Line Number to CTA-861 Picture Line Number Conversion Equations	
	Field 1 $L_{SMPTE} <= [L_n + (V_{total} / 2)]$	**Field 2** $L_{SMPTE} > [L_n + (V_{total} / 2)]$
480 Interlaced[a]	$L_{CTA} = 2 * [L_{SMPTE} - L_n - V_{sync} - V_{back}] - 1$	$L_{CTA} = 2 * [L_{SMPTE} - L_n - V_{front} - 2 * (V_{sync} + V_{back}) - (V_{active} / 2) - 1]$
Other Interlaced	$L_{CTA} = 2 * [L_{SMPTE} - L_n - V_{sync} - V_{back} + 1] - 1$	$L_{CTA} = 2 * [L_{SMPTE} - L_n - V_{front} - 2 * (V_{sync} + V_{back}) - (V_{active} / 2)]$

a. The 480 interlaced format is a special case, where the line number conversion equations are slightly modified due to the fact that SMPTE ST 2016-1 Vactive is offset relative to CTA-861 Vactive by one line (e.g., the bottom-most line in SMPTE ST 2016-1 Vactive ends on timing line number 525, while in CTA-861, it ends one line earlier on timing line number 524). All of the other interlaced Video Timings align perfectly.

Table 30 - Progressive Bar Data Line Number Normalization Equation

Format	SMPTE ST 2016-1 Coded Line Number to CTA-861 Picture Line Number Conversion Equation
All Progressive	$L_{CTA} = L_{SMPTE} - L_n - V_{sync} - V_{back} + 1$

The general procedure for converting a SMPTE ST 2016-1 Coded Pixel number (pixel_number_end_of_left_bar or pixel_number_start_of_right_bar) to an equivalent CTA-861 Picture Pixel number (ELB or SRB) and setting the CTA-861 vertical Bar Data bit (B0) is:

1) Determine if a left vertical Bar is present by inspecting the SMPTE ST 2016-1 left_bar_flag value. If the left vertical Bar is present, then use the equation in Table 28 (i.e., simply add one) to calculate ELB. If the left vertical Bar is not present, then use the special value zero for ELB.
2) Determine if a right vertical Bar is present by inspecting the SMPTE ST 2016-1 right_bar_flag value. If the right vertical Bar is present, then use the equation in Table 28 (i.e., simply add one) to calculate SRB. If the right vertical Bar is not present, then use the special value (Hactive + 1) for SRB.
3) If either vertical Bar is present, set B0 bit to one. Otherwise, if neither vertical Bar is present, set B0 bit to zero.

The general procedure for converting a SMPTE ST 2016-1 Coded Line number (line_number_end_of_top_bar or line_number_start_of_bottom_bar) to a CTA-861 Picture Line number (ETB or SBB) and setting the CTA-861 horizontal Bar Data bit (B1) is:

1) Determine if top horizontal Bar is present by inspecting the SMPTE ST 2016-1 top_bar_flag value. If the top horizontal Bar is not present, then use the special value zero for ETB. If the top horizontal Bar is present and the Video Format is progressive, then use the equation in Table 30 to calculate ETB. Otherwise, use one of the four equations as described in step 3 below to calculate ETB.
2) Determine if bottom horizontal Bar is present by inspecting the SMPTE ST 2016-1 bottom_bar_flag value. If the bottom horizontal Bar is not present, then use the special value (Vactive + 1) for SBB. If the bottom horizontal Bar is present and the Video Format is progressive, then use the equation in Table 30 to calculate SBB. Otherwise, use one of the four equations as described step 3 below to calculate SBB.
3) If a horizontal Bar is present and the Video Format is interlaced, then use one of the four equations in Table 29 as follows:
    a) Determine if the SMPTE ST 2016-1 line number is in the first field by comparing the number with the value [Ln + (Vtotal / 2)]. If the SMPTE ST 2016-1 number is less than or equal to the value [Ln + (Vtotal / 2)], then use one of the equations from the “Field 1” column of Table 29. Otherwise, use one of the equations from the “Field 2” column of Table 29.
    b) If the incoming Video Format is 480i, then use the appropriate equation from the “480 Interlaced” row of Table 29. Otherwise, use the appropriate equation from the “Other Interlaced” row of Table 29.

4) If either horizontal Bar is present, set B1 bit to one. Otherwise, if neither horizontal Bar is present, set B1 bit to zero.

Example Bar Data conversions are shown in Annex M.

Data Bytes 6 through 13 contain the location data for Bars. These 8 bytes are present in the AVI whether they contain the Bar Data or not. For the purposes of the Line Number and the Pixel Number, the pixel in the upper left hand corner of the Picture is considered to be in row 1, column 1. Lines and pixels are numbered consecutively as they would appear on a display.[12] All of the values are unsigned integers.

a) **Line Number of End of Top Bar (ETB) —** An unsigned integer value representing the last line of a horizontal letterbox Bar area at the top of the Picture. Zero means no horizontal Bar is present at the top of the Picture.

b) **Line Number of Start of Bottom Bar (SBB) —** An unsigned integer value representing the first line of a horizontal letterbox Bar area at the bottom of the Picture. If greater than the Maximum Vertical Active Lines of the known format, no horizontal Bar is present at the bottom of the Picture.

c) **Pixel Number of End of Left Bar (ELB) —** An unsigned integer value representing the last horizontal pixel of a vertical pillar-Bar area at the left side of the Picture. Zero means no vertical Bar is present on the left of the Picture.

d) **Pixel Number of Start of Right Bar (SRB) —** An unsigned integer value representing the first horizontal pixel of a vertical pillar-Bar area at the right side of the Picture. If greater than the Maximum Horizontal Pixels of the known format, no vertical Bar is present on the right side of the Picture.

6.4.5 Overscan/Underscan, Scaling and Picture Content

Data Byte 1 contains bits that describe overscan/underscan (e.g., computer graphics or video). Bits S1 and S0 (Table 31) contain information that defines the Picture composition of the next Video Field. A Source shall set S = 1 (S1 = 0, S0 = 1) or S = 2 (S1 = 1, S0 = 0) if it is confident of the accuracy of those values. Otherwise, it shall set S = 0 (no data). The Source shall follow these rules for setting S even in the absence of an indication that the Sink responds to the value of S.

A Sink should adjust its scan based on the value of S. A Sink would overscan if it received S = 1, and underscan if it received S = 2. If it receives S = 0, then it should overscan for a CE Video Format and underscan for an IT Video Format. A Sink should indicate its overscan/underscan behavior using a Video Capabilities Data Block (see Section 7.5.6).

[12] In this context, line numbers are not the same as the line numbers used in timing diagrams.

Table 31 - AVI InfoFrame Scan Information Field, Data Byte 1

S1	S0	Scan Information
0	0	No Data
0	1	Composed for an overscanned display, where some Active Pixels and lines at the edges are not displayed.
1	0	Composed for an underscanned display, where all Active Pixels and lines are displayed, with or without a border.
1	1	Reserved

Data Byte 3 contains the SC field as shown in Table 32.

Table 32 - AVI InfoFrame Scaling Fields, Data Byte 3

SC1	SC0	Non-Uniform Picture Scaling
0	0	No Known non-uniform Scaling
0	1	Picture has been scaled horizontally
1	0	Picture has been scaled vertically
1	1	Picture has been scaled horizontally and vertically

Bits SC1 and SC0 provide information on whether the Picture has been scaled in a non-uniform way (i.e., unequal along horizontal and vertical dimensions) prior to transmission to the Sink. The Non-uniform Picture Scaling bits shall be set if the Source scales the Picture or has determined that scaling has been performed in a specific direction. If the Picture has been stretched or shrunk in a uniform way (i.e., equally along both dimensions), then the bits should not be set.

The IT content bit in Data Byte 3 (ITC) indicates when Picture content is directly composed according to common IT practices or derived from a specific type of IT content. When the IT content bit of Data Byte 3 is set to 1 (ITC = 1), the content field (CN0, CN1) of Data Byte 5 is valid and downstream processors should process Pixel Data according to the definitions given in Table 33. When the IT content bit of Data Byte 3 is false (ITC = 0), the content field (CN0, CN1) of Data Byte 5 shall be ignored by the Sink and shall be set to 0 by the Source. The IT content bit is about the Picture content and should not be confused with IT vs CE video timings. The IT content bit can be used with both video timings.

Table 33 illustrates the meaning of Data Byte 5 Content Type bits CN1 and CN0. These bits should be used to signal delivery of IT content that is either classified as Graphics, Photo, Cinema, or Game.

Table 33 - AVI InfoFrame IT Contents Type, Data Bytes 3 and 5

ITC	IT content
0	No data
1	IT content (Data Byte 5 CN bits valid)

CN1	CN0	IT Content Type
0	0	Graphics
0	1	Photo
1	0	Cinema
1	1	Game

The Graphics type is indicated by the Source to flag content composed according to common IT practice (i.e., without regard to Nyquist criterion) and is unsuitable for analog reconstruction or filtering. In IT applications (e.g., involving bit mapped text), each pixel in the Source's frame buffer is most clearly displayed if it is directly mapped to a light-emitting pixel on the display device — such that adjacent pixels are completely independent and do not interact. When the IT content bit is set to 1 and the Graphics type is indicated, downstream processors should pass Pixel Data unfiltered and without analog reconstruction.

The Photo type is indicated by the Source (which might be a digital still camera, DVD player or other device) to flag content derived from digital still pictures. When the IT content bit is set to 1 and the Photo type is indicated, the Sink is expected to "pass-through" still pictures with minimal scaling and picture enhancement in order to avoid undesirable artifacts. The Photo type should not be associated with device type. For example, digital still cameras may support delivery of video.

The Cinema type is indicated by the Source to flag content derived from cinema material. Audio may be processed through an audio video amplifier (AV Amp) or Digital Television. When the IT content bit is set to 1 and the Cinema type is indicated, the Sink should "pass-through" cinema content with minimal scaling and picture enhancement in order to avoid undesirable artifacts. The Cinema type should not be associated with device type. For example, DVD players are capable of supplying various content types such as TV programs.

The Game type is indicated by the Source to flag content derived from game machine material. When the IT content bit is set to 1 and the Game type is indicated, the Sink should "pass-through" game content with minimal scaling and picture enhancement in order to avoid undesirable artifacts. Audio and video latency should also be minimized. The Game type should not be associated with device type. For example, game machines are capable of supplying various content types such as DVD movies.

6.5 Source Product Description (SPD) InfoFrame

The Source Product Description (SPD) InfoFrame communicates the name and product type of the Source. This allows the user to see which device is being selected when changing inputs on the Sink.

Including an appropriate VSDB in the Sink's EDID data structure indicates support of the SPD InfoFrame in the Sink. The transmission of this InfoFrame is optional for the Source. The use of the information by the Sink is also optional. It shall not be sent more than once per Video Frame. If used, it is recommended that it be sent once every second.

The format of the Source Product Description InfoFrame is shown in Table 34.

Table 34 - Source Product Description InfoFrame Format

InfoFrame Type Code	InfoFrame Type = 0x03	
InfoFrame Version Number	Version = 0x01	
Length of Source Product Description InfoFrame	Length of Source Product Description InfoFrame = 25	
Data Byte 1	0	Vendor Name Character 1 VN1 (7bit ASCII code)
Data Byte 2	0	Vendor Name Character 2 VN2
Data Byte 3	0	Vendor Name Character 3 VN3
Data Byte 4	0	Vendor Name Character 4 VN4
Data Byte 5	0	Vendor Name Character 5 VN5
Data Byte 6	0	Vendor Name Character 6 VN6
Data Byte 7	0	Vendor Name Character 7 VN7
Data Byte 8	0	Vendor Name Character 8 VN8
Data Byte 9	0	Product Description Character 1 PD1 (7-bit ASCII code)
Data Byte 10	0	Product Description Character 2 PD2
Data Byte 11	0	Product Description Character 3 PD3
Data Byte 12	0	Product Description Character 4 PD4
Data Byte 13	0	Product Description Character 5 PD5
Data Byte 14	0	Product Description Character 6 PD6
Data Byte 15	0	Product Description Character 7 PD7
Data Byte 16	0	Product Description Character 8 PD8
Data Byte 17	0	Product Description Character 9 PD9
Data Byte 18	0	Product Description Character 10 PD10
Data Byte 19	0	Product Description Character 11 PD11
Data Byte 20	0	Product Description Character 12 PD12
Data Byte 21	0	Product Description Character 13 PD13
Data Byte 22	0	Product Description Character 14 PD14
Data Byte 23	0	Product Description Character 15 PD15
Data Byte 24	0	Product Description Character 16 PD16
Data Byte 25	Source Information	

The Vendor Name consists of eight 7-bit ASCII characters. The name should be left justified (i.e., first character in Data Byte 1) and all unused characters should be Null (i.e., 0x00). The Vendor Name is intended to be the name of the company whose name appears on the product. The Product Description (contained in Data Bytes 9 – 24) consists of sixteen 7-bit ASCII characters. This code is meant to be the model number of the product and may contain a short description

also (e.g., CE524DVD Player). Data Byte 25 consists of a code that classifies the Source. Codes for the most common types of Sources are shown in Table 35.

Table 35 - Source Product Description InfoFrame Data Byte 25

Code	Source Information
0x00	unknown
0x01	Digital STB
0x02	DVD player
0x03	D-VHS
0x04	HDD Videorecorder
0x05	DVC
0x06	DSC
0x07	Video CD
0x08	Game
0x09	PC general
0x0A	Blu-Ray Disc (BD)
0x0B	Super Audio CD
0x0C	HD DVD
0x0D	PMP
0x0E ... 0xFF	Reserved

6.6 Audio InfoFrame

The Audio InfoFrame contains information that allows for the format of the digital audio streams to be identified more quickly via out-of-band information and, for multi-channel Uncompressed Audio (which does not otherwise give such information), provides channel allocation information for the Sink's speakers. The Audio InfoFrame format is shown in Table 36.

If the Sink supports any digital audio, it shall be capable of receiving the Audio InfoFrame and also capable of interpreting the audio identification information in Data Bytes 1 – 3. Support for digital audio other than Basic Audio is indicated in the Version 3 (or higher) CTA Extension (see Section 7.3.3).

If the Sink supports multi-channel (i.e., more than 2 channels) digital audio and has included speaker placement information in EDID (see Sections 7.5.3 and 7.5.15), it shall be able to interpret the speaker channel assignment information and down-mix information in Data Bytes 4 and 5.

If the Source supports the transmission of the Audio InfoFrame and if it determines that the Sink is capable of receiving the Audio InfoFrame (i.e., the Sink has included CTA Extension Version 3 in EDID) and digital audio, then the Audio InfoFrame, with Data Bytes 1 through 3 set correctly, shall be sent once per Video Field while digital audio is being sent across the interface. The data applies to the audio associated with the next full frame of video data.

If the Source is sending multi-channel Uncompressed Audio, then it shall also send valid speaker channel allocation information and down-mix information in Data Bytes 4 and 5 of this InfoFrame.

Table 36 - Audio InfoFrame Format When Data Byte 4 is 0x00 to 0x31

InfoFrame Type Code	InfoFrame Type = 0x04							
InfoFrame Version Number	Version = 0x01							
Length of Audio InfoFrame	Length of Audio InfoFrame = 10							
Data Byte 1	CT3	CT2	CT1	CT0	F13=0	CC2	CC1	CC0
Data Byte 2	F27=0	F26=0	F25=0	SF2	SF1	SF0	SS1	SS0
Data Byte 3	F37=0	F36=0	F35=0	CXT4	CXT3	CXT2	CXT1	CXT0
Data Byte 4	CA7	CA6	CA5	CA4	CA3	CA2	CA1	CA0
Data Byte 5	DM_INH	LSV3	LSV2	LSV1	LSV0	F52=0	LFEPBL1	LFEPBL0
Data Byte 6	F67=0	F66=0	F65=0	F64=0	F63=0	F62=0	F61=0	F60=0
Data Byte 7	F77=0	F76=0	F75=0	F74=0	F73=0	F72=0	F71=0	F70=0
Data Byte 8	F87=0	F86=0	F85=0	F84=0	F83=0	F82=0	F81=0	F80=0
Data Byte 9	F97=0	F96=0	F95=0	F94=0	F93=0	F92=0	F91=0	F90=0
Data Byte 10	F107=0	F106=0	F105=0	F104=0	F103=0	F102=0	F101=0	F100=0

Table 37 - Audio InfoFrame Format When Data Byte 4 is 0xFE

InfoFrame Type Code	InfoFrame Type = 0x04							
InfoFrame Version Number	Version = 0x01							
Length of Audio InfoFrame	Length of Audio InfoFrame = 10							
Data Byte 1	CT3	CT2	CT1	CT0	F13=0	CC2	CC1	CC0
Data Byte 2	F27=0	F26=0	F25=0	SF2	SF1	SF0	SS1	SS0
Data Byte 3	F37=0	F36=0	F35=0	CXT4	CXT3	CXT2	CXT1	CXT0
Data Byte 4	1	1	1	1	1	1	1	0
Data Byte 5	DM_INH	LSV3	LSV2	LSV1	LSV0	F52=0	LFEPBL1	LFEPBL0
Data Byte 6	FLw/FRw	F66=0[13]	FLc/FRc	BC	BL/BR	FC	LFE1	FL/FR
Data Byte 7	TpSiL/ TpSiR	SiL/SiR	TpBC	LFE2	LS/RS	TpFC	TpC	TpFL/ TpFR
Data Byte 8	F87=0	F86=0	F85=0	F84=0	F83=0[14]	BtFL/ BtFR	BtFC	TpBL/ TpBR
Data Byte 9	F97=0	F96=0	F95=0	F94=0	F93=0	F92=0	F91=0	F90=0
Data Byte 10	F107=0	F106=0	F105=0	F104=0	F103=0	F102=0	F101=0	F100=0

Table 38 - Audio InfoFrame Format When Data Byte 4 is 0xFF

InfoFrame Type Code	InfoFrame Type = 0x04							
InfoFrame Version Number	Version = 0x01							
Length of Audio InfoFrame	Length of Audio InfoFrame = 10							
Data Byte 1	CT3	CT2	CT1	CT0	F13=0	CC2	CC1	CC0
Data Byte 2	F27=0	F26=0	F25=0	SF2	SF1	SF0	SS1	SS0
Data Byte 3	F37=0	F36=0	F35=0	CXT4	CXT3	CXT2	CXT1	CXT0
Data Byte 4	1	1	1	1	1	1	1	1
Data Byte 5	DM_INH	LSV3	LSV2	LSV1	LSV0	F52=0	LFEPBL1	LFEPBL0
Data Byte 6	CID07	CID06	CID05	CID04	CID03	CID02	CID01	CID00
Data Byte 7	CID15	CID14	CID13	CID12	CID11	CID10	CID09	CID08
Data Byte 8	CID23	CID22	CID21	CID20	CID19	CID18	CID17	CID16
Data Byte 9	CID31	CID30	CID29	CID28	CID27	CID26	CID25	CID24
Data Byte 10	F107=0	F106=0	F105=0	F104=0	F103=0	F102=0	F101=0	F100=0

[13] Use of F66 for RLC/RRC has been deprecated. Sources shall direct audio intended for these channels to BL/BR, which are logically the same speaker positions. Legacy Sources shall not set both RLC/RRC and BL/BR and shall not transmit data in both pairs of channels simultaneously. Legacy Sinks shall support reception of both RLC/RRC and BL/BR data; if data is sent simultaneously to both pairs of channels, Sinks shall ignore the data in RLC/RRC. Future use of F66 for other speaker positions is reserved.

[14] Use of F83 for TpLS/TpRS has been deprecated. Future use of F83 for other speaker positions is reserved.

6.6.1 Audio Identification Information

The information in Data Bytes 1 – 3 may be useful in identifying the audio format, audio channel count, audio sampling frequency, and number of bits per audio sample. If the DTV and the Source support more than “Basic Audio,” as defined by the physical/link specification, then this information shall be sent and shall accurately identify the stream while digital audio is being sent. If the Source only supports Basic Audio, it is not required to send this information, but it is recommended. In most cases, it is possible to identify the audio by parsing the actual audio stream (e.g., as specified in IEC 60958-3 [12]). In cases where the audio information in the Audio InfoFrame does not agree with the actual audio stream being received, the conflicting information in the Audio InfoFrame shall be ignored.

> Note: HDMI requires the CT, SS, and SF fields to be set to 0 (“Refer to Stream Header”) when these items are indicated elsewhere. By extension the CXT field is also required to be set to 0.

Data Byte 1 bits CT3, CT2, CT1, and CT0, when coded, define the audio format type of the audio stream. These bits may be set according to Table 39. A Sink capable of receiving digital audio shall determine the audio format by parsing the audio stream header when CT = 0 (CT3 = 0, CT2 = 0, CT1 = 0, CT0 = 0). Audio format types shall be defined by the CXT field in Data Byte 3 when CT = 15 (CT3 = 1, CT2 = 1, CT1 = 1, CT0 = 1).

Data Byte 1, bits CC2, CC1, and CC0, when coded, indicate the audio channel count carried transmitted in the audio stream. These bits may be set according to Table 39. When CC = 0 (CC2 = 0, CC1 = 0, CC0 = 0), channel count may be determined either by examining Data Bytes 4 and 6 to 9 (in the case of L-PCM, L-PCM 3D Audio, or DSD Audio) or from information in the bitstream (in the case of compressed audio). In some cases (e.g., with Object Based Audio, or compressed streams containing multiple channel layouts) the channel count of the rendered audio is determined by the Sink device.

Table 39 - Audio InfoFrame Data Byte 1

CT3	CT2	CT1	CT0	Audio Coding Type	Audio Stream Encoding Standard	Audio Stream Transport Standard
0	0	0	0	**Refer to Stream Header**		
0	0	0	1	L-PCM	IEC 60958-3 [12]	
0	0	1	0	AC-3	ATSC A/52B [11], excluding Annex E	IEC 61937-3 [14]
0	0	1	1	MPEG-1	ISO/IEC 11172-3 [23] Layer 1 or Layer 2	IEC 61937-4 [15]
0	1	0	0	MP3	ISO/IEC 11172-3 [23] Layer 3	IEC 61937-4 [15]
0	1	0	1	MPEG2	ISO/IEC 13818-3 [24]	IEC 61937-4 [15]
0	1	1	0	AAC LC	ISO/IEC 14496-3 [25]	IEC 61937-6 [17]
0	1	1	1	DTS	ETSI TS 102 114 [36]	IEC 61937-5 [16]
1	0	0	0	ATRAC	IEC 61909 [13]	IEC 61937-7 [18]
1	0	0	1	DSD	ISO/IEC 14496-3 [25], subpart 10	
1	0	1	0	Enhanced AC-3	ATSC A/52B [11], with Annex E	IEC 61937-3 [14]
1	0	1	1	DTS-HD DTS-UHD	ETSI TS 102 114 [36] ETSI TS 103 491 [37]	IEC 61937-5 [16]
1	1	0	0	MAT	DVD Forum MLP [27]	IEC 61937-9 [20]
1	1	0	1	DST	ISO/IEC 14496-3 [25] subpart 10	
1	1	1	0	WMA Pro	WMA Pro Decoder Specification [29]	IEC 61937-8 [19]
1	1	1	1	**Refer to Audio Coding Extension Type (CXT) field in Data Byte 3**		

CC2	CC1	CC0	Audio Channel Count
0	0	0	**Refer to Stream Header**
0	0	1	2 channels
0	1	0	3 channels
0	1	1	4 channels
1	0	0	5 channels
1	0	1	6 channels
1	1	0	7 channels
1	1	1	8 channels

Data Byte 2, bits SF2, SF1, and SF0, when coded, indicate the audio sampling frequency in the audio stream. These bits shall be set according to Table 40. A Sink capable of receiving digital audio shall determine the audio sampling frequency by parsing the audio stream header when SF = 0 (SF2 = 0, SF1 = 0, SF0 = 0).

Data Byte 2, bits SS1 and SS0, when coded, indicate the number of bits per audio sample in the audio stream. These bits shall be set according to Table 40. A Sink capable of receiving digital audio shall determine the number of bits per audio sample by parsing the audio stream header when SS = 0 (SS1 = 0, SS0 = 0).

Table 40 - Audio InfoFrame Data Byte 2

SF2	SF1	SF0	Sampling Frequency
0	0	0	Refer to Stream Header
0	0	1	32 kHz
0	1	0	44.1 kHz (CD)
0	1	1	48 kHz
1	0	0	88.2 kHz
1	0	1	96 kHz
1	1	0	176.4 kHz
1	1	1	192 kHz

SS1	SS0	Bits / Sample
0	0	Refer to Stream header
0	1	16 bit
1	0	20 bit
1	1	24 bit

Data Byte 3, bits CXT4, CXT3, CXT2, CXT1, and CXT0, when coded and when the CT field in Data Byte 1 is set to 15, indicate the audio format type of the audio stream. The CXT4-CXT0 bits shall be set to 0x00 (CXT4 = 0, CXT3 = 0, CXT2 = 0, CXT1 = 0, CXT0 = 0) when the CT field in Data Byte 1 is set to a value other than 15. When the CT field in Data Byte 1 is set to 15 (CT3 = 1, CT2 = 1, CT1 = 1, CT0 = 1) the CXT bits may be set according to Table 41. When CXT = 0 (CXT4 = 0, CXT3 = 0, CXT2 = 0, CXT1 = 0, CXT0 = 0) a Sink capable of receiving digital audio shall determine the audio format by analyzing the value of the CT field in Data Byte 1 or by parsing the audio stream header.

Table 41 - Additional Audio Format Code Extension Values (Data Byte 3)

CXT	Audio Coding Extension Type	Audio Stream Encoding Standard	Audio Stream Transport Standard
0x00	**Refer to Audio Coding Type (CT) field in Data Byte 1**		
0x01	**Not in use**		
0x02	**Not in use**		
0x03	**Not in use**		
0x04	MPEG-4 HE AAC	ISO/IEC 14496-3 [25]	IEC 61937-11 [21]
0x05	MPEG-4 HE AAC v2	ISO/IEC 23003-1 [26]	IEC 61937-11 [21]
0x06	MPEG-4 AAC LC	ISO/IEC 14496-3 [25]	IEC 61937-11 [21]
0x07	DRA	GB/T 22726 [39]	IEC 61937-12 [22]
0x08	MPEG-4 HE AAC + MPEG Surround	ISO/IEC 14496-3 [25], ISO/IEC 23003-1 [26]	IEC 61937-11 [21]
0x09	**Reserved**		
0x0A	MPEG-4 AAC LC + MPEG Surround	ISO/IEC 14496-3 [25], ISO/IEC 23003-1 [26]	IEC 61937-11 [21]
0x0B	MPEG-H 3D Audio	ISO/IEC 23008-3 [44]	IEC 61937-13 [46]
0x0C	AC-4	ETSI TS 103 190 [45]	IEC 61937-14 [47]
0x0D	L-PCM 3D Audio	IEC 60958-3 [12]	
0x0E	Auro-Cx	Auro-Cx Functional Specification [141]	IEC 61937-15 [58]
0x0F	MPEG-D USAC	ISO/IEC 23003-3 [59]	IEC 61937-11 [21]
0x10 – 0x1F	**Reserved**		

6.6.2 Speaker Mapping and Down-mix Information

Data Bytes 4 and 5 apply only to multi-channel (i.e., more than two channels) Uncompressed Audio.

Data Byte 4 contains information that describes how various speaker locations are allocated to transmission channels. Data Byte 5 contains information that tells the Sink how much the Source attenuated the audio during a down-mixing operation. The down-mix inhibit flag (**DM_INH**) describes whether audio output is permitted to be down-mixed or not. This flag is used in DVD Audio applications (see Table 45).

The labels and placements of speakers used in CTA-861 are defined in Figure 6 and Table 42. Note that the speaker location names are informatively associated to the angular speaker locations as per Rec. ITU-R BS.2051 [139]. The Speaker labels and descriptions in Table 42 are consistent with those in ISO/IEC 62574 [43].

Annex K contains additional information concerning speaker placement relationships between CTA-861 and other standards, and Annex Q describes how the naming convention in Table 42 relates to prior revisions of CTA-861.

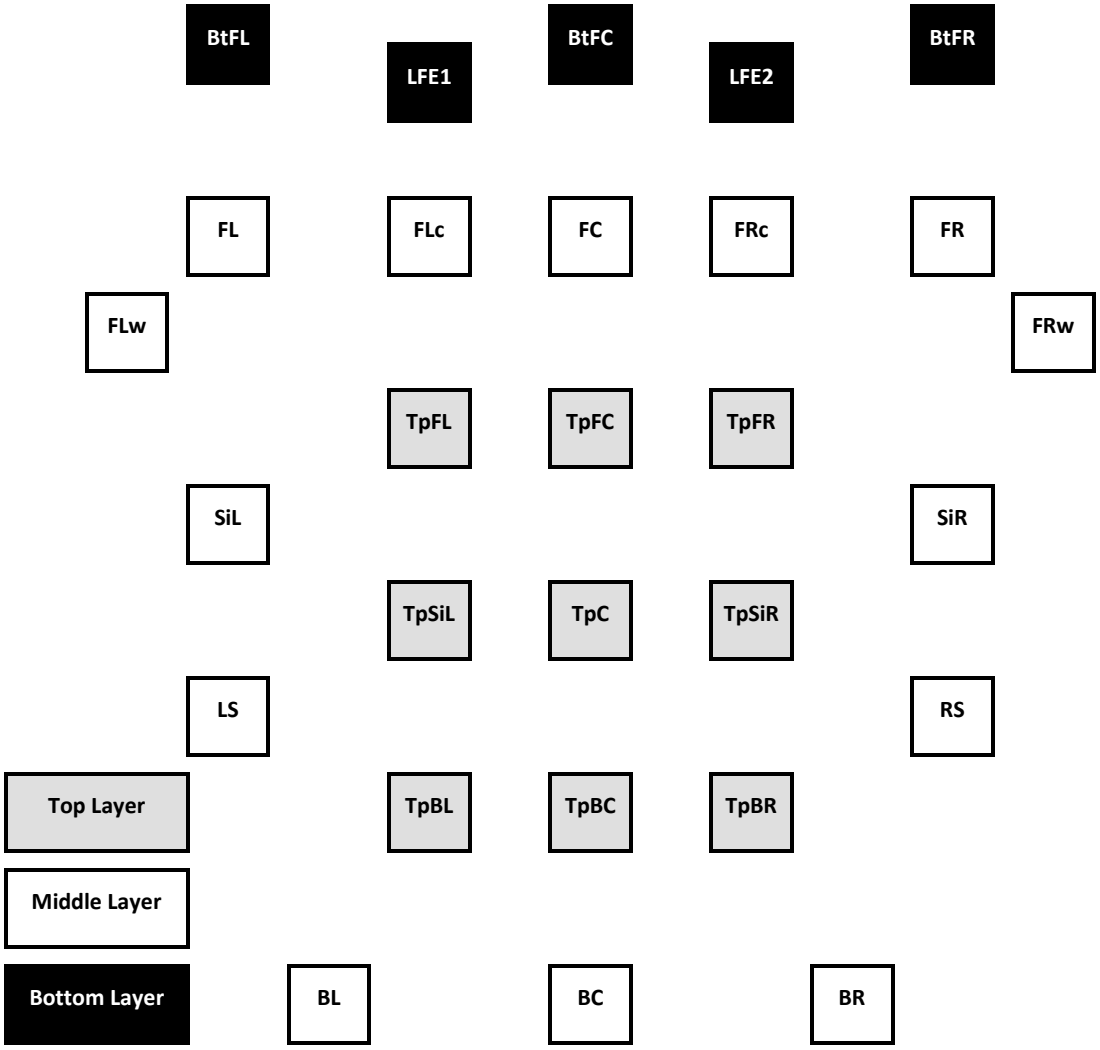


Figure 6 - Speaker Placement

Table 42 - Speaker Placement

Label	Position Description	Code
FL	Front Left	0x00
FR	Front Right	0x01
FC	Front Center	0x02
LFE1	Low Frequency Effects 1	0x03
BL	Back Left	0x04
BR	Back Right	0x05
FLc	Front Left of Center	0x06
FRc	Front Right of Center	0x07
BC	Back Center	0x08
LFE2	Low Frequency Effects 2	0x09
SiL	Side Left	0x0A
SiR	Side Right	0x0B
TpFL	Top Front Left	0x0C
TpFR	Top Front Right	0x0D
TpFC	Top Front Center	0x0E
TpC	Top Center	0x0F
TpBL	Top Back Left	0x10
TpBR	Top Back Right	0x11
TpSiL	Top Side Left	0x12
TpSiR	Top Side Right	0x13
TpBC	Top Back Center	0x14
BtFC	Bottom Front Center	0x15
BtFL	Bottom Front Left	0x16
BtFR	Bottom Front Right	0x17
FLw	Front Left Wide	0x18
FRw	Front Right Wide	0x19
LS	Left Surround	0x1A
RS	Right Surround	0x1B
	reserved	0x1C
	reserved	0x1D
	reserved	0x1E
	reserved	0x1F

NOTE —

F: indicates speakers forward of the primary listening position

B: indicates speakers behind the primary listening position

Si: indicates speaker to the side of the primary listening position

Tp: indicates speakers above the normal seating position of the listeners

Bt: indicates speakers below the normal seating position of the listeners

TpC: is directly over the primary listening position. All speakers are assumed to be generally pointing at the primary listening position.

Data Byte 4 contains information that describes how various speaker locations are allocated to transmission channels. Channel allocation is shown in Table 43 (see Annex K for additional information concerning audio channel allocation relationships between CTA-861 and other standards).

Codes 0xFE and 0xFF in Data Byte 4 designate an alternate delivery order of the L-PCM audio, described in Sections 6.6.3 and 6.6.4.

Table 43 - Audio InfoFrame Data Byte 4

CA (binary)								CA (hex)	Channel Number							
7	6	5	4	3	2	1	0		8	7	6	5	4	3	2	1
0	0	0	0	0	0	0	0	0x00	-	-	-	-	-	-	FR	FL
0	0	0	0	0	0	0	1	0x01	-	-	-	-	-	LFE1	FR	FL
0	0	0	0	0	0	1	0	0x02	-	-	-	-	FC	-	FR	FL
0	0	0	0	0	0	1	1	0x03	-	-	-	-	FC	LFE1	FR	FL
0	0	0	0	0	1	0	0	0x04	-	-	-	BC	-	-	FR	FL
0	0	0	0	0	1	0	1	0x05	-	-	-	BC	-	LFE1	FR	FL
0	0	0	0	0	1	1	0	0x06	-	-	-	BC	FC	-	FR	FL
0	0	0	0	0	1	1	1	0x07	-	-	-	BC	FC	LFE1	FR	FL
0	0	0	0	1	0	0	0	0x08	-	-	RS	LS	-	-	FR	FL
0	0	0	0	1	0	0	1	0x09	-	-	RS	LS	-	LFE1	FR	FL
0	0	0	0	1	0	1	0	0x0A	-	-	RS	LS	FC	-	FR	FL
0	0	0	0	1	0	1	1	0x0B	-	-	RS	LS	FC	LFE1	FR	FL
0	0	0	0	1	1	0	0	0x0C	-	BC	RS	LS	-	-	FR	FL
0	0	0	0	1	1	0	1	0x0D	-	BC	RS	LS	-	LFE1	FR	FL
0	0	0	0	1	1	1	0	0x0E	-	BC	RS	LS	FC	-	FR	FL
0	0	0	0	1	1	1	1	0x0F	-	BC	RS	LS	FC	LFE1	FR	FL
0	0	0	1	0	0	0	0	0x10	BR	BL[15]	RS	LS	-	-	FR	FL
0	0	0	1	0	0	0	1	0x11	BR	BL	RS	LS	-	LFE1	FR	FL
0	0	0	1	0	0	1	0	0x12	BR	BL	RS	LS	FC	-	FR	FL
0	0	0	1	0	0	1	1	0x13	BR	BL	RS	LS	FC	LFE1	FR	FL
0	0	0	1	0	1	0	0	0x14	FRc	FLc	-	-	-	-	FR	FL
0	0	0	1	0	1	0	1	0x15	FRc	FLc	-	-	-	LFE1	FR	FL
0	0	0	1	0	1	1	0	0x16	FRc	FLc	-	-	FC	-	FR	FL
0	0	0	1	0	1	1	1	0x17	FRc	FLc	-	-	FC	LFE1	FR	FL
0	0	0	1	1	0	0	0	0x18	FRc	FLc	-	BC	-	-	FR	FL
0	0	0	1	1	0	0	1	0x19	FRc	FLc	-	BC	-	LFE1	FR	FL
0	0	0	1	1	0	1	0	0x1A	FRc	FLc	-	BC	FC	-	FR	FL
0	0	0	1	1	0	1	1	0x1B	FRc	FLc	-	BC	FC	LFE1	FR	FL
0	0	0	1	1	1	0	0	0x1C	FRc	FLc	RS	LS	-	-	FR	FL
0	0	0	1	1	1	0	1	0x1D	FRc	FLc	RS	LS	-	LFE1	FR	FL
0	0	0	1	1	1	1	0	0x1E	FRc	FLc	RS	LS	FC	-	FR	FL
0	0	0	1	1	1	1	1	0x1F	FRc	FLc	RS	LS	FC	LFE1	FR	FL

[15] Speaker positions RLC/RRC for CA = 0x10 – 0x13 have been replaced by BL/BR, respectively. RLC/RRC have been deprecated as unique speaker locations as BL/BR are logically the same speaker locations. This does change the functionality of the Legacy mapping of speaker locations to transmission channels. For Table 43, this is a documentation change only.

Table 43 - Audio InfoFrame Data Byte 4 (continued)

CA (binary)								CA (hex)	Channel Number							
0	0	1	0	0	0	0	0	0x20	-	TpFC	RS	LS	FC	-	FR	FL
0	0	1	0	0	0	0	1	0x21	-	TpFC	RS	LS	FC	LFE1	FR	FL
0	0	1	0	0	0	1	0	0x22	TpC	-	RS	LS	FC	-	FR	FL
0	0	1	0	0	0	1	1	0x23	TpC	-	RS	LS	FC	LFE1	FR	FL
0	0	1	0	0	1	0	0	0x24	TpFR	TpFL	RS	LS	-	-	FR	FL
0	0	1	0	0	1	0	1	0x25	TpFR	TpFL	RS	LS	-	LFE1	FR	FL
0	0	1	0	0	1	1	0	0x26	FRw	FLw	RS	LS	-	-	FR	FL
0	0	1	0	0	1	1	1	0x27	FRw	FLw	RS	LS	-	LFE1	FR	FL
0	0	1	0	1	0	0	0	0x28	TpC	BC	RS	LS	FC	-	FR	FL
0	0	1	0	1	0	0	1	0x29	TpC	BC	RS	LS	FC	LFE1	FR	FL
0	0	1	0	1	0	1	0	0x2A	TpFC	BC	RS	LS	FC	-	FR	FL
0	0	1	0	1	0	1	1	0x2B	TpFC	BC	RS	LS	FC	LFE1	FR	FL
0	0	1	0	1	1	0	0	0x2C	TpC	TpFC	RS	LS	FC	-	FR	FL
0	0	1	0	1	1	0	1	0x2D	TpC	TpFC	RS	LS	FC	LFE1	FR	FL
0	0	1	0	1	1	1	0	0x2E	TpFR	TpFL	RS	LS	FC	-	FR	FL
0	0	1	0	1	1	1	1	0x2F	TpFR	TpFL	RS	LS	FC	LFE1	FR	FL
0	0	1	1	0	0	0	0	0x30	FRw	FLw	RS	LS	FC	-	FR	FL
0	0	1	1	0	0	0	1	0x31	FRw	FLw	RS	LS	FC	LFE1	FR	FL
0	0	1	1	0	0	1	0	0x32	Reserved							
…																
1	1	1	1	1	1	0	1	0xFD								
1	1	1	1	1	1	1	0	0xFE	**Channels delivered according to the Speaker Mask (see Section 6.6.3)**							
1	1	1	1	1	1	1	1	0xFF	**Channels delivered according to Channel Index (see Section 6.6.4)**							

The Sink's speaker allocation is not always the same as that contained within the Source's audio. In this case, the Source should down mix the audio in order to properly meet the Sink's speaker configuration. In actual implementations, all down-mix coefficients are equally attenuated to prevent calculation overflows. The total sound level becomes lower after down-mixing. For this reason, the Level Shift Value should also be transmitted to the Sink to ensure the proper sound level is achieved.

Data Byte 5 contains Level Shift Information, a Down-mix Inhibit Flag, and LFE playback level information.

The values of attenuation associated with the Level Shift Values (LSV0 – LSV3) are shown in Table 44.

Table 44 - Audio InfoFrame Data Byte 5, Level Shift Value

LSV3	LSV2	LSV1	LSV0	Level Shift Value
0	0	0	0	0dB
0	0	0	1	1dB
0	0	1	0	2dB
0	0	1	1	3dB
0	1	0	0	4dB
0	1	0	1	5dB
0	1	1	0	6dB
0	1	1	1	7dB
1	0	0	0	8dB
1	0	0	1	9dB
1	0	1	0	10dB
1	0	1	1	11dB
1	1	0	0	12dB
1	1	0	1	13dB
1	1	1	0	14dB
1	1	1	1	15dB

The Down-mix Inhibit Flag is shown in Table 45.

Table 45 - Audio InfoFrame Data Byte 5, Down-mix Inhibit Flag

DM_INH	Describes whether the down mixed stereo output is permitted or not.
0	Permitted or no information about any assertion of this
1	Prohibited

The LFE playback level information shown in Table 46 can be used to communicate one element of techniques used to balance low frequency audio information when audio is presented in a variety of speaker configurations combining speakers and subwoofers. One such technique uses a 10 dB boost to bring low frequency information in the LFE channel of 5.1 channel audio systems in acoustical balance with the low frequency information present at other speakers. This table is a simple way of communicating from Source-to-Sink that this element of low frequency balancing has been employed. If audio data does not contain a LFE signal, then the LFEPBL field shall be ignored.

Table 46 - Audio InfoFrame Data Byte 5, LFE Playback Level Information

LFEPBL1	LFEPBL0	Describes what value is used for LFE playback level comparing with other channel signal.
0	0	Unknown or refer to other information
0	1	0 dB playback
1	0	+ 10 dB playback
1	1	Reserved
Note: The Source may set the LFEPBL fields according to encoding rules of source content. Typically the Audio/Visual content use +10 dB and Audio contents use 0 dB. The Sink may adjust the LFE signal level and not the total output level for subwoofer in down mixing case. If the audio data does not contain a LFE signal, the LFEPBL field shall be ignored.		

6.6.3 Delivery According to the Speaker Mask (Data Byte 4 = 0xFE)

Support for this delivery method is indicated by the presence of the Room Configuration Data Block (RCDB). Data Bytes 6 to 8 of the InfoFrame shown in Table 37 correspond to the SPM Bytes defined in the RCDB (see Table 104) and these bytes constitute the Speaker Mask. The source shall not declare channels in the InfoFrame that were not declared as available in the Speaker Mask. All channels declared in the InfoFrame shall be present in the audio delivery.

In the Speaker Mask, many of the flags describe a pair of speakers (e.g., FL/FR), but some describe a single speaker (e.g., LFE1). When a source device is preparing audio for a room described in this manner, both speakers of a pair are always assumed to be present.

For systems conforming to ISO 60958, speaker feeds are always packed in channel pairs. For example, in a 5.1 speaker system, LFE1 and FC are packed together in a L-PCM transmission.

The ordering of the L-PCM channels feeding into the speakers shall be according to the following rules:

1) Channels shall be sent consecutively in the order indicated in the Speaker Mask, starting from bit 0, Data Byte SPM1.
2) Channel pairs shall always be sent together in the order of Left/Right
3) Single channels shall be sent in the order of the first flag/next flag, e.g., LFE1/Center.
4) If one or more channel pairs exists between two single channels, then the second single channel shall be brought forward to fill the vacancy ahead of the pairs.
5) If an odd number of channels are being presented, then an inactive channel shall occupy the second channel of the channel pair carrying the single channel.

Example:

A source is sending the following channels:

FL, FR, LFE1, FC, BL, BR, BC, TpFL, TpFR, LFE2

The order of the channel pairs shall be as follows:

FL/FR,

LFE1/FC,

BL/BR,

BC/LFE2,

TpFL/TpFR

6.6.4 Delivery by Channel Index (Data Byte 4 = 0xFF)

A Source shall only deliver audio data by channel index when a Speaker Location Data Block is available for all channels being utilized by the Source.

If the delivery by Channel Index is used, it will be used exclusively, and delivery by speaker mask (as described in Section 6.6.3) will not be used.

When Delivery by Channel Index is being used, Data Bytes 6, 7, 8, and 9 in the Audio InfoFrame, shown in Table 38, indicate which channels are being delivered. Bits CID00 to CID31 correspond to Channel Index 0 to Channel Index 31 respectively, as assigned in the Speaker Location Descriptors.

The Source shall only deliver audio to Channel Indices that were declared in a Speaker Location Data Block. For each Channel Index that is represented in the audio transmission, the corresponding CID flag of the InfoFrame shall be set to 1. All other CID flags shall be set to 0.

The ordering of the L-PCM channels in the audio transmission shall directly correspond to the Channel Index from the lowest value to the highest, not make any exception for paired (L/R) channels, and only include channels indicated as Active in the corresponding Speaker Location Descriptor (see Table 106).

When Delivery by Channel Index is used, then the Sink shall not apply spatial mapping according to the Speaker Locations defined in the SLDB.

6.6.5 Additional Audio InfoFrame Information

The value of the LFEPBL field, in Audio InfoFrame Data Byte 5, shall apply to all LFE channels in use (i.e., LFE1 and LFE2).

6.7 MPEG Source InfoFrame

The MPEG Source InfoFrame describes aspects of the compressed video stream that were used to produce the uncompressed video. In many cases, the compressed source is MPEG2, although this InfoFrame can be applied to any similar compressed format. Some Sinks may use this information to improve the displayed Picture.

> Note: Implementation of the MPEG Source InfoFrame is not recommended due to issues that have been reported and not resolved. The information contained in this section is reserved for future use and enhancement.

Transmission of this information by the Source is optional. Use of this information by the Sink is also optional.

If the Source supports the transmission of the MPEG Source InfoFrame and if it determines that the Sink is capable of receiving the MS InfoFrame (i.e., the Sink has included CTA Extension Version 3 in EDID), then this information should be sent once per Video Frame when applicable. The data applies to the next full frame of video data.

The format of the MPEG Source InfoFrame is shown in Table 47.

Table 47 - MPEG Source InfoFrame format

InfoFrame Type Code	InfoFrame Type = 0x05							
InfoFrame Version Number	Version = 0x01							
Length of MPEG Source InfoFrame	Length of MPEG Source InfoFrame (10)							
Data Byte 1	MB#0 (MPEG Bit Rate: Hz Lower → Upper)							
Data Byte 2	MB#1							
Data Byte 3	MB#2							
Data Byte 4	MB#3 (Upper Byte)							
Data Byte 5	F57=0	F56=0	F55=0	FR0	F53=0	F52=0	MF1	MF0
Data Byte 6	F67=0	F66=0	F65=0	F64=0	F63=0	F62=0	F61=0	F60=0
Data Byte 7	F77=0	F76=0	F75=0	F74=0	F73=0	F72=0	F71=0	F70=0
Data Byte 8	F87=0	F86=0	F85=0	F84=0	F83=0	F82=0	F81=0	F80=0
Data Byte 9	F97=0	F96=0	F95=0	F94=0	F93=0	F92=0	F91=0	F90=0
Data Byte 10	F107=0	F106=0	F105=0	F104=0	F103=0	F102=0	F101=0	F100=0

Data Bytes 1 – 4 give the MPEG bit rate. The MPEG Bit Rate is stored as a 32-bit number and is expressed in Hertz. MB#0 contains the least significant byte while MB#3 contains the most significant byte. If the MPEG Bit Rate is unknown or this field does not apply, then all of the bits in Data Bytes 1 – 4 shall be set to 0.

Example:

10 Mbps → 10,000,000 Hz (dec.) → 0x00989680 Upper … Lower Byte

Byte 1 MB#0 0x80 Lower Byte

Byte 2 MB#1 0x96

Byte 3 MB#2 0x98

Byte 4 MB#3 0x00 Upper

MF1 and MF0 in Data Byte 5 (see Table 48) designate whether the current field/frame was generated from an I, B, or P picture from the source MPEG stream. If this is unknown or does not apply, then the field shall be set to “unknown.”

In some cases, the Source creates 60 field/second video from 24 frames/second source material. 3:2 pull-down is commonly used. FR0 can be used to designate whether a field is a repeated field or not. The Sink can use this information to improve the picture. If 3:2 pull-down

does not apply to the current video decoding, then all of the fields/frames should be marked as "New field."

Table 48 - MPEG Source InfoFrame Data Byte 5

FR0	Field Repeat (for 3:2 pull-down)
0	New field (Picture)
1	Repeated Field

MF1	MF0	MPEG Frame
0	0	Unknown (No Data)
0	1	I Picture
1	0	B Picture
1	1	P Picture

6.8 NTSC VBI InfoFrame

The NTSC VBI InfoFrame provides for the carriage of SCTE 127 [28] payloads containing VBI data. Transmission of this information by the Source is optional. Use of this information by the Sink is also optional. However, when present, Sinks can extract this information for direct use, or when analog NTSC outputs are present, regenerate relevant VBI data along with the video and audio.

This InfoFrame should be sent once per Video Frame when applicable. The data applies to the next full frame of video data.

The format of the NTSC VBI InfoFrame is shown in Table 49.

Table 49 - NTSC VBI InfoFrame

InfoFrame Type Code	InfoFrame Type = 0x06
InfoFrame Version Number	Version = 0x01
Length of NTSC VBI	Length of NTSC VBI InfoFrame — total number of bytes following this field
Data Bytes 1…Length	The PES_data_field() structure of SCTE 127, Table 2 [28]

The stuffing_bytes should be omitted before transmission by the Source.

In order to maximize the possibility of operation with existing silicon, Sources should constrain the NTSC VBI InfoFrame's payload to 27 bytes or less (e.g., a 31-byte HDMI InfoFrame less 3-byte header and 1-byte checksum leaves 27 bytes for payload). This means the PES_data_field() structure would be constrained (modified if necessary) to 27 bytes. NABTS (which requires 37 bytes) should not be encoded.

> Note: 27-byte payload is adequate, for example, for two fields of AMOL96 and one field of TVG2X per frame.

6.9 Dynamic Range and Mastering InfoFrame

The Dynamic Range and Mastering InfoFrame carries data such as the Transfer Function and the Static Metadata associated with the dynamic range of the video signal.

If the Source supports the transmission of the Dynamic Range and Mastering InfoFrame and if the Source determines that the Sink is capable of receiving that InfoFrame:

- The Source shall send the Dynamic Range and Mastering InfoFrame once per Video Field if the Source is sending a video signal that requires any Data Byte 1 to *n* in Table 50 to be non-zero (i.e., HDR metadata or other uses requiring non-zero data).
- The Source may send the Dynamic Range and Mastering InfoFrame once per Video Field when the Data Bytes 1 to *n* in Table 50 are all zero.

The Source shall not send a Dynamic Range and Mastering InfoFrame to a Sink that does not have at least one of the TF_*n* bits in the HDR Static Metadata Data Block (Section 7.5.13) set to '1'. The Source shall not send EOTF (Transfer_Function) = n to a Sink that has the corresponding TF_*n* bit set to '0'.

The format of the Dynamic Range and Mastering InfoFrame is shown in Table 50.

Table 50 - Dynamic Range and Mastering InfoFrame

InfoFrame Type Code	InfoFrame Type = 0x07					
InfoFrame Version number	Version = 0x01					
Length of InfoFrame	Length of following HDR Metadata InfoFrame					
Data Byte 1	F17=0	F16=0	F15=0	F14=0	F13=0	Transfer_Function (3 bits)
Data Byte 2	F27=0	F26=0	F25=0	F24=0	F23=0	Static_Metadata_ Descriptor_ID (3 bits)
Data Byte 3	Static_Metadata_Descriptor					
…	…					
Data Byte n	…					

Data Byte 1 Transfer_Function identifies the Electro-Optical Transfer Function (EOTF) or Opto-Electronic Transfer Function (OETF) used in the video signal.

Table 51 - Data Byte 1 — Transfer Function

Transfer_Function	Transfer Function used in the signal
0	Traditional gamma — SDR Luminance Range
1	Traditional gamma — HDR Luminance Range
2	Perceptual Quantization (PQ) based on SMPTE ST 2084 [41][16]
3	Hybrid Log-Gamma (HLG) based on Rec. ITU-R BT.2100 [48]
4 - 7	Reserved for future use

"Traditional Gamma" indicates that the Transfer Function used in the video signal is consistent with the requirements in CTA-861. If the Colorimetry bits C0 and C1 in the AVI InfoFrame are both zero (indicating "No Data"), then the transfer function shall be consistent with the requirements in Section 5.1, "Default Encoding Parameters". If either of bits C0 and C1 in the AVI InfoFrame are non-zero, then the transfer function shall be consistent with the colorimetry standard indicated by the Colorimetry (C0 and C1), Extended Colorimetry (EC0 to EC2) and Additional Colorimetry Extension (ACE0 to ACE3) bits in the AVI InfoFrame. See Table 5 for the transfer characteristics of supported colorimetries.

The SMPTE ST 2086 [42] metadata contained in Data Bytes 3 – 22 of Table 53 - Static Metadata Descriptor Type 1 may be used by the Source to provide information about the mastering display color volume characteristics associated with the video signal.

If "Traditional Gamma — SDR Luminance Range" is indicated, then the maximum encoded luminance is typically mastered to 100 cd/m^2.

If "Traditional Gamma — HDR Luminance Range" is indicated, then the maximum encoded luminance is understood to be the maximum luminance of the Sink device.

Data Byte 2 Static_Metadata_Descriptor_ID identifies the structure used in Data Byte 3 and higher.

Table 52 - Data Byte 2 – Static_Metadata_Descriptor_ID

Static_Metadata_ Descriptor_ID	Metadata Descriptor
0	Static Metadata Type 1
1 – 7	Reserved for future use

6.9.1 Static Metadata Type 1

When Static_Metadata_Descriptor_ID = 0, Static_Metadata_Descriptor uses the structure defined in Table 53 that was defined at the request of the Blu-ray Disc Association, see [138].

[16] The Transfer Function 2 is equivalent to the Transfer Function in Rec. ITU-R BT.2100, where it is referred to as Perceptual Quantization (PQ).

Table 53 - Static Metadata Descriptor Type 1

Data Byte number	Contents	Group
Data Byte 3	display_primaries_x[0], LSB	1
Data Byte 4	display_primaries_x[0], MSB	
Data Byte 5	display_primaries_y[0], LSB	
Data Byte 6	display_primaries_y[0], MSB	
Data Byte 7	display_primaries_x[1], LSB	
Data Byte 8	display_primaries_x[1], MSB	
Data Byte 9	display_primaries_y[1], LSB	
Data Byte 10	display_primaries_y[1], MSB	
Data Byte11	display_primaries_x[2], LSB	
Data Byte 12	display_primaries_x[2], MSB	
Data Byte 13	display_primaries_y[2], LSB	
Data Byte 14	display_primaries_y[2], MSB	
Data Byte 15	white_point_x, LSB	2
Data Byte 16	white_point_x, MSB	
Data Byte 17	white_point_y, LSB	
Data Byte 18	white_point_y, MSB	
Data Byte 19	max_display_mastering_luminance, LSB	3
Data Byte 20	max_display_mastering_luminance, MSB	
Data Byte 21	min_display_mastering_luminance, LSB	4
Data Byte 22	min_display_mastering_luminance, MSB	
Data Byte 23	Maximum Content Light Level, LSB	5
Data Byte 24	Maximum Content Light Level, MSB	
Data Byte 25	Maximum Frame-average Light Level, LSB	6
Data Byte 26	Maximum Frame-average Light Level, MSB	

Data Bytes 3 – 22 contain the Display Mastering data defined in SMPTE ST 2086 [42].

Data Bytes 3 – 18 are coded as unsigned 16-bit values in units of 0.00002, where 0x0000 represents zero and 0xC350 represents 1.0000.

Data Bytes 3 – 18 describe the chromaticity of the Red, Green, and Blue color primaries and of the white point of the mastering display.

All possible mappings of the chromaticity of Red, Green, and Blue color primaries to indices 0, 1, and 2 are allowed and shall be supported by the sink.

The correspondence between Red, Green, and Blue color primaries and indices 0, 1, or 2 are determined by the following relationship:

> The Red color primary corresponds to the index of the largest display_primaries_x[] value.
>
> The Green color primary corresponds to the index of the largest display_primaries_y[] value.
>
> The Blue color primary corresponds to the index with neither the largest display_primaries_y[] value nor the largest display_primaries_x[] value.

Data Bytes 19 – 20 specify a value for the max_display_mastering_luminance. This value is coded as an unsigned 16-bit value in units of 1 cd/m^2, where 0x0001 represents 1 cd/m^2 and 0xFFFF represents 65535 cd/m^2.

Data Bytes 21 – 22 specify a value for the min_display_mastering_luminance. This value is coded as an unsigned 16-bit value in units of 0.0001 cd/m^2, where 0x0001 represents 0.0001 cd/m^2 and 0xFFFF represents 6.5535 cd/m^2.

Data Bytes 23 – 24 contain the Maximum Content Light Level (MaxCLL). This value is coded as an unsigned 16-bit value in units of 1 cd/m^2, where 0x0001 represents 1 cd/m^2 and 0xFFFF represents 65535 cd/m^2.[17] The algorithm used to calculate MaxCLL is defined in Annex P section P.1.

Data Bytes 25 – 26 contain the Maximum Frame-Average Light Level (MaxFALL). This value is coded as an unsigned 16-bit value in units of 1 cd/m^2, where 0x0001 represents 1 cd/m^2 and 0xFFFF represents 65535 cd/m^2.[18] The algorithm used to calculate MaxFALL is defined in Annex P section P.2.

The data in Data Bytes 3 – 26 are arranged into groups, as indicated in Table 53 - Static Metadata Descriptor Type 1 above. When all of the Data Bytes in a group are set to zero, then the Sink shall interpret the data for that group as unknown[19]. It is permissible to send a Static Metadata Descriptor (Data Bytes 3 through n) containing a subset of data that is constructed with partial zeroes.

If a Source does not have SMPTE ST 2086 [42] metadata to send, then Data Bytes 3–22 shall be populated with zeroes to indicate the data is not provided.

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[17] For MaxCLL, the unit is equivalent to cd/m^2 when the brightest pixel in the entire video signal has the chromaticity of the white point of the encoding system used to represent the video signal. Since the value of MaxCLL is computed with a max() mathematical operator, it is possible that the true CIE Y Luminance value is less than the MaxCLL value. This situation may occur when there are very bright blue saturated pixels in the video signal, which may dominate the max(R,G,B) calculation, but since the blue channel is an approximately 10% contributor to the true CIE Y Luminance, the true CIE Y Luminance value of the example blue pixel would be only approximately 10% of the MaxCLL value.

[18] For MaxFALL, the unit is equivalent to cd/m^2 when the maximum frame average of the entire video signal corresponds to a full-screen of pixels that has the chromaticity of the white point of the encoding system used to represent the video signal. The frame-average computation used to compute the MaxFALL value is performed only on the active image area of the image data. If the video signal is a "letterbox" format (e.g. where a 2.40:1 aspect ratio is put inside a 16:9 image container with black bars on the top and bottom of the image), the black bar areas are not part of the active image area and therefore are not included in the frame-average computation. This allows the MaxFALL value to remain an upper bound on the maximum frame-average light level even if image zooming or pan/scan is performed as a post-processing operation.

[19] For MaxCLL and MaxFALL, this may occur when information about the content light level has not been, or cannot be, provided – for example, content that is rendered or broadcast in real-time, or pre-processed content that was delivered without information about the content light level.

6.10 Extended InfoFrame

The Extended InfoFrame is intended to carry larger amounts of data than is allowed by the InfoFrame mechanism described in Sections 6.1 to 6.9.

The content of the Extended InfoFrame is shown in Table 54:

Table 54 - Extended InfoFrame

Extended InfoFrame Type Code	Extended InfoFrame Type LSB
	Extended InfoFrame Type MSB
Length (n) of Extended InfoFrame	Length of following Extended InfoFrame Data LSB
	Length of following Extended InfoFrame Data MSB
Data Byte 1	Extended InfoFrame Data Byte 1
…	…
Data Byte *n*	Extended InfoFrame Data Byte *n*

Extended InfoFrame Type Code — a 2-byte number identifying the data carried in this Extended InfoFrame. Extended InfoFrame Type Codes are shown in Table 55:

Table 55 - Extended InfoFrame Type Codes

Value	Extended InfoFrame Type
0x0000	Reserved
0x0001	HDR Dynamic Metadata according to the syntax specified in Annex R
0x0002	HDR Dynamic Metadata carried in Supplemental Enhancement Information (SEI) messages according to ETSI TS 103 433-1 [50]
0x0003	HDR Dynamic Metadata carried in Colour Remapping Information SEI message according to Rec. ITU-T H.265 [52]
0x0004	HDR Dynamic Metadata carried according to the syntax specified in Annex S
0x0005-0x00FF	Reserved
0x0100	Graphics Overlay Flag carried according to the syntax specified in Annex T
All other values	Reserved

Length of Extended InfoFrame — a 2-byte value indicating the length of the following Extended InfoFrame data.

Data Byte 1 to *n* — The Extended InfoFrame Data identified by the Extended InfoFrame Type Code.

A Source shall not send the same Extended InfoFrame Type more than once within an MTW.

The data contained in the HDR Dynamic Metadata Extended InfoFrame applies to the Active Pixels that follow the MTW.

If the Source supports transmission of a type of HDR Dynamic Metadata Extended InfoFrame (as indicated by the relevant Extended InfoFrame Type) and if it determines that the Sink is capable of receiving that information, the Source may send the HDR Dynamic Metadata

Extended InfoFrame in conjunction with the video encoded according to the rules of the Extended InfoFrame Type.

6.10.1 HDR Dynamic Metadata Extended InfoFrame

When the Extended InfoFrame Type Code is set to 0x0001, 0x0002, 0x0003, or 0x0004, the Extended InfoFrame carries HDR Dynamic Metadata. The HDR Dynamic Metadata Extended InfoFrame contains the HDR Dynamic Metadata that might typically be carried in Supplemental Enhancement Information (SEI) messages.

When the Extended InfoFrame Type Code is set to 0x0100, the Extended InfoFrame carries the Graphics Overlay Flag, which applies to the HDR Dynamic Metadata as specified in Annex T.

Figure 7 depicts the transfer timing and applicability of the HDR Dynamic Metadata Extended InfoFrame data relative to the video lines. The Metadata Transmission Window (MTW) is defined as the period of time beginning with the first Active Pixel of a Video Frame and ending with the final Blank Pixel of the final Blanking Line in a Vblank period. As shown by the arrow in Figure 7, HDR Dynamic Metadata transmitted during the MTW applies to the Active Pixels that immediately follow the MTW. Standards and specifications that incorporate CTA-861 may reduce the period of the MTW by delaying the start of the MTW and/or ending the MTW on an earlier Blanking Line.

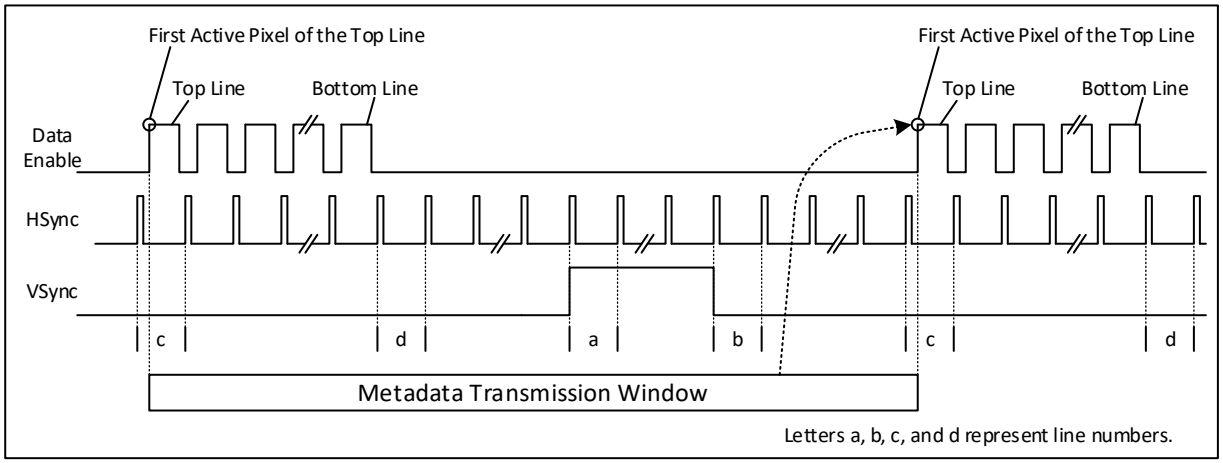

Figure 7 - HDR Dynamic Metadata Transmission Window and Metadata Applicability

The HDR Dynamic Metadata Extended InfoFrame is shown in Table 56 below:

Table 56 - HDR Dynamic Metadata Extended InfoFrame common structure

Extended InfoFrame Type Code	Extended InfoFrame Type LSB
	Extended InfoFrame Type MSB
Length (n) of Extended InfoFrame	Length of following Extended InfoFrame Data LSB
	Length of following Extended InfoFrame Data MSB
Data Byte 1	Application-specific Data 1
…	…
Data Byte *n*	Application-specific Data *n*

When Extended InfoFrame Type Code is set to 0x0001, the Application-specific Data in Data Bytes 1 – *n* is the display management message data that is specified in Annex R.

When Extended InfoFrame Type Code is set to 0x0002, the Application-specific Data in Data Bytes 1 – *n* is a concatenation of the Supplemental Enhancement Information (SEI) messages described in the Annexes of ETSI TS 103 433-1 [50] in any order. The data bytes defining payloadType and payloadSize as defined in Rec. ITU-T H.264 [51] and Rec. ITU-T H.265 [52] are included in the Application-specific Data. The processing mode to be used by the Sink with the HDR Dynamic Metadata is indicated by sl_hdr_mode_value_minus1 as described in ETSI TS 103 433-1 [50] Annex G.3.

When Extended InfoFrame Type Code is set to 0x0003, the Application-specific Data in Data Bytes 1 – *n* is the Colour Remapping Information (CRI) Supplemental Enhancement Information (SEI) message described in Rec. ITU-T H.265 [52]. The data bytes defining payloadType and payloadSize as defined in Rec. ITU-T H.265 [52] are included in the Application-specific Data.

When Extended InfoFrame Type Code is set to 0x0004, the Application-specific data in Data Bytes 1 – *n* is the Dynamic Metadata for Color Volume Transform — Application #4 of SMPTE ST 2094-40 [55] standard that is specified in Annex S.

A Source shall not send an Extended InfoFrame Type 0x0001, 0x0002, 0x0003, or 0x0004 to a Sink that does not indicate support for that Extended InfoFrame Type in the Sink's HDR Dynamic Metadata Data Block. A Source shall not send more than one type of HDR Dynamic Metadata Extended InfoFrame Types 0x0001, 0x0002, 0x0003, 0x0004 within the same MTW. A Source may send a Graphics Overlay Flag Extended InfoFrame when an HDR Dynamic Metadata Extended InfoFrame Type 0x0001, 0x0002, 0x0003, or 0x0004 is present.

When used, the HDR Dynamic Metadata Extended InfoFrame shall be sent every MTW and shall contain the data relevant to the immediately following Active Video period. Data in each HDR Dynamic Metadata Extended InfoFrame may be the same as that sent in previous HDR Dynamic Metadata Extended InfoFrames.

Where the information contained in the Dynamic Range and Mastering InfoFrame conflicts with the same data in the HDR Dynamic Metadata Extended InfoFrame, then the data in the HDR Dynamic Metadata Extended InfoFrame shall take priority.

6.10.2 Graphics Overlay Flag Extended InfoFrame

When the Extended InfoFrame Type Code is set to 0x0100, the Extended InfoFrame carries the Graphics Overlay Flag, which applies to the HDR Dynamic Metadata as specified in Annex T. The Application-specific Data in Data Bytes 1 – *n* is the Graphics Overlay Flag carried according to the syntax specified in Table 180.

A Source shall not send an Extended InfoFrame Type 0x0100 to a Sink that does not indicate support for that Extended InfoFrame Type in the Sink's HDR Dynamic Metadata Data Block.

A Source shall not send a Graphics Overlay Flag Extended InfoFrame containing Extended InfoFrame Type 0x0100 unless it is also sending an HDR Dynamic Metadata Extended InfoFrame containing an Extended InfoFrame Type 0x0001, 0x0002, 0x0003, or 0x0004 that applies to the same frame.

The Graphics Overlay Flag follows the same transfer timing and applicability as shown in Figure 7.

The format of the Graphics Overlay Flag Extended InfoFrame is shown in Table 56.

7 EDID DATA STRUCTURE

Extended Display Identification Data (EDID) was created by VESA to enable plug and play capabilities of displays (Sinks). This data, which is stored in the Sink, describes Video Formats that the DTV (display) is capable of receiving and rendering. The information is supplied to the Source, over the interface, upon the request of the Source. The Source then chooses its output format, taking into account the format of the original video stream and the formats supported by the Sink. Format conversions necessary to supply video to the Sink should be determined according to recommendations in Annex F.

The EDID data structures version 1, revision 3 [9] and newer are known as Enhanced EDID (E-EDID). Sources should interpret and Sinks should implement the EDID data structure according to the VESA E-EDID Implementation Guide [119]. Sink implementers should verify a Sink's EDID data structure using the VESA E-EDID Verification Guide [122]. The Sink shall support E-DDC [133] as the method of transporting EDID information. A Source shall use E-DDC [133] to read EDID Blocks since critical information may not otherwise be readable if the Sink contains a large EDID.

Some Sinks may contain more than 2 blocks of EDID data. For example, the Sink may include a second CTA-861 Extension Block, a VESA DI-EXT Block (as defined in VESA DI-EXT [117]) or a VTB-EXT Block (as defined in VESA VTB-EXT [125]). In this case the Sink is required to support E-DDC Addressing (using the Segment Pointer) as defined in VESA E-DDC [133]. It is also recommended that the Source be capable of reading and parsing more than 2 blocks of EDID data. For more information on E-DDC addressing refer to the VESA E-DDC Standard [133].

EDID Block 0 contains both version and revision numbers. The version number indicates the data structure of Block 0, which remains backwards compatible as the revision number changes. Therefore, a Source should continue parsing a recognized structure version — even if it encounters an unexpected revision number.

The Sink shall protect its EDID from accidental corruption resulting from communication errors by write-protecting its contents.

See Annex A and Annex D for example EDIDs.

7.1 Use of CTA Extensions

Two of the four 18-byte descriptor slots contained in EDID Block 0 are designated for a Monitor Range Limits Descriptor and a Monitor Name Descriptor. Users of CTA-861 should note that future alternate usage of these descriptors is possible, including replacing them with additional Detailed Timing Descriptors; therefore, dependency upon data in these descriptors should be avoided. Consequently, the E-EDID standard provides a method for including only two Detailed Timing Descriptors.

To accommodate additional Detailed Timing Descriptors, the CTA Extension has been defined. The tag (0x02) for this extension, previously reserved within VESA, has now been assigned to CTA for the purposes of CTA-861. Therefore, further changes to this structure are under the

control of CTA. It is referred to in CTA-861 as the CTA Extension. Section 7.3 defines the structure of the CTA Extension.

7.2 Describing Video Formats in EDID

Two methods of describing Video Formats are used in CTA-861: VESA-standard data structures and CTA Short Video Descriptors. The former includes the following:

- Established Timings in Block 0 (IT Timing Formats only)
- Standard Timings in Block 0 (IT Timing Formats only)
- 18-byte Detailed Timing Descriptors in Block 0 and CTA Extensions
- 20-byte Detailed Timing Descriptors in CTA Extensions (IT Timing Formats only)
- 1- and 2-byte Code indices in CTA Extensions (IT Timing Formats only)
- 6- or 7-byte CVT-based descriptors in CTA Extensions (IT Timing Formats only)

The Sink shall declare support in its EDID for all of the DTV formats that it supports. The 640x480@60Hz flag, in the Established Timings area, shall always be set, since the 640x480p format is a mandatory default timing: both Sinks and Sources shall support this video format at 24bpp RGB as a minimum point of interoperability. The Source shall read the EDID to determine if a specific format is supported. The Source shall not transmit video formats that are not contained within the EDID to which it is attached, if no format supported by the Source exists in the EDID, or the user manually overrides the format selection. In either of these exceptions, the Source should implement a UI test that verifies good operation with the user. This methodology should also be used in the case of a bad or missing EDID when switching video formats. If the Source does not otherwise know the Sink's capabilities, and if the Source cannot successfully read a Sink’s EDID, the Source shall output 640x480p 24bpp RGB.

When using CTA Extension Version 1, all of the CTA Video Formats listed in E-EDID are described using Detailed Timing Descriptors. No matter which CTA Extension is used, there is also room for two Detailed Timing Descriptors in EDID Block 0. CTA Extension Version 3 can include a combination of Detailed Timing Descriptors and Short Video Descriptors.

If a Version 3 CTA Extension has been included in EDID, all CTA Video Formats shall be advertised using Short Video Descriptors or Video Format Descriptors, even if they are also advertised using any VESA-standard data structures (see Section 7.2.1).

Even though Short Video Descriptors are available in the Version 3 CTA Extension, there is still a need to use Detailed Timing Descriptors if full backward compatibility with legacy Sources is desired. Formats with video ID codes of 2 to 5 and 17 to 20 should be advertised using the Detailed Timing Descriptors for any Video Formats that the DTV designer wishes to guarantee are available to Sources that cannot interpret the Short Video Descriptors and that require Detailed Timing Descriptors for proper operation. If sufficient room is not available in the first two blocks of the EDID for all of the supported Video Formats, the DTV designer may choose to declare support for some of the formats in Short Video Descriptors only.

A Sink that supports any Dual-Aspect Ratio Timing shall, in its EDID, list the DTD and SVD with the Preferred Picture Aspect Ratio before the DTD and SVD with the other Picture Aspect Ratio.

7.2.1 Use of EDID Detailed Timing Descriptors

As required in Section 4, a DTV that declares it is capable of displaying a Video Timing with a vertical frequency that is either an integer multiple of 6 Hz (excluding 300 Hz) or an integer multiple of 6 Hz adjusted by a factor of 1000/1001 shall be capable of displaying both versions of the Video Timing. DTVs capable of displaying 59.94/60 Hz versions of Video Timings shall declare in the EDID structure the 60 Hz version of the Video Timing for all Video Formats, except the 240-line and 480-line formats, which shall declare the 59.94 Hz version of the Video Timing.

Note that the EDID 18-byte Detailed Timing Descriptor allows for the designation of an interlaced format. However, there are no provisions to specify separate vertical blanking/sync for Field 1 and Field 2. Therefore, for the purposes of CTA-861, the following rules apply for interlaced formats:

a) The Field 1 Vertical Blanking Interval shall equal the Vertical Blanking Lines in the Detailed Timing Descriptor.

b) The Field 2 Vertical Blanking Interval shall equal the Vertical Blanking Lines in the Detailed Timing Descriptor + 1.

c) The Field 1 Vertical Sync Offset shall equal the Vertical Sync Offset in the Detailed Timing Descriptor.

d) The Field 2 Vertical Sync Offset shall equal the Vertical Sync Offset in the Detailed Timing Descriptor + 1/2.

A Sink capable of receiving a Video Format with a Video Identification Code greater than 7 or capable of receiving a Dual-Aspect Ratio Timing shall declare different Detailed Timing Descriptors in its EDID for each supported Video Timing with a different Picture Aspect Ratio. The vertical and horizontal image size parameters in the EDID shall contain numbers that describe the aspect ratio of the displayed video (actual dimensions are preferred, but not required).

A special interlaced Video Timing exists (see Figure 5) that modifies the Field 2 Vertical Blanking Interval (b) and Vertical Sync Offset (d) values presented here. When all DTD parameters match those of Video Identification Code 39 (see Table 57) and a SVD indicating support for code 39 Video Format also exists, the Field 2 Vertical Blanking Interval (b) and Vertical Sync Offset (d) shall instead equal the DTD's "Vertical Blanking Lines" and the DTD's "Vertical Sync Offset" - 1/2, respectively.

Table 57 - Video Timing Code 39 Detailed Timing Descriptor

Byte #	Data Hex	Dec	Description	Remarks
1	0x20	32	Pixel Clock	72.00 MHz
2	0x1C	28		
3	0x80	128	H Active	1920 pixels
4	0x80	128	H Blanking	384 pixels
5	0x71	113	H Active: H Blanking	
6	0x1C	28	V Active	540 lines
7	0x55	85	V Blanking	85 lines
8	0x20	32	V Active: V Blanking	
9	0x20	32	H Sync Offset	32 pixels
10	0xA8	168	H Sync Pulse Width	168 pixels
11	0x75	117	VS Offset: VS Pulse Width	23 lines, 5 lines
12	0x04	4	HS Offset: HS Pulse Width: VS Offset: VS Pulse Width	
13	(any)	(any)	H Image Size	(any)
14	(any)	(any)	V Image Size	(any)
15	(any)	(any)	H&V Image Size	
16	0x00	0	H Border	0 lines
17	0x00	0	V Border	0 pixels
18	0x9A, 0x9B, 0xBA, 0xBB, 0xDA, 0xDB, 0xFA, or 0xFB	154, 155, 186, 187, 218, 219, 250, or 251	Flags	Interlaced, digital separate, Vsync polarity is negative, Hsync polarity is positive (NOTE — stereo mode bits 0, 5, and 6 may have any value)

Examples of Detailed Timing Descriptors for the Video Formats are contained in A.2.10.

7.2.2 Preferred Video Formats and Native Video Resolution

Sinks should include a Video Format Preference Data Block (VFPDB, Section 7.5.12) to indicate preferential ordering of Video Formats. Preference-by-location, such as ordering of Detailed Timing Descriptors (DTD) within the EDID, or VICs within Video Data Blocks, has been deprecated. However, some legacy Sources may consider the first 18-byte DTD in Block 0 and the first SVD to be preferred, and Sinks may want to take this into consideration when populating those items.

CTA Extension Version 1 is only capable of indicating preference using the first DTD.

Sources can use the following pseudocode to determine the preferred video formats:

```
If the CTA Extension is Version 1
        The first DTD in Block 0 is preferred
Elseif a Video Format Preference Data Block (VFPDB) is present
        Video Formats are listed in order of preference
Else
        The first DTD in Block 0 and the first SVD are preferred
End If
```

Sinks should include a Native Video Resolution Data Block (NVRDB, Section 7.5.18) to indicate the native video resolution of the Sink.

Sources can use the following pseudocode to determine the native video resolution:

```
If the CTA Extension is Version 1
        The native video resolution of the Sink is unknown
Elseif a Native Video Resolution Data Block (NVRDB) is present
        The NVRDB defines the native video resolution
Else
        The first N 18-byte DTDs (N = CTAX_NATIVE_DTDS, see Table 60) and any SVDs
        with the Native bit indicate native video resolutions

        If N = 0 and none of the SVDs have the native bit set
                        The native video resolution is unknown
        Elseif all provided native video resolutions are the same
                        This is the native video resolution
        Else
                        Sources may assume the native video resolution of the Sink is
        unknown
        End If
End If
```

A Source shall override any other native video resolution indicators if the Source supports the NVRDB and the NVRDB was found in the E-EDID.

The legacy method of marking multiple DTDs and/or SVDs with different video resolutions as representing the native resolution is deprecated[20]. When the Sink indicates multiple different video resolutions as native (e.g., 720p and 1080i), Sources may assume the native resolution is unknown.

[20] Until CTA-861-C a Native Format was defined as any Video Format that a display device could handle without timing or scaling conversion. So a Video Format that required a pillar or letter box but otherwise needed no conversion was still considered a Native Format. This definition made it difficult to determine the Native Video Resolution.

From CTA-861-D onwards the Native Format definition was narrowed to the current definition, which requires that Native Formats have the Native Video Resolution. So Video Formats that require a pillar or letter box are no longer considered a Native Format.

7.3 CTA Extension

Three versions of the CTA Extension exist. If more than one CTA Extension is included in EDID, they shall all be the same version.

To maintain backward compatibility, newer versions of the CTA Extension include all of the fields that were present in the previous versions. Additionally, length fields are provided on internal data structures to convey block size and to aid the Source in interpreting the data. Having block sizes should help Source devices move from block-to-block, when blocks contain corrupt or Vendor-Specific data. Future versions of the CTA Extension are expected to have the version number incremented and be backward compatible with previous versions. A current generation Source is capable of parsing these future EDIDs exactly as it does existing EDIDs, if it ignores the version number. Sources should continue parsing the EDID structure even if an unexpected version number is encountered.

CTA Extension Version 1 only provides a way to supply extra Detailed Timing Descriptors. It is still permitted to be used for some Sinks (e.g., limited format DVI displays) but Version 3 is more applicable for most devices.

CTA Extension Version 2 is no longer supported and shall not be included in Sinks.

CTA Extension Version 3 includes all of the fields and capabilities of Versions 1 and 2, but also includes the ability to specify any of the CTA Video Formats using “CTA Short Video Descriptors" or "CTA Video Format Descriptors.” It provides the ability for the Sink to specify what types of advanced audio it supports using “CTA Short Audio Descriptors.” It also provides a way for the Sink to specify its speaker configuration. This information is correlated to the speaker channel allocation information that is sent in the Audio InfoFrame.

If a Sink supports any Video Format with a format code greater than 7, YC_BC_R color space, InfoFrames, or digital audio (e.g., is an HDMI monitor), then it shall include the version 3 (or higher) CTA Extension in its EDID data structure.